Job Preferences, Labor Market Power, and Inequality

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Abstract

This paper studies how a firm's labor market power shapes, and is shaped by, its workforce, and evaluates the consequences for wage inequality and welfare. Using matched worker-firm panel data from Norway (1995-2018), I develop, identify, and estimate an equilibrium model of the labor market where firms compete for workers who are heterogeneous in both skills and preferences over wages versus non-wage job amenities. I allow workers' wage-amenity trade-offs to correlate with skills, while also varying among equally skilled workers. When a firm adjusts its wages, its workforce composition shifts, and this in turn affects the shape of its labor supply curve. Thus, a firm's wage setting power varies based on which workers it employs. I use the model to draw inference about the incidence of labor market power within and across firms and the implications for wage inequality and sorting. Eliminating market power widens within-firm skill premia while compressing wage differences between firms, leading to a 4% reduction in overall wage inequality and a 3.3 percentage-point (23% of the baseline) decline in the gender pay gap. Variation in wage setting power also generates substantial allocative inefficiency, with welfare losses from imperfect competition estimated to be 9.6% relative to the competitive benchmark.

I. Introduction

The role of firms in shaping wage inequality is a central topic for labor policy. Three growing empirical literatures analyze how the characteristics of firms influence workers' earnings. One body of research (Manning, 2021; Card, 2022; Azar & Marinescu, 2024; Kline, 2025) studies the role of imperfect competition in labor markets, where firms exert wage setting power due to facing upward-sloping labor supply curves. A second literature, building on work by Abowd, Kramarz, & Margolis (1999), focuses on sorting, using linked employer-employee panel data to uncover the relationship between worker and firm productivity. A third literature examines

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workers' willingness to trade higher wages for better non-wage aspects of a job, finding that the trade-offs vary substantially among workers (Mas & Pallais, 2017; Wiswall & Zafar, 2018).

While often studied in isolation, the topics explored in these literatures are interconnected. In a labor market with heterogeneous workers and firms, employers with different characteristics attract different types of workers: some who are highly responsive to wages and others who are less responsive, prioritizing non-wage factors. Because of this sorting, the composition of a firm's workforce can affect its wage setting power by shaping the elasticity of labor supply that it faces. This raises a key question: *How is a firm's wage setting power linked to its workforce?* Addressing this question would offer new insight into the determinants of worker sorting and the incidence of market power, which have key consequences for inequality and welfare.

This paper develops an empirical framework to analyze how workers choose jobs based on heterogeneous wage-amenity trade-offs, and the consequences of this sorting for firms' wage setting power. I show that the market power of a firm—as measured by wage markdowns and rents—depends directly on the composition of its workforce, which is influenced by the wages and amenities the firm provides. Moreover, by failing to account for this relationship, one would overlook a key source of allocative inefficiency that arises from differences in wage markdowns among heterogeneous firms. I show how to recover key economic quantities under my framework using linked worker-firm panel data, while allowing for unobserved heterogeneity in worker skills, firm productivity, technology, and amenities. I then apply my framework to administrative data from Norway, where I draw inference about the determinants of worker sorting, the concentration of market power, and the welfare impacts of imperfect competition.

I begin my analysis by presenting an equilibrium model of the labor market. Building on previous models of job differentiation, such as Card et al. (2018) and Lamadon et al. (2022), I assume workers' idiosyncratic tastes are not fully priced into wages, leading firms to face imperfectly elastic labor supply curves. Yet, unlike previous work, I allow workers to have varying marginal rates of substitution between wages and amenities. Drawing on the literature on the willingness to pay for amenities, I allow the wage-amenity trade-offs to correlate with workers' skills, while also differing among equally skilled workers. These differences can reflect a variety of factors such as household circumstances, financial constraints, or social ties, and I allow the distribution of workers' trade-offs to be fully nonparametric and unobserved to the analyst.

This extension has key implications for the shape of the labor supply curve faced by a firm. It establishes a link between worker and firm characteristics, so that the elasticity of labor supply the firm faces depends on the composition of its workforce. As a result, the elasticities vary across firms and are also endogenous to changes in the environment: if a firm adjusts its wages, then its workforce composition shifts, and this in turn influences the elasticity of labor supply it faces. This relaxes with the standard assumption of homogenous and isoelastic labor supply

curves, which implies uniform wage markdowns across firms. Moreover, to the extent that wage-amenity trade-offs correlate with workers' skills (on which firms can condition wages), markdowns also differ within a firm, as well as across firms. These implications do not rely on firms competing strategically for workers, as in Berger et al. (2022). Even if firms are strategically small in the labor market, their wage setting power still varies based on *who* they employ.

To account for many sources of worker sorting, I allow firms to be differentiated along several dimensions. Specifically, I allow for match-specific productivity, so a worker's skills may be valued differently at different firms. I also allow for firm-specific production technology, so that firms can exhibit different returns to scale and differ in the degree of substitutability between labor inputs. Moreover, I allow firms to have distinct non-wage amenities, which can vary among workers within a firm.[1] I examine how each of these firm characteristics interacts with worker preferences in equilibrium and their consequences for the incidence of market power.

I then turn to the question of identification, demonstrating how to learn about imperfect competition under the model using linked worker-firm panel data. In my analysis, identification is challenging for two reasons. First, as is typical in most contexts, a worker's wage markdown is not observed in the data. In particular, observationally similar workers can earn different wages due to unobserved skill differences, not market power.[2] Wages may also not fully reflect a worker's compensation, as firms may have amenities that researchers do not observe. Indeed, as Rosen (1986) discusses, compensating differentials can exist even in competitive markets. A second identification challenge stems from the generality of my framework, which allows workers' wage markdowns to vary both within and across firms, and to be endogenous to wage changes within a firm. This added feature complicates my analysis because instead of recovering a single wage markdown, I need to recover entire functions of firm characteristics.

I show how to overcome these challenges to achieve identification of the model through the use of panel data by leveraging an economic shock that shifts the labor demand of multiple firms in the economy. These labor demand shocks can be recovered either from existing data using internal instruments or from external data sources, and I leverage both methods in my analysis. One novelty of my approach is that I allow the labor demand shocks to shift wages in a way that is reflected in market aggregates, thus impacting workers' reservation wages, skill investments, and preferences. This generates spillovers on the labor supply curves faced by *all* firms, including those that do not directly experience a shock. Such spillovers prevent me from recovering the elasticity of labor supply to a firm by directly comparing firm-level outcomes before and after the shocks. Nevertheless, I show that, under weak conditions on the instrum-

[1] In my analysis, I do not take a stance on which amenities matter most to workers. Rather, I allow firms to have a set of amenities that are unobserved to the analyst and are assumed to remain fixed over the estimation window. This neither imposes nor precludes the possibility that firms initially choose amenities to maximize profits. Indeed, it is straightforward to show that letting firms choose their amenities initially would not impact the estimates.

[2] For additional discussion, see Murphy & Topel (1990), Gibbons & Katz (1992), and Gibbons et al. (2005).

ent, the elasticities are identified from a difference-in-differences comparison of wage and labor outcomes. This method involves comparing mean changes in wages and labor among firms with different exposures to labor demand shocks, while controlling for a firm's initial wages.

An additional methodological contribution is that, by recovering the labor supply elasticities faced by firms, I can infer the distribution of workers' wage-amenity trade-offs. This result is grounded in a revealed preference argument: holding all else fixed, firms that offer higher wages to a given skill group disproportionately attract workers who respond more strongly to wages. This compositional effect is reflected in the labor supply elasticities firms face. Leveraging this relationship, I prove there exists a one-to-one mapping between a firm's labor supply curve and the density of workers' wage-amenity trade-offs. I then derive a nonparametric estimator of this density, which can be implemented separately for different population subgroups.

Using my framework, I show how to recover the model parameters that characterize firm productivity, technology, amenities, and workers' skills and preferences. From these parameters, I can quantify the division of rents between workers and firms, and conduct counterfactual analyses that isolate the role of imperfect competition on wage inequality and worker sorting.

Lastly, I estimate the model using a linked employer-employee panel dataset covering all workers and firms in Norway for the period 1995-2018. My estimates yield three key findings. First, many employers have substantial wage setting power over workers, and this power varies considerably both within and across firms. While on average, a worker's wage constitutes 86% of her marginal product, the wage-to-marginal-product ratio ranges from 70% to 95% between the 1st and 99th percentiles. Moreover, many workers and firms earn sizable rents: the average worker is willing to forgo 16% of her wage to be at her current firm, while the average firm derives 23% of its profit from wage setting power. Rents are highly dispersed in the economy, with worker rent shares ranging from below 10% to over 25% of earnings, and firm rent shares ranging from below 5% to over 50% of profit. I document systematic differences in wage markdowns and rents across industries and regions and by worker skills and demographics, giving a detailed account of which workers are most affected by market power. In a counterfactual that shuts down variation in workers' wage-amenity trade-offs, one fails to recover these differences, as the restricted model implies a constant wage markdown and little scope for rents to vary.

Second, given how wage markdowns vary within and across firms, I find that labor market power has sizable effects on wage inequality. Within firms, higher skilled workers tend to face larger wage markdowns, whereas, across firms, markdowns are largest at firms that pay lower wages to a given skill group. Consistent with these patterns, and using the variance of log earnings to measure inequality, I find that eliminating labor wedges raises within-firm inequality by 1.3% and reduces between-firm inequality by 16%, yielding a 4% decline in total inequality. Moreover, I find that women are disproportionately employed at high-markdown firms, and

that removing labor wedges would reduce the gender wage gap by 3.3 percentage points from 14.3% to 11.0%. This result highlights the role of monopsony power in sustaining gender pay differences, building on work dating back to Robinson (1933).[3] These effects would not be captured by a model that does not allow for rich heterogeneity in workers' labor supply responses.

Third, I find that labor market power leads to substantial misallocation of workers to firms. Eliminating labor wedges would raise total welfare by 9.6% in dollar terms. In the competitive allocation, more workers would be employed at lower-productivity firms. This finding reflects that labor wedges are largest at firms that pay low wages to a given skill group, and those firms tend to be less productive. Thus, removing wedges leads to larger wage and employment gains at the bottom of the firm productivity distribution.[4] At the sector level, I find that employment would shift on net away from high-premium industries such as finance toward retail, education, and healthcare. Employment would also rise in rural labor markets. This reallocation is highly heterogeneous across skill groups, in both size and direction. I also show that it would be overlooked in a version of the model that rules out variation in workers' wage-amenity trade-offs.

This paper contributes to three literatures. First, it contributes to a large literature on monopsony in labor markets; see Manning (2021), Card (2022), Azar & Marinescu (2024), and Kline (2025) for recent surveys.[5] Within this literature, I build on a class of static equilibrium models studied by Card et al. (2018), Azar et al. (2022), Lamadon et al. (2022), Kroft et al. (2025), Chan et al. (2025), Roussille & Scuderi (2026), and others, where imperfect competition arises because workers' idiosyncratic tastes are not fully priced into wages. This work, which Manning (2021) terms "new classical monopsony," draws on differentiated demand models commonly used in Industrial Organization. A limitation of these models, however, is that they often specify utility functions that strongly restrict how worker and firm characteristics interact. These restrictions constrain worker sorting in a way that limits the scope for firm market power to vary.[6] I relax these restrictions to allow for fully unrestricted and unobserved

[3]For recent empirical work on monopsony and gender pay gaps, see Barth & Dale-Olsen (2009), Card et al. (2016), Sharma (2023), Caldwell & Oehlsen (2023), and Morchio & Moser (2024). This work often infers gender differences in monopsony power by estimating firm-level labor supply elasticities separately for men and women. I extend this approach by allowing elasticities to vary within gender and, for each gender, estimating the full distribution of workers' wage-amenity trade-offs. This provides a richer characterization of gender wage gaps by revealing whether gender differences in monopsony are widespread or concentrated in certain parts of the economy.

[4]This result runs counter to the common presumption that larger, more productive firms will necessarily have more wage setting power, an idea rooted in single-buyer monopsony models dating back to Robinson (1933). However, when labor markets contain many firms, wage-setting power need not increase with firm size (Manning, 2021). Indeed, my estimates show that firm characteristics, rather than size per se, play the central role in shaping wage setting power, by determining which workers the firm attracts and how responsive they are to wage changes.

[5]See also Boal & Ransom (1997), Bhaskar et al. (2002), Manning (2011), and Sokolova & Sorensen (2021).

[6]Abstracting from strategic interactions, it implies uniform wage markdowns. As discussed in Section II.E, this restriction can be relaxed by adopting a nested logit model (Lamadon et al., 2022; Azar et al., 2022) or by letting markdowns vary within a firm based on observed worker characteristics (Deb et al., 2024; Chan et al., 2025). Roussille & Scuderi (2026) also relax this assumption while restricting preference heterogeneity to a finite set of

heterogeneity in workers' wage-amenity trade-offs. This extension is in the spirit of consumer demand models with rich preference heterogeneity (Berry et al., 1995). As I show, it greatly expands the scope for worker sorting, yielding a richer characterization of firm market power.

Second, this paper contributes to an empirical literature on sorting, drawing on the additive worker and firm effects wage model introduced by Abowd, Kramarz, & Margolis (1999). This work largely focuses on skill-based sorting, seeking to uncover the relationship between worker and firm productivity (e.g. Andrews et al., 2008; Card et al., 2013; Song et al., 2018). While my model allows for sorting on workers' skills, e.g., through production complementarities, I also study a second type of sorting based on workers' wage-amenity trade-offs. I show that this feature introduces a firm-skill interaction term—the wage markdown—in the worker's wage equation, such that the same firm can apply a larger markdown to some workers and a smaller markdown to others. In doing so, my paper provides new insights into the assortativeness of the allocation of workers to firms. In particular, I find evidence that wage markdowns are match-specific: firms are not uniformly "high-markdown" or "low-markdown" for all workers, so market power distorts the allocation of labor differently for different types of workers.

Third, this paper contributes to a growing literature on the willingness to pay for non-wage amenities. Both experimental studies and survey data show that workers value a wide array of non-wage job attributes (Hamermesh, 1999; Pierce, 2001; Maestas et al., 2017). Furthermore, the willingness to pay for these amenities varies across workers (Mas & Pallais, 2017; Wiswall & Zafar, 2018), contributing to wage inequality (Maestas et al., 2023).[7] This paper takes the literature in a new direction by exploring how differences in workers' amenity valuations affect firms' wage-setting power in imperfectly competitive labor markets. I show that workers' wage-amenity trade-offs help to shape the labor supply elasticities faced by employers. Moreover, by recovering these elasticities, one can infer the distribution of workers' trade-offs. This delivers a new revealed-preference approach to learn about workers' willingness to pay for amenities, building on work by Sorkin (2018), Hall & Mueller (2018), Lagos (2026), and others.

This paper is organized as follows. Section II presents the model along with equilibrium properties related to sorting and rent sharing. Section III describes the data and sample selection. Section IV develops the identification analysis. Section V outlines the estimation and reports parameter estimates. Section VI presents the main empirical results on the incidence of labor market power and the consequences for inequality and welfare. Section VII concludes.

types based on observables. In my estimates, I find that allowing for fully unobserved heterogeneity in workers' preferences and productivities is essential for evaluating how wage markdowns and rents vary in the economy.

[7] For a recent comprehensive review of the empirical literature on non-wage amenities, I refer to Mas (2025).

II. Equilibrium Model of the Labor Market

In this section, I develop a theoretical framework to characterize how workers sort into jobs and the implications of these sorting patterns for a firm's wage-setting power. This framework extends prior equilibrium models of the labor market, studied by Lamadon et al. (2022), Azar et al. (2022), Kroft et al. (2025) and others, by allowing workers to differ in the marginal rate of substitution between wages and non-wage aspects of jobs. I show that this extension leads firms to face heterogeneous labor supply elasticities, which reflect differences in the composition of workers at different firms. As a result, this model generates variation in the incidence of wage markdowns and rents, thus introducing a potential source of allocative inefficiency.

II.A. Environment

The economy comprises a unit measure of workers, indexed by i, and a finite number of firms, indexed by j. Every firm operates in a local labor market $m(j)$, where $\mathcal{J}_m$ is the set of firms in market m. Workers are differentially productive, and they have heterogeneous preferences over employers. Meanwhile, firms provide differentiated work environments, and they vary in their productivities and technologies. Note that, while distinguishing between labor markets will be important for the empirical analysis, it does not meaningfully impact the theoretical results. I therefore simplify notation by focusing on a single market and suppressing the label m.

Each worker has a vector of skills $X_i = (\chi_i, \varphi_i)$. These skills contain two components: a categorical *skill type* $\chi_i \in \mathcal{X}$ and a continuous *skill level* $\varphi_i \in \mathbb{R}_+$. The skill type χ_i captures non-ordinal aspects of human capital, such as training and specialization. The skill level φ_i, which I treat as unobservable to the researcher in the empirical analysis, measures productivity within a given skill type. By modeling skills as a composite of types and levels, the framework incorporates insights from the literature on multidimensional skills (e.g., Lise & Postel-Vinay, 2020), while also allowing human capital to vary in ways that a researcher cannot observe.

Throughout the analysis, I will maintain two assumptions about the information structure. First, I abstract from search frictions by assuming that workers are fully informed about job opportunities and that, given posted wages, a worker freely chooses where to work.[8] Second, I assume that firms can observe workers' skills, but they only know the distribution of workers' preferences conditional on those skills. Therefore, while firms can set wages based on X_i, they cannot further price-discriminate with respect to each worker's idiosyncratic preferences.[9]

[8] When specifying firm production, I allow worker-firm matches to be nonproductive. This effectively narrows the relevant choice set for many workers, while maintaining the assumption that workers are fully informed.

[9] This assumption reflects information asymmetries between workers and firms. It may also account for laws against pay discrimination, which are common in many labor markets. See Bhaskar et al. (2002) for discussion.

II.B. Worker Preferences

Workers have heterogeneous preferences over wages and non-wage amenities. Let $W_j(X)$ be the wage that a firm j offers to workers with skills X, and let $a_j(X)$ be a firm's skill-specific amenity. For any worker i with skills X_i, the indirect utility from working at firm j is:

$$u_{ij}\big(W_j(X_i), a_j(X_i)\big) = \beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij}. \tag{1}$$

In this utility function, ϵ_{ij} is a worker's idiosyncratic taste for firm j, reflecting non-pecuniary match effects such as distance from work and personal preferences over work environment. I assume $\{\epsilon_{ij}\}_{i,j}$ are i.i.d. with a continuous distribution F_ϵ taking strictly positive density on $\mathbb{R}$.

The coefficient β_i is the worker's marginal utility of log earnings. This parameter governs the marginal rate of substitution between wages and amenities, thereby capturing the trade-off between higher pay and better working conditions. Workers with lower β_i will be more willing to sacrifice earnings in exchange for better non-wage job characteristics. In Appendix A.9, I provide an explicit micro-foundation for the utility function (1), where I show that β_i can be reinterpreted as the worker's marginal rate of substitution between consumption and leisure.

In my analysis, I allow β_i to vary freely among workers. In particular, I treat β as a random variable with density f_β, and I do not impose any parametric restrictions on this density, such as assuming normality. One important aspect of this framework is that I allow for two distinct forms of heterogeneity in β. First, I allow β to correlate with a worker's skills X, so that the trade-offs workers make when choosing jobs can depend on factors that impact productivity. Second, I allow β to vary conditionally within each skill group X, so that equally skilled workers can have different wage-amenity trade-offs. This within-skill heterogeneity is particularly important, as it allows worker's trade-offs to reflect considerations beyond productivity. For example, β may depend on factors such as age, gender, household characteristics, and wealth. It may also be correlated among workers in the same household, neighborhood, or peer group.

II.C. Firm Production

Let $D_j(\chi, \varphi)$ denote the mass of workers with skills $X = (\chi, \varphi)$ that is demanded by firm j. For each skill type $\chi \in \mathcal{X}$, the total efficiency units of labor at the firm are defined as follows:

$$L_j^{\text{eff}}(\chi) = \int \varphi D_j(\chi, \varphi) d\varphi. \tag{2}$$

Define Y_j to be the firm's value added from hiring workers. The firm's production function is a constant elasticity of substitution (CES) composite of the effective labor from different skill types. This specification permits imperfect substitutability between skill types and allows for decreasing returns to scale, so that output need not increase in direct proportion to labor inputs.

$$Y_j = T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(L_j^{\text{eff}}(\chi) \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}}. \tag{3}$$

In this production function, $T_j \in \mathbb{R}_{++}$ is the firm's total factor productivity (TFP); $\theta_{j\chi} \in \mathbb{R}_{++}$ is the firm-specific efficiency of skill type χ; $(1-\rho_j)^{-1} \in [0, \infty)$ is the elasticity of substitution between skill types at the firm; and $1 - \alpha_j \in (0, 1]$ is the firm's returns to scale. For simplicity, I abstract from capital and intermediate inputs, as well as from the nature of competition in the product market. Nevertheless, as I show in Appendix A.10, this production function can be rationalized under either a perfectly competitive or an imperfectly competitive product market.

By allowing the parameters $(T_j, \{\theta_{j\chi}\}_{\chi \in \mathcal{X}}, \rho_j, \alpha_j)$ to vary freely across firms, the model accommodates a wide range of production technologies. First, firms can differ in their overall productivity through T_j. Second, firms can differ in the relative efficiency of each skill type through $\theta_{j\chi}$; for example, some firms may place more value on a college degree than others. Third, firms can differ in the substitutability of skill types through ρ_j; for example, some firms may have specific task requirements that affect their ability to interchange high- and low-skill workers. Fourth, firms can differ in their returns to scale through α_j. Throughout the analysis, I place no restrictions on the joint distribution of firm productivity, technology, and amenities.

II.D. Characterization of an Equilibrium

In equilibrium, each worker chooses a firm to maximize utility. Given wage offers $\{W_k(X)\}_{k=1}^J$, a worker with preference parameter β and skill profile X would select firm j with probability:

$$\mathrm{P}\left(j(i) = j | \beta, X\right) = \int_{-\infty}^{\infty} \prod_{k \neq j} F_\epsilon \left(\beta \log \left(\frac{W_j(X)}{W_k(X)} \right) + a_j(X) - a_k(X) + \tilde{\epsilon} \right) f_\epsilon(\tilde{\epsilon}) d\tilde{\epsilon}. \tag{4}$$

Averaging over realizations of β, the labor that is supplied to firm j by workers with skills X is:

$$S_j(X) = \int \left(\int_{-\infty}^{\infty} \prod_{k \neq j} F_\epsilon \left(\beta \log \left(\frac{W_j(X)}{W_k(X)} \right) + a_j(X) - a_k(X) + \tilde{\epsilon} \right) f_\epsilon(\tilde{\epsilon}) d\tilde{\epsilon} \right) f_{\beta,X}(\beta, X) d\beta. \tag{5}$$

Given these labor supply curves, each firm sets wages to maximize profit. Since a firm cannot condition wages on individual preferences (β_i, ϵ_{ij}), its wage-setting decisions are based on the distributions of these parameters, $\{F_{\beta|X}\}_X$ and F_ϵ. The firm's profit maximization problem is:

$$\max_{\{W_j(\chi,\varphi)\}_{\chi,\varphi}} T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} - \sum_{\chi \in \mathcal{X}} \left(\int W_j(\chi, \varphi) D_j(\chi, \varphi) d\varphi \right), \tag{6}$$

subject to the constraint that labor demand equals supply: $D_j(X) = S_j(X)$ for all $X = (\chi, \varphi)$.

I consider an equilibrium in which firms view themselves as strategically small in the economy. Specifically, the firm does not internalize the effect of changing its own wages on the labor supplied to other firms. It is worth noting that this condition does not imply that workers regard firms as infinitesimal. On the contrary, there is a finite number of firms, and each worker chooses a firm by considering the entire set of offered wages and amenities. Instead, this condition applies only to firm conduct; it assumes that a firm sees itself as a marginal player, thus ruling out strategic interactions in wage-setting. This assumption also appears in Card et al. (2018), Lamadon et al. (2022), Azar et al. (2022), and Kroft et al. (2025), among others.[10] In Section V, I derive a statistical test of this assumption. Applying this test to Norwegian administrative data, I find no evidence to reject that firms view themselves as strategically small.

Definition. Given worker skill and preference distributions $(F_{\chi,\varphi}, \{F_{\beta|\chi,\varphi}\}_{\chi,\varphi}, F_\epsilon)$, firm amenities $\{a_j(\chi,\varphi)\}_{\chi,\varphi}$ and production parameters $(T_j, \{\theta_{j\chi}\}_\chi, \rho_j, \alpha_j)$, an equilibrium is defined by the employment outcomes $j(i)$, labor supply curves $S_j(\chi,\varphi)$, and wages $W_j(\chi,\varphi)$ where:

(i) Each worker chooses the firm that maximizes utility, with utility given by equation (1).

(ii) Labor supply curves $S_j(\chi,\varphi)$ are consistent with workers' optimal decisions, as in (5).

(iii) Each firm sets wages $W_j(\chi,\varphi)$ for all $\chi \in \mathcal{X}$ and $\varphi \in \mathbb{R}_+$ to maximize profit, as defined in equation (6), ensuring that labor demand $D_j(\chi,\varphi)$ is equal to labor supply $S_j(\chi,\varphi)$.

In Lemma 1 of Appendix C.1, I establish the existence of an equilibrium with strictly positive wages and employment. Lemmas 2 and 3 further characterize properties concerning equilibrium uniqueness. Additionally, I analyze properties such as the concavity of the profit function and dynamic stability of the equilibrium. These results are useful for counterfactual analysis, as they enable me to compute equilibrium outcomes from the underlying model parameters.

II.E. Properties of the Labor Supply Curve to a Firm

A primary motivation for incorporating random coefficients in the worker's utility function is that it enables the model to capture rich substitution patterns that reflect sorting on individual heterogeneity. This subsection examines how this heterogeneity expands the scope of worker behavior in the model and what it implies about the shape of the labor supply curve to a firm.

To streamline the exposition, I focus on a version of the model where ϵ_{ij} follows a Type 1 Extreme Value (logit) distribution, $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$. This is the most common parameterization of ϵ_{ij} used in the literature, and it yields transparent expressions for the firm's labor supply curve and its elasticities. The labor supply curve, as specified in equation (5), becomes:

[10] Alternatively, Berger et al. (2022), Deb et al. (2024), and Chan et al. (2025) study models of job differentiation where firms leverage their market shares when setting wages, and Jarosch et al. (2024) studies a search model with large firms. Roussille & Scuderi (2026) develop a method to adjudicate between models of firm conduct and find, in their context, that models that abstract from strategic interactions outperform those that include them.

$$S_j(X) = \int \frac{\exp\big(\beta \log W_j(X) + a_j(X)\big)}{\sum_{k=1}^{J} \exp\big(\beta \log W_k(X) + a_k(X)\big)} f_{\beta,X}(\beta, X) d\beta. \tag{7}$$

Define $I(\beta, X) = \sum_{k=1}^{J} \exp\big(\beta \log W_k(X) + a_k(X)\big)$ to be the wage index for workers with preference parameter β and skills X. From the perspective of a strategically small firm, its wage has a negligible effect on $I(\beta, X)$. Note that, although I work with the logit error structure for expositional clarity, the main insights in this subsection do not depend on this specific distribution. Appendix Properties A.9 and A.10 show the results extend to more general error structures.

II.E.1. Labor Supply under Homogeneous Effects

The standard assumption in models of job differentiation is that β is constant across workers. Under this assumption, firm wages and amenities do not interact with worker-specific preferences in the utility function.[11] This separability between worker and employer characteristics delivers a relatively simple and tractable labor supply system. Yet, it imposes a strong form of homogeneity on aggregate substitution patterns, which significantly restricts worker sorting. This critique has previously been raised in the context of product markets (Berry et al., 1995). Moreover, as I will now demonstrate, it also holds significant implications for labor markets.

To see how this assumption restricts economic behavior, consider two firms, j and j', with different technologies and work environments. In a model with constant β, these firms exhibit identical substitution effects. That is, if a worker at j and a worker at j' both receive a wage cut, each is just as likely to substitute toward any third firm k, regardless of whether k is more similar to j or to j'. This contradicts the usual intuition that similar firms attract similar types of workers. It is also hard to reconcile with the observed sorting patterns in most labor markets.

One key economic quantity that is impacted by this restriction is the elasticity of labor supply faced by a firm, $\frac{\partial \log S_j(X)}{\partial \log W_j(X)}$. If β is homogeneous, then this elasticity is identical across firms:

$$\frac{\partial \log S_j(X)}{\partial \log W_j(X)} = \frac{\partial \log S_{j'}(X)}{\partial \log W_{j'}(X)}, \quad \text{for all } j, j' \in \{1, \ldots, J\}. \tag{8}$$

This property underscores the limitations of assuming homogeneous effects. It implies that all firms, regardless of their characteristics, face the same employment response to a wage change.

One way to relax these restrictions on worker sorting patterns is to allow ϵ_{ij} to be correlated across firms in a local labor market, for example, via a nested logit structure. This approach is adopted by Azar et al. (2022), Lamadon et al. (2022), and others, and it implies that workers are more likely to substitute toward firms within markets than across markets. Nevertheless,

[11] Specifically, conditional on skills, workers differ in how they rank firms only due to i.i.d. taste shocks ϵ_{ij}.

in any given labor market, the same restrictions on worker sorting patterns would still apply.[12]

A second extension could be to allow β to vary based on observable worker characteristics such as education or experience.[13] Yet, a concern with this strategy is that firms are also likely to observe these characteristics and would thus factor them into a worker's skills X. Indeed, as long as β is constant conditional on skills, the implications of equation (8) remain unchanged.

II.E.2. Labor Supply under Heterogeneous Effects

If β is heterogeneous, then a firm's labor supply curve reflects differences in the composition of workers at different firms. For example, a firm offering higher wages and worse amenities is more likely to attract workers who place greater value on wages (higher β), whereas a firm offering lower wages and better amenities tends to attract workers who place greater value on amenities (lower β). Thus, firms with similar wage-amenity profiles are more likely to attract workers who make similar trade-offs, which leads these firms have larger substitution effects.

This variation in workforce composition shapes the elasticity of labor supply to a firm. In particular, this elasticity can be expressed as the average value of β within the firm's workforce:

$$\frac{\partial \log S_j(X)}{\partial \log W_j(X)} = \mathrm{E}\left(\beta_i \middle| X_i = X, j(i) = j\right). \tag{9}$$

When β is heterogeneous, this elasticity varies across firms, because firms with different wage-amenity profiles attract workers who make different trade-offs. As a result, each firm employs a workforce with a distinct distribution of β. The labor supply elasticities are therefore linked to firm characteristics in a way that cannot be captured in a model with homogeneous effects.

Another consequence of heterogeneous effects is that the firm's labor supply elasticity is endogenous to wages. If a firm posts a higher wage, then it attracts workers who are more responsive to wages (higher β), and this in turn leads the firm to face a more elastic labor supply. To formalize this property, I analyze the second derivative of the labor supply curve to a firm:

$$\frac{\partial^2 \log S_j(X)}{\partial [\log W_j(X)]^2} = \mathrm{Var}\left(\beta_i \middle| X_i = X, j(i) = j\right). \tag{10}$$

This second derivative is strictly positive if and only if $\mathrm{Var}(\beta_i | X_i = X) > 0$. Additionally, it is increasing in $\mathrm{Var}(\beta_i | X_i = X)$, since greater heterogeneity in β expands the scope for sorting.[14]

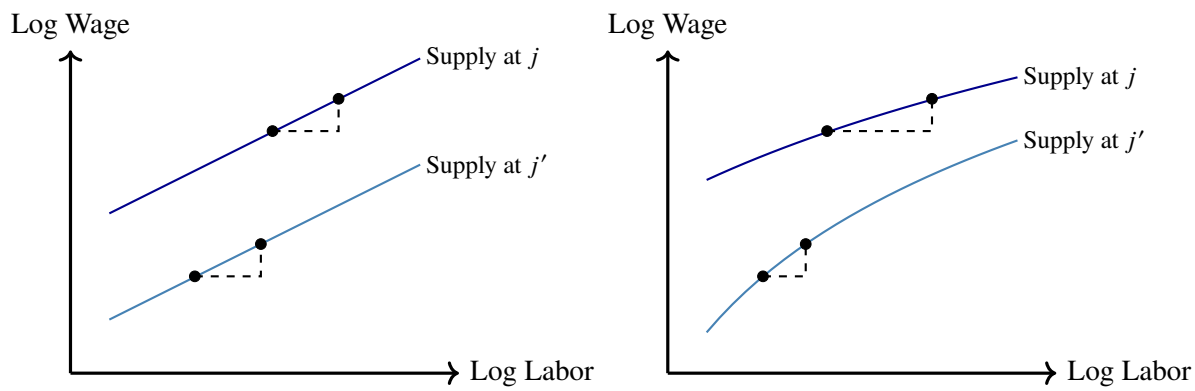
[12] In principle, nests can be defined more broadly so they do not coincide with markets. However, any nesting structure would still be based on factors fully observable to the analyst, a restriction that is typically undesirable.

[13] Examples of this approach include Deb et al. (2024) and Chan et al. (2025). The former uses a multinomial logit discrete choice model and the latter uses a representative agent CES model. As shown in Verboven (1996), Proposition 2, both modeling approaches lead to demand functions that are isomorphic at the market level.

[14] In Property A.1 of Appendix A, I derive the third, fourth, and fifth derivatives of a firm's labor supply curve under the logit model. I show that these quantities depend on the higher-order moments of the distribution of β.

Figure 1 illustrates these implications by contrasting firm-level labor supply curves under homogeneous and heterogeneous effects. Consider two firms, j and j', where firm j pays high wages but has worse amenities, and firm j' pays low wages but has favorable amenities. Under homogeneous effects (Figure 1.a), both firms would face the same employment response to an internal wage change, meaning that an X% change in wages yields the same percent change in labor at each firm. Under heterogeneous effects (Figure 1.b), these firms would face different employment responses. For the same X% wage change, the high-wage, low-amenity firm sees a larger employment response, while the low-wage, high-amenity firm sees a smaller response.

Figure 1: Illustration of Firm-Specific Labor Supply Responses



(a) Homogeneous Effects (b) Heterogeneous Effects

Notes. This figure depicts the labor supply curves $\log S_j(X)$ and $\log S_{j'}(X)$ for two firms, j and j', respectively. Figure 1.a shows the case where $\text{Var}(\beta_i \mid X_i = X) = 0$. Figure 1.b shows the case where $\text{Var}(\beta_i \mid X_i = X) > 0$.

II.F. Wage Determination and Sorting in Equilibrium

In equilibrium, firms face upward-sloping labor supply curves, allowing them to set wages below workers' marginal products. Let $L_j(X)$ be the equilibrium employment of skill group X at firm j, and let $\boldsymbol{\varepsilon}_j(X) = \frac{\partial \log L_j(X)}{\partial \log W_j(X)}$ be the labor supply elasticity firm j faces. A worker's wage is:

$$W_j(\chi, \varphi) = \underbrace{\frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)}}_{\textit{Markdown}} \times \underbrace{\varphi T_j (1 - \alpha_j) \theta_{j\chi} \left(L_j^{\text{eff}}(\chi)\right)^{\rho_j - 1} \left(\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi')\right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j} - 1}}_{\textit{Marginal Product of Labor}} . \quad (11)$$

There are three main factors that drive systematic worker sorting in the model. One factor is that workers' marginal products differ across firms. These differences arise due to variation in firm productivity $(T_j, \{\theta_{j\chi}\}_\chi)$ and technology (ρ_j, α_j), which have the potential to generate large dispersion in worker productivity within and across firms. Note that, with production co-

mplementarities, $\rho_j < 1$, and match-specific efficiency, $\theta_{j\chi}$, wages need not be log-additively separable into worker and firm components, as assumed in much of the sorting literature following Abowd et al. (1999). Relaxing this restriction allows for richer sorting, since the same firm can have a high marginal product for some workers and a low marginal product for others.

A second factor driving sorting is the non-wage amenity $a_j(X)$. These amenities generate vertical firm differentiation, as some firms can have strictly better amenities than others. They also create horizontal differentiation, as workers in the same firm can encounter different amenities. Since the amenities can depend arbitrarily on firm production parameters T_j, $\{\theta_{j\chi}\}_\chi$, ρ_j, and α_j, the model imposes no restriction on how they relate to a worker's marginal product.

A third factor driving sorting is the wage markdown $\frac{\varepsilon_j(X)}{1+\varepsilon_j(X)}$. These markdowns vary across firms due to differences in workforce composition, reflecting an interaction between worker preferences and firm characteristics. A low-productivity, high-amenity firm is more likely to attract workers who are less responsive to wages, and this allows it to set a larger markdown. Note that, in the model, wage markdowns are skill-specific, since firms can condition wages on skills. Therefore, workers with different skills can face different markdowns even at the same firm. This further relaxes the log-additivity restriction of two-way fixed effects models, since a given firm can apply a larger markdown to some workers and a smaller markdown to others.

II.G. Rents, Pass-throughs, and Allocative Inefficiency

I conclude this section by showing how the structural parameters in the model map into key economic quantities, such as rents earned by workers and firms, the pass-through of productivity shocks to earnings, and measures of welfare loss due to allocative inefficiency. To simplify these derivations, I assume a logit error structure for ϵ_{ij}, setting $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$. In each of the derivations in the Appendix, I carefully highlight which parts of the analysis rely on this assumption, and I also show how the results extend to alternative parameterizations of F_ϵ.

Worker Rents

In the economy, firms do not observe workers' individual preferences, β_i and ε_{ij}, which means they cannot price-discriminate based on each worker's reservation wage. Thus, an equilibrium involves surpluses—or rents—for inframarginal workers. These rents represent the earnings that workers are willing to forgo to be indifferent between their current firm and their next-best alternative. Let R_i^w denote a worker i's rents at her current firm. I define these rents as follows:

$$u_{ij}\big(W_j(X_i) - R_i^w, a_j(X_i)\big) = \max_{j' \neq j(i)} u_{ij'}\big(W_{j'}(X_i), a_{j'}(X_i)\big). \tag{12}$$

In a model where β is constant, the scope for worker rents to vary is sharply limited. Property A.4 in Appendix A shows that under homogeneous effects, a worker's rents as a share of wages,

$R_i^w / W_{j(i)}(X_i)$, are statistically independent of both her skills X and firm j. Thus, a worker's employer and skills would have no impact on the fraction of wages she earns as rents. This places a strong restriction on the incidence of rents in the economy, and it runs counter to the idea that workers at different firms, or with different skills, benefit differently from their wage contracts.

When β is heterogeneous, the share of wages that workers earn as rents will systematically vary across firms and skill groups. To characterize this variation, I derive an expression for the average rent share, $\mathrm{E}(R_i^w | j(i) = j, X_i = X)/W_j(X)$, among workers with skills X at firm j.

$$\frac{\mathrm{E}(R_i^w | j(i) = j, X_i = X)}{W_j(X)} = \left(\frac{1}{1+\beta_i} \,\middle|\, j(i) = j, X_i = X \right). \tag{13}$$

As shown in Property A.5 of Appendix A, this average rent share declines with the firm's wage:

$$\frac{\partial \left(\frac{\mathrm{E}(R_i^w | j(i)=j, X_i=X)}{W_j(X)} \right)}{\partial \log W_j(X)} = \mathrm{Cov} \left(\beta_i, \frac{1}{1+\beta_i} \,\middle|\, j(i) = j, X_i = X \right) \leq 0. \tag{14}$$

This negative relationship reflects how workers sort based on individual heterogeneity. Holding skills fixed, higher-wage firms attract a larger share of workers who are more responsive to wages. These workers would be more likely to switch firms in response to small pay differences. Therefore, even though they earn higher wages, a smaller share of those wages would constitute rents. As I demonstrate empirically in Section VI, neglecting this compositional effect leads to an inaccurate assessment of where worker rents are concentrated in the economy.

Firm Rents

Firms also capture rents in equilibrium. These rents come in the form of excess profit that firms obtain by exerting wage-setting power. To measure firm rents, I study a counterfactual where the firm is a price-taker, facing a perfectly elastic labor supply curve. The firm then maximizes:

$$\max_{\{D_j^{\mathrm{pt}}(\chi,\varphi)\}_{\chi,\varphi}} T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j^{\mathrm{pt}}(\chi,\varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} - \sum_{\chi \in \mathcal{X}} \left(\int W_j^{\mathrm{pt}}(\chi,\varphi) D_j^{\mathrm{pt}}(\chi,\varphi) d\varphi \right), \tag{15}$$

subject to the labor supply constraint: $D_j^{\mathrm{pt}}(X) = S_j^{\mathrm{pt}}(X)$ for all $X = (\chi, \varphi)$. In this counterfactual, the only difference in the firm's behavior is that it does not exert its wage-setting power. Hence, this counterfactual is distinct from an equilibrium where all firms act as price takers.

I define a firm j's rents as the difference between its realized profits and its counterfactual profits, given by $R_j^e = \Pi_j - \Pi_j^{\mathrm{pt}}$. In Property A.6 of Appendix A, I show that these rents equal:

$$R_j^e = Y_j \times \left[\sum_{\chi \in \mathcal{X}} \int \left(\frac{1 + \alpha_j \, \boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \right) \left(\frac{\theta_{j\chi} (L_j^{\text{eff}}(\chi))^{\rho_j}}{\sum_{\chi'} \theta_{j\chi'} (L_j^{\text{eff}}(\chi'))^{\rho_j}} \right) \left(\frac{\varphi L_j(\chi, \varphi)}{\int \varphi' L_j(\chi, \varphi') d\varphi'} \right) d\varphi - \alpha_j \left(\frac{Y_j^{\text{pt}}}{Y_j} \right) \right], \quad (16)$$

where Y_j^{pt} is the firm's value added as a price-taker. In this model, Y_j^{pt} does not appear to have a closed-form expression. Nevertheless, I show how to solve for this counterfactual quantity from the structural parameters in the model using a gradient descent algorithm. This procedure relies on the concavity of the firm's profit function, which is established in Appendix A.4.

In a model with homogeneous β, there is little scope for firm rents to vary. As shown in Property A.6, homogeneous effects implies that all firms will capture the same rents as a share of profit, R_j^e / Π_j, provided that they share the same returns to scale and that labor is perfectly substitutable.[15] Thus, for a simplified production technology (common α_j, $\rho_j = 1$), all firms receive the same fraction of profits as rents, regardless of how their characteristics might differ.

When β is heterogeneous, a firm's rents depend on its workforce composition and therefore on the full set of firm characteristics that influence sorting. Moreover, because the firm's market power is endogenous to wages, the relationship between firm productivity T_j and rents R_j^e becomes ambiguous. On the one hand, a more productive firm can capture higher profits. On the other hand, it can raise wages and attract workers with more wage-elastic labor supply. This erodes firm market power and makes labor more costly. Thus, while higher productivity can raise a firm's profit, it may also reduce the share of that profit that constitutes rents. In the limit, as $\boldsymbol{\varepsilon}_j(X) \to \infty$ for all X, a firm's profits converges to that of a price-taker, so $R_j^e \to 0$.

In my analysis, it is difficult to empirically distinguish between a firm's ex-ante and ex-post rents. This task would require detailed information about how firms choose and pay for amenities, which is not available in the administrative data. So, for identification, I assume that firms cannot adjust their amenities, at least during the time frame that the model is estimated. This assumption does not restrict how amenities correlate with firm productivity or technology. It also does not rule out the possibility that firms initially choose amenities to maximize profits.

Pass-through of TFP Shocks to Wages

Next, I examine the impact of a hypothetical shock to firm productivity on workers' wages, as measured by the elasticity $\frac{\partial \log W_j(X)}{\partial \log T_j}$. This object is commonly used to learn about rent sharing, as it reflects how an increase in a firm's value added would be passed on to workers in the form of higher pay. In Property A.7 of Appendix A, I show that $\left[\frac{\partial \log W_j(X)}{\partial \log T_j} \right]_X = (I - \mathbf{K}_j)^{-1} \mathbf{1}$, where:

[15] As the firm approaches constant returns to scale, $\alpha_j \to 0$, the profit that it could earn as a price-taker tends to zero. Therefore, any profit that the firm manages to capture would come in the form of rents: $\lim_{\alpha_j \to 0} \Pi_j = R_j^e$.

$$\mathbf{K}_j[X, X'] = \boldsymbol{\varepsilon}_j(\chi', \varphi')\Bigg[(1-\alpha_j-\rho_j)\left(\frac{\theta_{j\chi}(L_j^{\text{eff}}(\chi))^{\rho_j}}{\sum_{\chi''}\theta_{j\chi''}(L_j^{\text{eff}}(\chi''))^{\rho_j}}\right)\left(\frac{\varphi L_j(\chi,\varphi)}{\int \varphi'' L_j(\chi,\varphi'')d\varphi''}\right)$$
$$- \mathbb{1}\{\chi = \chi'\}(1-\rho_j)\frac{\varphi L_j(\chi,\varphi)}{\int \varphi'' L_j(\chi,\varphi'')d\varphi''}\Bigg] + \mathbb{1}\{X = X'\}\frac{\frac{\partial^2 \log L_j(X)}{\partial[\log W_j(X)]^2}}{\boldsymbol{\varepsilon}_j(X)[1+\boldsymbol{\varepsilon}_j(X)]}. \quad (17)$$

In a model where β is homogenous, the pass-through of productivity shocks to wages can only vary due to differences in firm technology. Indeed, as shown in Property A.7, homogeneous effects implies that $\frac{\partial \log W_j(X)}{\partial \log T_j}$ is constant whenever firms exhibit the same returns to scale and labor is perfectly substitutable. In other words, all workers, regardless of their firm or skills, would experience the same proportional wage change in response to a firm productivity shock.

When β is heterogeneous, the pass-through $\frac{\partial \log W_j(X)}{\partial \log T_j}$ varies systematically across firms and skill groups because employers face different labor supply elasticities $\boldsymbol{\varepsilon}_j(X)$. Moreover, given how workers sort on individual heterogeneity, the wage markdown can amplify the impact of firm productivity on earnings. To see why, consider that when a productivity shock drives up wages at a firm, it shifts the firm's workforce composition toward workers with more elastic labor supply. This reduces the firm's market power and drives wages up further. As a result, wage changes at a firm can be self-reinforcing.[16] Note that, in principle, the pass-through may be greater than one, although my empirical results show that, in practice, it is well below one.

Welfare and Allocative Inefficiency

I consider a measure of social welfare that incorporates firm profits. Specifically, I assume that profits are redistributed to workers in direct proportion to their wages. This approach ensures that the redistribution does not distort workers' labor supply decisions. Welfare is defined as:

$$\mathcal{W} = \mathrm{E}\left(\max_j \left\{u_{ij}\big(W_j(X_i), a_j(X_i)\big)\right\}\right) + \log\left(\sum_{j=1}^{J}\Pi_j\right). \quad (18)$$

[16] Another way to visualize this effect is to analyze the wage elasticity of the markdown $\frac{\boldsymbol{\varepsilon}_j(X)}{1+\boldsymbol{\varepsilon}_j(X)}$, which equals:

$$\frac{\partial \log \frac{\boldsymbol{\varepsilon}_j(X)}{1+\boldsymbol{\varepsilon}_j(X)}}{\partial \log W_j(X)} = \frac{\frac{\partial^2 \log L_j(X)}{\partial[\log W_j(X)]^2}}{\boldsymbol{\varepsilon}_j(X)[1+\boldsymbol{\varepsilon}_j(X)]}, \quad \text{where} \quad \frac{\partial^2 \log L_j(X)}{\partial[\log W_j(X)]^2} \geq 0.$$

This elasticity is positive, implying that wage changes at a firm are amplified through the markdown. In principle, this can generate multiple solutions to (11), potentially yielding two local optima—one with lower wages and employment but larger markdowns, and another with higher wages and employment but smaller markdowns. This phenomenon is discussed by Manning (2010) in the context of spatial agglomeration. In my setting, I show that multiplicity does not arise as long as $\mathrm{Var}(\beta \mid X)$ is bounded relative to $1-\alpha_j$. I also show that, even if multiple solutions to (11) exist, there is almost surely a unique global profit maximum (Appendix C.1, Lemma 2).

To measure the welfare cost of market power, I compare welfare in the monopsonistic economy to what would be achieved by a social planner who optimally assigns workers to firms, solving: $\mathcal{W}^* = \max_{\{j(i)\}_i} \mathcal{W}$. I assume that the planner has full knowledge of worker and firm fundamentals, but firms continue to set wages based on their limited information of (β_i, ϵ_{ij}). In Property A.8 of Appendix A, I prove that the planner's allocation coincides with the competitive (Walrasian) allocation, where firms act as price-takers facing perfectly elastic labor supply curves. I also show how to solve for welfare under the optimal allocation from the model's structural parameters, allowing me to evaluate the welfare cost of imperfect competition in the economy.

In an economy with constant markdowns, there would be no misallocation of labor across firms due to imperfect competition, because workers would sort into the same firms that they would choose in the absence of monopsony power. However, if markdowns vary across firms, then there is a potential for allocative inefficiency. This inefficiency occurs, for example, when a worker who is most productive at one firm j earns a higher wage at another firm k because firm j has a larger markdown than firm k. Such discrepancies can alter workers' rankings of firms and distort their labor supply decisions. Therefore, heterogeneous markdowns introduce a welfare cost of imperfect competition that is not present in the case of constant markdowns.[17]

III. Data and Empirical Context

I now describe the data sources and the key variables that are used for my empirical analysis, as well as the criteria for sample selection. Additional details are provided in the Appendix.

III.A. Data Sources

For my analysis, I use a matched employer-employee panel dataset from Norway, containing information about workers and firms over the time period 1995-2018. The data is constructed by linking annual business tax records to a national employment register from the Norwegian Labour and Welfare Administration. It encompasses all workers who are not self-employed, and it covers the universe of firms that are required to file a tax return. Additional information about worker demographics and financial assets comes from individual tax filings and social security registers. All administrative data are maintained by Statistics Norway and are subject to comprehensive quality control measures. These data are considered to be highly reliable.[18]

At the worker-level, the data contains information about annual earnings, scheduled work hours, and the duration of a job spell. Earnings include a worker's salary, bonuses, and other payments that employers make such as overtime, vacation, and severance pay.[19] All monetary variables are adjusted for inflation and expressed in 2014 U.S. dollars. The data also reports a

[17] Of course, with a nonemployment margin, the existence of any markdown can distort workers' labor-leisure trade-offs, creating underemployment; yet, constant markdowns would still imply no misallocation across firms.

[18] See Fagereng et al. (2021) for discussion. Bruun et al. (2021) also give an independent quality evaluation.

[19] In-kind benefits are also included in earnings; however, parental leave and sickness benefits are excluded.

worker's education, experience, and specialization (e.g., college major and vocational training). These variables are used to define a worker's skill type χ. In the baseline specification, I define χ to be a combination of education and experience, where education is divided into three categories (no high school degree, high school degree but no college degree, and college degree) and experience is divided into two categories (less than 10 years in the workforce and more than 10 years in the workforce). This yields six distinct skill types. Alternate definitions of χ are considered in the Appendix. Finally, the data contains rich demographic information about workers and their families. These data are used for making cross-sectional comparisons.

In the data, workers are linked to establishments, which represent sub-units (or branches) of a firm. To address this dimension, I measure employment at the establishment-level, and I assume that a firm's production function aggregates labor in a way that allows for imperfect substitutability between establishments.[20] This approach allows for more flexible modeling of sorting and wage determination. For example, if an oil and gas firm operates a processing plant in rural Norway and a corporate office in Oslo, then these two establishments are treated as distinct entities—both by workers and by the managing firm. Establishments are assigned to local labor markets based on industry and location. The Central Register of Establishments and Enterprises classifies them into 10 broad industries and 46 commuting zones, as defined in Bhuller (2009). I define a market as the intersection of an industry and a commuting zone.

Accounting data is measured at the firm-level and is collected from corporate income statements and balance sheets. A firm's value-added is defined as the difference between revenue and non-labor expenses, which include operating costs, spending on intermediate inputs, and capital depreciation. A firm's profit is defined as revenue minus overall costs, where overall costs account for wages paid to workers. All variables are measured prior to interest and taxes.

III.B. Sample Construction

Because my framework studies labor supply on the extensive margin, I restrict the estimation sample to full-time workers, defined as those who work at least 30 hours per week at a given establishment. Throughout my analysis, labor will be measured as full-time employment. To calculate earnings, I further restrict the sample to individuals aged 25 to 55 who work at the same establishment for an entire year and who do not have a higher-paying job elsewhere. Following Song et al. (2018), I also require that a worker's earnings are above some minimum threshold. In the baseline sample, I set this threshold to be \$15,000 per year. In the Appendix, I show that the empirical results are robust to changes in this minimum earnings threshold.

[20] Specifically, if the firm j operates multiple establishments $k \in \mathcal{K}_j$, then I extend the production function so that $Y_j = T_j[\sum_{k\in\mathcal{K}_j}\sum_{\chi\in\mathcal{X}}\theta_{jk\chi}(L^{\text{eff}}_{jk}(\chi))^{\rho_j}]^{(1-\alpha_j)/\rho_j}$, where $L^{\text{eff}}_{jk}(\chi) = \int \varphi D_{jk}(\chi,\varphi)d\varphi$ is the efficiency units of labor of skill type χ demanded by firm j for establishment k. This extension does not meaningfully affect the theoretical results or identification analysis. Thus, to ease notation, the ongoing discussion will continue to focus on a case where each firm operates only one establishment ($|\mathcal{K}_j| = 1$). As a robustness check in the Appendix, I assess the sensitivity of the parameter estimates to restricting the sample to single-establishment firms.

Part of my analysis relies on utilizing a balanced panel of firms. To achieve this, I restrict the estimation sample to firms that remain operational for nine consecutive years. During this period, the firm must consistently report positive revenue and maintain at least five full-time employees. In addition, the firm must continue operating the same set of establishments, and each establishment must stay in the same industry and commuting zone for the full nine years.

Table 1 provides descriptive statistics for the main estimation sample. More information about the distribution of wages, labor, value added, and profit can be found in Appendix D.1.

Table 1: Summary Statistics—Main Estimation Sample

	Unique Identifiers	Unique Observations
Panel A. Population Counts		
Number of Full-time Workers	1,325,863	8,303,075
Number of Establishments	31,252	475,614
Number of Firms	23,066	274,687
Number of Local Markets	457	10,563
	Share of Workforce	Mean Log Earnings
Panel B. Worker Skill Types		
Lower Secondary, <10 Years Exp.	6.82%	10.92
Upper Secondary, <10 Years Exp.	19.03%	11.12
College Graduate, <10 Years Exp.	14.76%	11.33
Lower Secondary, ≥10 Years Exp.	21.37%	11.10
Upper Secondary, ≥10 Years Exp.	23.74%	11.30
College Graduate, ≥10 Years Exp.	14.28%	11.57
	Mean	Std. Dev.
Panel C. Accounting Variables		
Log Revenues	15.82	1.29
Log Total Costs	15.78	1.27
Log Value Added	14.64	1.17
Log Profit	12.82	1.60

Notes. This table presents descriptive statistics for the baseline estimation sample. All monetary quantities are expressed in log 2014 US dollars and are adjusted for inflation.

IV. Identification Analysis

In this section, I demonstrate how to bring the model to data. In particular, I show how to recover key economic quantities of interest—such as the distribution of wage markdowns and rents, and measures of allocative inefficiency—from the matched employer-employee dataset. While some of these quantities can be recovered without full knowledge of the model's structural parameters, others—particularly those based on counterfactual outcomes—rely on the full set of parameters. For this reason, my goal is to achieve full point identification of the model.

My identification analysis follows two main steps. First, I show how to recover the firm-specific labor supply elasticities $\boldsymbol{\varepsilon}_j(X)$ by using panel data together with an economic shock

that shifts the labor demand of multiple firms in the economy. Second, with these elasticities in hand, I prove identification of firms' production parameters $(T_j, \{\theta_{j\chi}\}_\chi, \rho_j, \alpha_j)$ and amenities $a_j(X)$, as well as the joint density $f_{\beta,X}$ governing workers' skills and wage-amenity trade-offs.

IV.A. Identification of Firm-Specific Labor Supply Elasticities

IV.A.1. Ideal Experiment

I begin by laying out the comparative statics that identify the labor supply elasticities. Consider a hypothetical, firm-specific productivity shock that shifts firm j's TFP in logs from $\log T_j$ to $\log T_j + \delta_j$. Assume all other parameters in the economy remain unchanged. For every skill group X, let $\big(W_j(X), L_j(X)\big)$ and $\big(W'_j(X), L'_j(X)\big)$ represent the firm's equilibrium wage and employment before and after the shock, respectively. As $\delta_j \to 0$, these equilibrium outcomes can be used to recover the elasticity $\boldsymbol{\varepsilon}_j(X)$. In particular, $\boldsymbol{\varepsilon}_j(X)$ is defined by the condition:

$$\lim_{\delta_j \to 0} \Big[\big(\log L'_j(X) - \log L_j(X) \big) - \boldsymbol{\varepsilon}_j(X) \times \big(\log W'_j(X) - \log W_j(X) \big) \Big] = 0. \tag{19}$$

This condition follows from two key properties of the model. First, under the equilibrium characterization, firms view each other to be strategically small. Therefore, they do not internalize wage changes that are confined to a single rival firm in the economy. This delivers an exclusion restriction, whereby an exogenous shock to T_j affects firm j's labor supply $L_j(X)$ only through changes to its own wage $W_j(X)$. The second property of the model, established in Lemma 2 of Appendix C.1, is that the solution to the firm's first order condition is locally stable, which means that iterating on the equilibrium wage condition (11) will converge to a fixed point for nearby initial values. This property is important for the comparative statics because it ensures that a small, isolated productivity shock does not lead to large jumps in wages or employment.

Based on equation (19), an ideal experiment to recover the elasticities $\{\boldsymbol{\varepsilon}_j(X)\}_{j,X}$ would be to introduce isolated, infinitesimal shocks to each firm's TFP and then measure the resulting change in each firm's wages and employment. This type of experiment is not feasible in practice, however, because I cannot independently manipulate each firm's productivity while holding all other aspects of the labor market fixed. Moreover, even if it were possible to do so, this would require measuring a firm's wages and employment conditional on each skill group $X = (\chi, \varphi)$. Since workers' skill levels φ are not observed in the data, the researcher would not be able to directly distinguish between a firm's labor inputs. In the remainder of this subsection, I show how to overcome these challenges using panel data and instrumental variables.

IV.A.2. Econometric Model

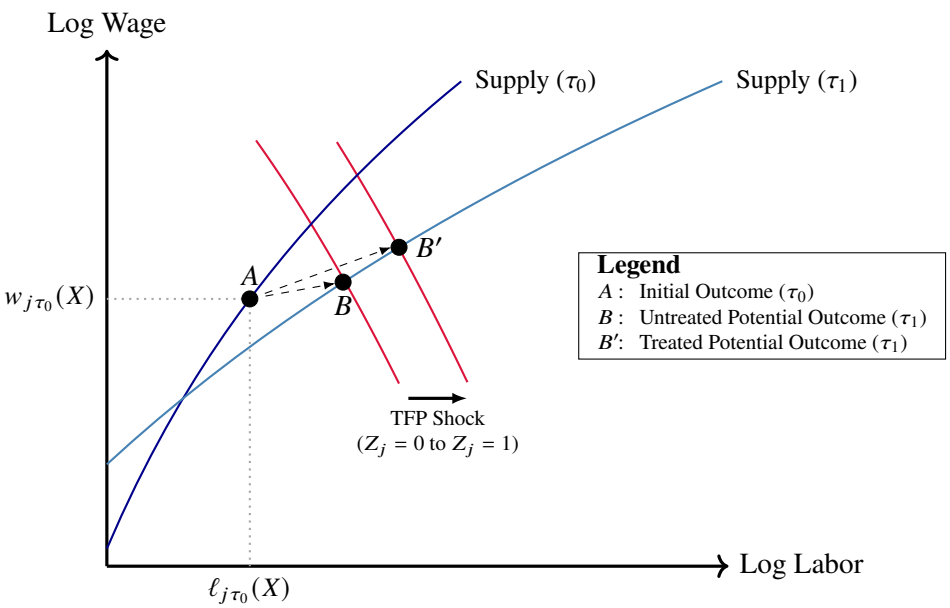
I now describe the empirical framework and assumptions that form the basis for my identification results. Throughout this discussion, I adopt the logit error structure, where I assume

$F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$. Under this specification, the firm-specific labor supply curves are defined by equation (7). To simplify notation, I use lowercase letters to denote variables that are expressed in log form, defining $w_j(X) = \log W_j(X)$, $\ell_j(X) = \log L_j(X)$, and $y_j = \log Y_j$.

Description of Empirical Framework for a Generic Instrument

Consider the evolution of equilibrium outcomes over two time periods: τ_0 and τ_1. After the initial period τ_0, an event occurs in which a subset of firms in the economy experiences a TFP shock, while the productivity of other firms remains unchanged. Following this event, wages and labor adjust to a new equilibrium by period τ_1. Let $w_{j\tau_0}(X)$ and $\ell_{j\tau_0}(X)$ denote the log wages and employment for skill group X at firm j in period τ_0, and let $w_{j\tau_1}(X)$ and $\ell_{j\tau_1}(X)$ be the corresponding outcomes in τ_1. Define Z_j as a binary variable indicating whether firm j experiences a TFP shock. Firms are classified as *treated* if $Z_j = 1$ and *untreated* if $Z_j = 0$.[21]

Figure 2: Illustration of Potential Outcomes for a Firm



Notes. This figure depicts how a firm j's potential wage and labor outcomes respond to an event that shifts TFP for multiple firms in the labor market. At τ_0, workers' wage indices, skills and preferences are characterized by $\{I_{\tau_0}(\beta, X)\}_{\beta,X}$ and $\{F_{\beta,X|\tau_0}\}_X$, respectively, and at τ_1, they are characterized by $\{I_{\tau_1}(\beta, X)\}_{\beta,X}$ and $\{F_{\beta,X|\tau_1}\}_X$, respectively. Apart from the TFP parameter T_j, which changes only for treated firms, all firm characteristics $(\{\theta_{jX}\}_X, \rho_j, \alpha_j, \{a_j(X)\}_X)$ remain fixed between τ_0 and τ_1.

Figure 2 illustrates the evolution of wages and labor for a given firm j, delineating which aspects of the model remain constant and which vary over time. Importantly, between periods τ_0 and τ_1, wages and labor adjust for all firms, including those that are untreated. This occurs because, unlike in the ideal experiment, the productivity shocks are not confined to just one

[21] While I focus on a binary instrument for exposition, the subsequent identification results extend to multivalued or continuous Z_j, where identification comes from comparing firms exposed to shocks at different intensities.

firm. Rather, they take place at multiple firms in the economy. Consequently, wage adjustments among the treated firms would affect market aggregates by shifting workers' wage indices, $I(\beta, X)$. This, in turn, reshapes the labor supply curves faced by every firm, including untreated firms. Thus, the TFP shocks generate spillover effects that impact all firms in the economy.

Contributing to these dynamics, I allow workers' skills and preferences to evolve over time in response to the productivity shocks affecting firms. In particular, I permit the distribution of workers' skills F_X and the distributions $\{F_{\beta|X}\}_X$ governing workers' wage-amenity tradeoffs to change from period τ_0 to τ_1. I place no restrictions on the evolution of these distributions, imposing no constraints on how $\left(F_{X|\tau_0}, \{F_{\beta|X,\tau_0}\}_X\right)$ relates to $\left(F_{X|\tau_1}, \{F_{\beta|X,\tau_1}\}_X\right)$. While I also allow workers' idiosyncratic tastes to evolve over time, I assume that the cross-sectional distribution of these tastes retains a logit structure in each period, $F_\epsilon(\epsilon_{ij\tau}) = \exp(-\exp(-\epsilon_{ij\tau}))$. To achieve this, I assume that $\epsilon_{ij\tau}$ is a Markov process with independent innovations across i.

Given the evolution of workers' wage indices, skills, and preferences over time, firms face different labor supply curves in periods τ_0 and τ_1. Therefore, in order to measure the elasticity of labor supply to any firm j, it is necessary to compare the firm's potential outcomes within the same time period. In period τ_0, the treated and untreated potential outcomes are the same, since no TFP shocks have yet occurred. However, in period τ_1, the potential outcomes differ, as illustrated by points B and B' in Figure 2. This difference reflects the comparative statics from the ideal experiment, capturing how a firm j's wages and labor would respond to a TFP shock that is confined just to that firm. If it were possible to compare both potential outcomes for a firm in period τ_1, then one could compute the firm's labor supply elasticity in that period.

Assumptions on Treatment Assignment

Throughout my identification analysis, I impose an assumption about how the treatment status of a firm, denoted by Z_j, depends on that firm's characteristics. Specifically, I assume that Z_j is statistically independent of the firm's technology (ρ_j, α_j), the relative efficiencies $\{\theta_{j\chi}\}_\chi$ of each skill type, and the differences in amenities $\{a_j(X) - a_j(X')\}_{X,X'}$ between skill groups.

Assumption I (*Instrument Independence*)**.** $Z_j \perp \left(\rho_j, \alpha_j, \{\theta_{j\chi}\}_\chi, \{a_j(X) - a_j(X')\}_{X,X'}\right)$.

Note that this assumption does not require Z_j to be fully idiosyncratic. In particular, I allow Z_j to depend on a firm's productivity through the baseline TFP $T_{j\tau_0}$. Additionally, while Z_j must be independent of the amenity differences within a firm, I do not restrict how it relates to the firm's average amenity $\mathrm{E}(a_j(X)|j)$. Consequently, exposure to the productivity shock need not be random: because a firm's treatment status can depend on its productivity and its amenities, treated and untreated firms may exhibit systematically different wages, labor, and output.

IV.A.3. Identification Results

I now establish the identification of firm-specific labor supply elasticities. As described above, these elasticities can vary over time due to changes in workers' wage indices, skills, and preferences. In any period τ, I can write $\boldsymbol{\varepsilon}_{j\tau}(X)$ as an (X,τ)-specific function of the log wage $w_{j\tau}(X)$:

$$\boldsymbol{\varepsilon}_{j\tau}(X) = \int \beta \times \frac{\left[\exp\left(\beta w_{j\tau}(X)\right)/I_\tau(\beta,X)\right] f_{\beta|X,\tau}(\beta|X,\tau)}{\int \left[\exp\left(\beta' w_{j\tau}(X)\right)/I_\tau(\beta',X)\right] f_{\beta|X,\tau}(\beta'|X,\tau)d\beta'} d\beta. \tag{20}$$

My analysis proceeds in two parts. First, I prove identification in the case where workers' skills X are fully observed. This allows me to focus on the main identification challenge: recovering elasticities $\boldsymbol{\varepsilon}_{j\tau}(X)$ from labor demand shocks at multiple firms, while accounting for how these shocks affect market aggregates. Second, I show how my results extend to the case where workers' skills are not fully observed, in particular where the skill level φ is unknown to the researcher. In this part, I provide assumptions that allow me to recover φ using panel data, and I derive an estimand for the labor supply elasticities that can be applied even with unobserved skills.

Identification with Observed Skills

When workers' skills X are observed in the data, the elasticities $\boldsymbol{\varepsilon}_{j\tau_1}(X)$ are identified through a difference-in-differences (DiD) comparison of firms' wage and employment outcomes. To motivate this design, it is important to recognize that one could not recover these elasticities from either a purely cross-sectional comparison of treated and untreated firms or from a comparison of outcomes within the same firm over time. A cross-sectional comparison fails because treated and untreated firms are likely to face systematically different labor supply curves, for example if Z_j correlates with $T_{j\tau_0}$ or $\mathrm{E}(a_j(X)|j)$. A within-firm panel regression fails because Z_j creates spillovers that can affect workers' wage indices, skills, and preferences. These spillovers imply that the same firm faces different labor supply curves in the pre- and post-periods.

I propose to recover the elasticity $\boldsymbol{\varepsilon}_{j\tau_1}(X)$ from a ratio of DiD estimands that condition on a firm's pre-period wage. Let $\left(\ell_{j\tau_0}(X), w_{j\tau_0}(X)\right)$ and $\left(\ell_{j\tau_1}(X), w_{j\tau_1}(X)\right)$ be the log employment and log wage for skill group X at firm j in periods τ_0 and τ_1, respectively. I define the estimand:

$$\mathrm{DiD}_{\tau_0,\tau_1}(w|X) = \frac{\mathrm{E}\left[\ell_{j\tau_1}(X)-\ell_{j\tau_0}(X)\middle|Z_j=1, w_{j\tau_0}(X)=w\right] - \mathrm{E}\left[\ell_{j\tau_1}(X)-\ell_{j\tau_0}(X)\middle|Z_j=0, w_{j\tau_0}(X)=w\right]}{\mathrm{E}\left[w_{j\tau_1}(X)-w_{j\tau_0}(X)\middle|Z_j=1, w_{j\tau_0}(X)=w\right] - \mathrm{E}\left[w_{j\tau_1}(X)-w_{j\tau_0}(X)\middle|Z_j=0, w_{j\tau_0}(X)=w\right]}. \tag{21}$$

The numerator of this expression is a DiD estimand of changes in log labor, while the denominator is a DiD estimand of changes in log wages. Figure 3 illustrates how the observed movements in wages and labor among treated and untreated firms are used to construct $\mathrm{DiD}_{\tau_0,\tau_1}(w|X)$.

To show that this estimand recovers the labor supply elasticity, I first prove that a common trends assumption holds. Specifically, I show that, conditional on a firm's initial wage $w_{j\tau_0}(X)$, the change in untreated potential outcomes, $w_{j\tau_1,0}(X)-w_{j\tau_0,0}(X)$ and $\ell_{j\tau_1,0}(X)-\ell_{j\tau_0,0}(X)$, is a deterministic function of the parameters Γ_j, where $\Gamma_j = \left(\rho_j, \alpha_j, \{\theta_{jX}\}_X, \{a_j(X)-a_j(X')\}_{X,X'}\right)'$.

Assumption I guarantees that a firm's treatment status Z_j does not depend on Γ_j. Therefore, treated and untreated firms with the same initial wages must, on average, see the same change in their untreated potential outcomes. By this reasoning, I establish the following proposition.

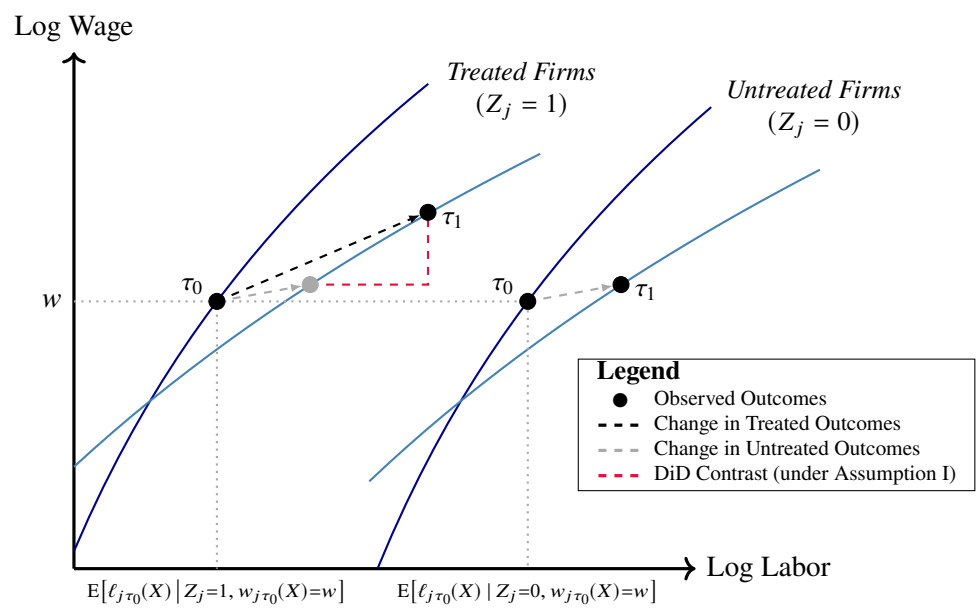
Proposition 1. Suppose Assumptions I holds. Then, given a firm's pre-period wage $w_{j\tau_0}(X)$, the change in untreated potential outcomes is mean independent of the firm's treatment status:

$$\mathrm{E}\left[\ell_{j\tau_1,0}(X) - \ell_{j\tau_0,0}(X) \middle| Z_j = 1, w_{j\tau_0}(X) = w\right] = \mathrm{E}\left[\ell_{j\tau_1,0}(X) - \ell_{j\tau_0,0}(X) \middle| Z_j = 0, w_{j\tau_0}(X) = w\right].$$

$$\mathrm{E}\left[w_{j\tau_1,0}(X) - w_{j\tau_0,0}(X) \middle| Z_j = 1, w_{j\tau_0}(X) = w\right] = \mathrm{E}\left[w_{j\tau_1,0}(X) - w_{j\tau_0,0}(X) \middle| Z_j = 0, w_{j\tau_0}(X) = w\right].$$

Proposition 1 is central to my analysis. It implies that, by using a DiD research design, I can reproduce the comparative statics from the ideal experiment. In particular, I can recover the mean difference between treated and untreated potential outcomes in the post-period for any treated firm offering initial wage w. Using this result, together with the fact that the elasticity $\boldsymbol{\varepsilon}_{j\tau_1}(X)$ is itself an (X, τ_1)-specific function of the firm's wage (see equation (20)), I establish that $\boldsymbol{\varepsilon}_{j\tau_1}(X)$ is identified from the estimand $\mathrm{DiD}_{\tau_0,\tau_1}(w_{j\tau_0}(X)|X)$ for any firm j and skill group X. Thus, by evaluating $\mathrm{DiD}_{\tau_0,\tau_1}(w|X)$ over the entire wage distribution, I can recover the labor supply elasticity faced by every firm in the economy. In practice, as I discuss in Section V, this task relies on an *overlap* condition, which requires the presence of both treated and untreated firms throughout the wage distribution: $\mathrm{P}\left(Z_j = 1 | w_{j\tau_0}(X) = w\right) \in (0, 1)$ for all $w \in \mathrm{supp}(w_{j\tau_0}(X))$.

Figure 3: Illustration of DiD Comparison of Firm-Level Outcomes



Notes. This figure illustrates how mean changes in log wages and log labor for treated and untreated firms are used to define the estimand $\mathrm{DiD}_{\tau_0,\tau_1}(w|X)$ for a given log wage w and skill group X. Dark blue curves display labor supply in the pre-period; light blue curves display labor supply in the post-period. Under Assumption I, the changes in untreated potential outcomes, $w_{j\tau_1,0}(X) - w_{j\tau_0,0}(X)$ and $\ell_{j\tau_1,0}(X) - \ell_{j\tau_0,0}(X)$, are mean independent of Z after controlling for the pre-period wage $w_{j\tau_0}(X) = w$.

Before proceeding, it is useful to review how this identification result differs from that of previous studies, such as Lamadon et al. (2022) and others, which also use the pass-through of labor demand shocks to measure firm-specific labor supply elasticities. First, while previous work typically assumes that the elasticities are constant (or constant conditional on observables), I instead allow them to vary within and across firms, and to depend endogenously on a firm's own wages. This expands the identification problem: rather than recovering a single elasticity $\boldsymbol{\varepsilon}$, I recover the entire distribution of elasticity vectors $[\boldsymbol{\varepsilon}_{j\tau}(X)]_X$ across firms j, and I allow these distributions to evolve over time τ. Second, while most previous studies consider isolated firm-level shocks that leave market aggregates unaffected, I allow multiple firms to be shocked simultaneously, which could generate spillovers to workers' wage indices, skills, and preferences. These spillovers violate the stable unit treatment value assumption (SUTVA) and thus complicate identification. I overcome this challenge by showing that, under the equilibrium framework, the spillovers can be absorbed into a time trend within a well-specified DiD estimand. Third, unlike in most prior work, I allow a firm's exposure to a shock to depend on its characteristics, via its TFP and average amenity. Thus, Z_j need not be randomly assigned.

Identification with Unobserved Skills

I now extend my results to allow workers' skills to have unobserved components. In particular, I assume that the researcher cannot directly measure a worker's skill level φ and thus cannot fully distinguish between a firm's labor inputs. To recover the labor supply elasticities in this case, I derive conditions under which φ can be inferred from matched worker-firm panel data. These conditions imply that φ is identified, allowing me to apply the same IV design as above.

Going forward, I make an assumption about the relationship between a worker's skill level φ and the other parameters in the model. This assumption consists of three main restrictions.

Assumption II (*Restrictions on Unobserved Skill Levels*)**.**

II.1. $\varphi \perp \beta | \chi, \tau$.
II.2. $a_j(\chi, \varphi) = a_{j\chi} + a_{\chi\varphi}$.
II.3. $\mathrm{E}(\log \varphi | \chi, \tau) = \mathrm{E}(\log \varphi | \chi)$.

Assumption II.1 states that, conditional on a worker's skill type χ and time period τ, the trade-off between wages and amenities, as summarized by β, does not depend on the skill level φ. This assumption does not require that β is fully independent of workers' skills, as I still allow β to depend on the skill type χ. Additionally, it does not rule out the possibility that β varies along unobserved skill dimensions; instead, it only precludes systematic variation based on φ.

Assumption II.2 asserts that the non-wage amenity $a_j(\chi, \varphi)$ can be expressed as the sum of two components: one that is specific to the firm and the skill type, $a_{j\chi}$, and another that

is specific to the skill level, $a_{\chi\varphi}$. This separability implies that amenities do not vary across firms with respect to unobserved aspects of worker skills. To better interpret the two amenity components, $a_{j\chi}$ and $a_{\chi\varphi}$, I employ the normalization $\mathrm{E}(a_{\chi\varphi}|\chi) = 0$ without loss of generality. This normalization allows me to define $a_{j\chi}$ as the average amenity at firm j for skill type χ.

Assumption II.3 requires that the average log skill level $\mathrm{E}(\log\varphi|\chi,\tau)$ among workers of a given skill type χ is invariant over time τ. This does not restrict how the variance or higher-order moments of $\log\varphi$ evolve. It also does not restrict the evolution of the distribution $F_{\chi|\tau}$, which governs workers' observable skill types. In my subsequent analysis, it will be useful to normalize φ by setting $\mathrm{E}(\log\varphi|\chi) = 0$ for each χ. This normalization is without loss of generality, as it is always possible to redefine the relative efficiencies $\theta_{j\chi}$ to absorb $\mathrm{E}(\log\varphi|\chi)$.[22]

To establish identification, I first examine the implications of Assumption II for workers' labor supply decisions and wage determination. The key implication of this assumption is that the labor supply elasticities faced by a firm do not vary based on unobserved skill dimensions: $\boldsymbol{\varepsilon}_{j\tau}(\chi,\varphi) = \boldsymbol{\varepsilon}_{j\tau}(\chi)$. Consequently, wage markdowns at a firm would differ across skill types χ, but not across skill levels φ within a given χ. This property implies the following result.

Proposition 2. Suppose that Assumption II holds. Then, for any worker at firm j with skills (χ,φ) in period τ, the wage can be represented as $W_{j\tau}(\chi,\varphi) = \varphi\,\psi_{j\chi\tau}$, where $\psi_{j\chi\tau}$ is a firm-skill-time specific term, and φ is a worker-specific component satisfying $\mathrm{E}(\log\varphi|j,\chi,\tau) = 0$.

Proposition 2 provides the necessary structure for computing worker skill levels φ from linked worker-firm panel data. In particular, I can recover φ from cross-sectional wage differences:

$$\log\varphi = \log W_{j\tau}(\chi,\varphi) - \mathrm{E}\left[\log W_{j\tau}(\chi,\varphi)\;\middle|\; j,\chi,\tau\right]. \tag{22}$$

To simplify notation, it will be useful to measure wages in efficiency units. Define the effective wage for workers with skill type χ at firm j in period τ as $W^{\mathrm{eff}}_{j\tau}(\chi) = W_{j\tau}(\chi,\varphi)/\varphi$. Using this notation, I can reinterpret the elasticity of labor supply to a firm in terms of the effective wage:

$$\boldsymbol{\varepsilon}_{j\tau}(\chi) = \int \beta \times \frac{\left[\exp\left(\beta w^{\mathrm{eff}}_{j\tau}(\chi)\right)/I^{\mathrm{eff}}_{\tau}(\beta,\chi)\right] f_{\beta|\chi,\tau}(\beta|\chi,\tau)}{\int\left[\exp\left(\beta' w^{\mathrm{eff}}_{j\tau}(\chi)\right)/I^{\mathrm{eff}}_{\tau}(\beta',\chi)\right] f_{\beta|\chi,\tau}(\beta'|\chi,\tau)d\beta'}d\beta, \tag{23}$$

where $I^{\mathrm{eff}}_{\tau}(\beta,\chi) = \sum_{k=1}^{J}\exp(\beta w^{\mathrm{eff}}_{k\tau}(\chi) + a_{k\chi})$ is the effective wage index for workers with preference parameter β and skill type χ at time τ. The elasticity $\boldsymbol{\varepsilon}_{j\tau_1}(\chi)$ is identified from a DiD estimand $\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi)$ evaluated at the effective log wage $w^{\mathrm{eff}}_{j\tau_0}(\chi)$. I define this estimand as:

$$\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi) = \frac{\mathrm{E}\left[\ell_{j\tau_1}(\chi)-\ell_{j\tau_0}(\chi)\middle|Z_j=1,\, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[\ell_{j\tau_1}(\chi)-\ell_{j\tau_0}(\chi)\middle|Z_j=0,\, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]}{\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1}(\chi)-w^{\mathrm{eff}}_{j\tau_0}(\chi)\middle|Z_j=1,\, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1}(\chi)-w^{\mathrm{eff}}_{j\tau_0}(\chi)\middle|Z_j=0,\, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]}. \tag{24}$$

[22] Specifically, if $\mathrm{E}(\log\varphi|\chi) \neq 0$ for some skill type χ, then I can redefine $\log\theta_{j\chi}$ as $\log\theta_{j\chi} + \rho_j\,\mathrm{E}(\log\varphi|\chi)$.

IV.A.4. Sources of Instruments

Up to this point, I have not yet specified how to construct the instrument Z_j. In my analysis, I consider both internal instruments, constructed from existing data using the model's implied restrictions, and external instruments, which draw on additional data sources. Both types of instruments may be used within my framework. However, they rely on different assumptions.

Internal Instruments

My main specification uses internal instruments, which are constructed by specifying a process by which a firm experiences productivity shocks. This approach follows Guiso et al. (2005) and Lamadon et al. (2022) by assuming that a firm's TFP evolves according to an autoregressive process with a unit root. Specifically, in each time period τ, I assume that log TFP equals:

$$t_{j\tau} = \bar{t}_j + \tilde{t}_{j\tau}, \quad \text{where} \quad \tilde{t}_{j\tau} = \tilde{t}_{j\tau-1} + u_{j\tau}. \tag{25}$$

To ensure that the productivity shocks do not lead to large jumps in wages and labor, I assume that $u_{j\tau}$ is small relative to the log wage levels at firm j. I also make the following assumption.

Assumption III *(Identification Assumptions for Internal Instruments).*
III.1. *(Instrument Relevance)* $\text{Var}(u_{j\tau}) > 0$.
III.2. *(Instrument Independence)* $u_{j\tau} \perp \left(\rho_j, \alpha_j, \{\theta_{j\chi}\}_\chi, \{a_j(X) - a_j(X')\}_{X,X'}\right)$.

Assumption III.1 requires that the productivity shocks exhibit variation across firms, implying that the internal instrument is relevant. Assumption III.2 requires that the internal instrument satisfies Assumption I, which is sufficient to ensure that the common trends assumption holds.

Under this setup, I construct the internal instrument from short-run changes in a firm's log value added, $\Delta y_{j\tau}$, defined within a given calendar year τ. For integers $p, p' \geq 2$, I define the pre-period as $\tau_0 = \tau - p$ and the post-period as $\tau_1 = \tau + p'$, corresponding to p years before and p' years after τ, respectively. Under Assumption III, I derive the following moment condition:

$$\text{E}\left[\Delta y_{j\tau}\left[\left(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\right) - \text{DiD}_{\tau_0,\tau_1}(w|\chi)\left(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\right)\right] \;\middle|\; w^{\text{eff}}_{j\tau_0}(\chi) = w\right] = 0. \tag{26}$$

This moment condition corresponds to an IV regression of long run differences in log labor on long run differences in log effective wages, instrumented by short run differences in log value added, while controlling for effective wages in the pre-period. In implementing this procedure, I define the treatment status Z_j by splitting firms at the median of the distribution of $\Delta y_{j\tau}$.[23]

External Instruments

To complement the analysis using internal instruments, I also use an external instrument based

[23] In the Appendix, I show how to extend this setup to allow for transitory measurement error in a firm's value added and labor. I require that the errors are both mutually independent and independent of firm-level parameters.

on plausibly exogenous product demand shocks from public procurement auction outcomes. This supplementary analysis draws on data covering the universe of product, construction, and service contracts procured by the Norwegian government from 2003 to 2018. This data comprises about 30,000 contract awards across all industries and regions. The estimation sample is defined using the same dataset and following similar protocols as those used to analyze the internal instrument. Further details about the procurement data are provided in Appendix D.2.

To construct the external instrument, I adopt the research design of Kroft et al. (2025) and de Frahan et al. (2024), where a firm is classified as treated ($Z_j = 1$) in the year τ in which it wins its first procurement contract. A firm is untreated ($Z_j = 0$) if it wins its first contract in some later year $\tau' > \tau_1$ or never wins. Because the instrument is defined from outside data sources, it does not require making assumptions about the evolution of firm TFP. However, its main limitation is that the auction outcomes only pertain to a subset of firms and are only observed over a limited time period, from 2003 to 2018. For common trends to hold, I maintain Assumption I, which requires that a firm's auction outcome is independent of the parameters $\left(\rho_j, \alpha_j, \{\theta_{j\chi}\}_\chi, \{a_j(X) - a_j(X')\}_{X,X'}\right)$, although it can depend freely on $T_{j\tau_0}$ and $\mathrm{E}(a_j(X)|j)$.

IV.B. Identification of Technology, Amenities, and Preferences

I now show how to recover the structural parameters characterizing firm productivity, technology, and non-wage amenities, as well as the distribution of worker preferences. These quantities are needed to compute worker and firm rents, and they also allow me to run counterfactuals.

Firm Productivity and Technology

I begin by proving the identification of the production parameters $(T_{j\tau}, \{\theta_{j\chi}\}_\chi, \rho_j, \alpha_j)$ for any arbitrary firm j and any time period τ. Consistent with my previous analysis, I assume that the firm's TFP parameter $T_{j\tau}$ can evolve over time, while the parameters $(\{\theta_{j\chi}\}_\chi, \rho_j, \alpha_j)$ remain constant. In addition, I normalize the firm-specific efficiencies $\{\theta_{j\chi}\}_\chi$ by setting $\theta_{j\chi^*} = 1$ for some skill type $\chi^* \in \mathcal{X}$, as I can only recover these terms up to scale, relative to the firm's TFP.

Identification of firm technology relies on the ability to recover the labor supply elasticities faced by the firm at two different time periods. This task is feasible in my setting because I have access to a 28-year panel, enabling me to replicate the analysis from Section IV.A over multiple years. For periods τ and τ', I define $\boldsymbol{\varepsilon}_{j\tau}(\chi)$ and $\boldsymbol{\varepsilon}_{j\tau'}(\chi)$ as the labor supply elasticities for skill type χ at firm j. If these elasticities are identified, then I can compute a worker's marginal product from the structural wage equation (11). Specifically, for $\tilde{\tau} \in \{\tau, \tau'\}$, I define:

$$\mathrm{MPL}^{\mathrm{eff}}_{j\tilde{\tau}}(\chi) = \frac{1 + \boldsymbol{\varepsilon}_{j\tilde{\tau}}(\chi)}{\boldsymbol{\varepsilon}_{j\tilde{\tau}}(\chi)} \times W^{\mathrm{eff}}_{j\tilde{\tau}}(\chi) \tag{27}$$

as the effective marginal product for skill type χ at firm j, adjusting for workers' skill levels φ.

I can then recover the substitution parameter ρ_j from changes over time in the relative effective marginal products and labor shares of different skill types. This result is formalized below.

Proposition 3. For each firm j, the elasticity of substitution between different skill types is:

$$(1-\rho_j)^{-1} = \frac{\log\left(\frac{L^{\text{eff}}_{j\tau}(\chi)}{L^{\text{eff}}_{j\tau}(\chi')}\right) - \log\left(\frac{L^{\text{eff}}_{j\tau'}(\chi)}{L^{\text{eff}}_{j\tau'}(\chi')}\right)}{\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau}(\chi')}\right) - \log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau'}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau'}(\chi')}\right)}, \quad \text{for any } \chi, \chi' \in \mathcal{X}.$$

Proposition 3 expresses the elasticity of substitution $(1-\rho_j)^{-1}$ in terms of identified model quantities. This expression reflects how the relative demand for two labor inputs depends on the relative productivity of those inputs, as measured by the elasticity of $L^{\text{eff}}_{j\tau}(\chi)/L^{\text{eff}}_{j\tau}(\chi')$ with respect to $\text{MPL}^{\text{eff}}_{j\tau}(\chi)/\text{MPL}^{\text{eff}}_{j\tau}(\chi')$. In my framework, this relationship may be firm-specific.

Once ρ_j is identified, I show that $\{\theta_{j\chi}\}_\chi$, α_j, and T_j can also be recovered from the firm's effective marginal products, labor shares, and value added. I establish the following result.

Proposition 4. For any firm j, the parameters $\{\theta_{j\chi}\}_\chi$, α_j, and T_j are defined by the equations:

$$\theta_{j\chi} = \exp\left[\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau}(\chi^*)}\right) + (1-\rho_j)\log\left(\frac{L^{\text{eff}}_{j\tau}(\chi)}{L^{\text{eff}}_{j\tau}(\chi^*)}\right)\right].$$

$$1-\alpha_j = \exp\left[\log \text{MPL}^{\text{eff}}_{j\tau}(\chi) - y_{j\tau} - \log(\theta_{j\chi}) + (1-\rho_j)\log L^{\text{eff}}_{j\tau}(\chi) + \log\sum_{\chi'\in\mathcal{X}} \theta_{j\chi'}\left(L^{\text{eff}}_{j\tau}(\chi')\right)^{\rho_j}\right].$$

$$T_{j\tau} = \exp\left[y_{j\tau} - \frac{1-\alpha_j}{\rho_j}\log\sum_{\chi\in\mathcal{X}} \theta_{j\chi}\left(L^{\text{eff}}_{j\tau}(\chi)\right)^{\rho_j}\right].$$

Note that recovering the substitution parameter ρ_j is the only step that requires knowledge of labor supply elasticities for two periods. If ρ_j is already known, or if one assumes perfect substitutability ($\rho_j = 1$), then $\{\theta_{j\chi}\}_\chi$, α_j, and T_j are identified from a single period of estimates.

Firm Amenities

Next, I prove identification of the parameters $a_{j\chi}$, which correspond to the average amenities at each firm j for any skill type χ. In my analysis, these terms are only identified up to a location normalization, because common shifts do not affect workers' labor supply decisions. Therefore, I normalize the amenities by defining a reference firm j^* at which $a_{j^*\chi} = 0$ for all $\chi \in \mathcal{X}$.

To recover firm amenities, I use a revealed preference approach. This method rests on the

idea that—holding wages fixed—firms that offer better amenities to a given skill type attract more workers of that type. Therefore, by comparing employment across firms in a counterfactual where all firms pay the same wages, I can infer the cross-firm differences in amenity values.

To implement this strategy, I use the labor supply elasticity curves $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$, which are identified for each skill type χ by the estimand $\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi)$. Integrating along these curves allows me to recover the labor supplied to any firm j under any counterfactual wage w, holding all other wages fixed. In doing so, I can then compare the employment at firm j to that of the reference firm j^* under a counterfactual in which both pay the same wages. This comparison allows me to infer the amenity value $a_{j\chi}$. The following proposition formalizes this argument.

Proposition 5. Under Assumptions I and II, the average amenity for skill type χ at firm j is:

$$a_{j\chi} = \ell_{j\tau}(\chi) + \int_{w^{\text{eff}}_{j\tau}(\chi)}^{w^{\text{eff}}_{j^*\tau}(\chi)} \boldsymbol{\varepsilon}_{j\tau}(\chi, w)dw - \ell_{j^*\tau}(\chi).$$

Worker Preferences

I conclude this section by demonstrating the identification of the densities $f_{\beta|\chi,\tau}$, which characterize the distribution of workers' marginal rates of substitution between wages and amenities. This identification result is achieved through revealed preference, by establishing a direct link between the distribution of workers' preferences and the shape of a firm's labor supply curve.

To understand the source of identification, first consider how the distribution of β impacts a firm's labor supply curve. If β is homogeneous, then the curve is isoelastic, and the elasticity of labor supply to a firm reduces to β. By contrast, if β is heterogeneous, then the curve is nonlinear, as its elasticity depends on a firm's wage. As equations (9) and (10) reveal, the nature of this wage-dependence carries information about the moments of the distribution of β. For example, greater curvature in the firm's labor supply curve indicates greater dispersion in β.

In the following proposition, I demonstrate that this relationship extends even further. In particular, I prove that the elasticity curve $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ uniquely determines the density $f_{\beta|\chi,\tau}$.

Proposition 6. Given wages and labor $\{W^{\text{eff}}_{j\tau}(\chi), L_{j\tau}(\chi)\}_{j=1}^{J}$, there is a one-to-one mapping between the labor supply elasticity curve $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ and the density function $f_{\beta|\chi,\tau}(\beta|\chi, \tau)$.

Proposition 6 implies that the density $f_{\beta|\chi,\tau}$ is identified under same the assumptions needed to recover the labor supply elasticity curve $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$. The proof leverages the fact that a firm's labor supply curve can be represented as a Laplace transform whose inverse is proportional to $f_{\beta|\chi,\tau}$.[24] This property ensures that the labor supply curve can be inverted to recover the density of β. Note that this identification result is fully nonparametric with respect to this density.

[24] Any two functions share the same Laplace transform only if they differ on a set of Lebesgue measure zero.

V. Estimation Strategy and Results

In this section, I outline the estimation procedure and present estimates of firm-specific labor supply elasticities and structural parameters. My results show that there is large variation in labor supply elasticities both within and across firms, implying that wage-setting power varies considerably in the economy. I also document sizable dispersion in firm productivity, technology, and non-wage amenities, providing insight into how these factors shape worker sorting.

My estimates reveal two key patterns. First, within firms, labor supply is less elastic for more educated and experienced workers. This suggests that workers with observably higher skills face larger wage markdowns. Second, across firms, labor supply is less elastic for firms that pay lower wages to a given skill group. This second pattern aligns with the model prediction that, holding skills fixed, lower-paying firms disproportionally attract workers who are less responsive to wages. Note that the estimation nowhere imposes this relationship, so the result offers substantive support for the mechanism rather than reflecting any built-in structure.

V.A. Estimation of Firm-Specific Labor Supply Elasticities

Estimation Procedure and Specifications

My estimation procedure is based on the identification strategy in Section IV.A, and it consists of two main steps. First, I estimate worker skill levels by implementing a two-way fixed effects estimator, where I regress log wages on a worker fixed effect γ_i and a fixed effect $\psi_{\chi(i,\tau),j(i,\tau),\tau}$ for the skill type, firm, and year. From this regression, I define the permanent component of the skill level as $\exp(\gamma_i)$ and the effective wage $W^{\text{eff}}_{j\tau}(\chi)$ as $\exp(\psi_{\chi(i,\tau),j(i,\tau),\tau})$.[25] Table 2 presents summary statistics about worker skill levels, with further details provided in the Appendix.

Second, I estimate the elasticities of labor supply to each firm by computing a sample analogue of the estimand $\text{DiD}_{\tau_0,\tau_1}(w|\chi)$. This estimand corresponds to a ratio of DiD estimands for wages and labor, controlling for a firm's pre-period effective wage. Since there are finitely many firms in the data, I rarely observe multiple firms that pay exactly the same effective wage to workers. Thus, for the purposes of implementation, I use a nonparametric kernel estimator:

$$\frac{\sum_j K_{1,j}(w)\mathbf{1}\{Z_j = 1\}\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big) - \sum_j K_{0,j}(w)\mathbf{1}\{Z_j = 0\}\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big)}{\sum_j K_{1,j}(w)\mathbf{1}\{Z_j = 1\}\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big) - \sum_j K_{0,j}(w)\mathbf{1}\{Z_j = 0\}\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big)}, \quad (28)$$

where $K_{1,j}(w)$ and $K_{0,j}(w)$ are Kernel weights for treated and untreated firms, respectively. This estimator allows me to compare firms that offer similar, though not identical, pre-period wages, thus ensuring tractability in finite samples. In Appendix B.8, I prove that the estimator is consistent, and I derive its asymptotic properties, establishing that it is capable of inference.

[25] In the Appendix, I show that defining $W^{\text{eff}}_{j\tau}(\chi)$ as the average wage in each (χ, j, τ) cell yields similar results.

In the baseline specification, I use a Gaussian kernel function to define the weights $K_{z,j}$.[26] As shown in the Appendix, the estimates are robust to alternative kernels, including the Uniform kernel, which corresponds to a standard binning estimator. I also assess the sensitivity of the estimates to the bandwidth choice, finding that the results are stable across varying bandwidths. These robustness checks validate the estimator's ability to approximate $\text{DiD}_{\tau_0,\tau_1}(w|\chi)$.

My primary estimates are based on internal instruments, following the identification analysis in Section IV.A.4. For each year τ that TFP shocks occur, I define Z_j by splitting firms at the median of the distribution of $\Delta y_{j\tau}$. I then define the pre- and post-periods to be $\tau_0 = \tau - 2$ and $\tau_1 = \tau + 3$, respectively. I estimate the model repeatedly using all years of data, with the post-period τ_1 ranging from 2002 to 2017. As a robustness check, I also report estimates from the external instruments, defined from public procurement auction wins in Norway. These supplementary results draw on the research design of de Frahan et al. (2024), who also use Norwegian administrative data. I report the external IV estimates in the Appendix, along with a description of the data sources. Both internal and external instruments deliver similar results.

In implementing my approach, I include fixed effects for local labor markets m, where m represents the interaction between an industry and a commuting zone, as defined in Section III. For comparison, I also estimate the model without fixed effects, treating Norway as one single labor market; those results are reported in the Appendix. I average my estimates across all available years of data to improve precision. Since I allow the labor supply elasticities to evolve over time, my estimates can be interpreted as an average of these quantities over multiple years.

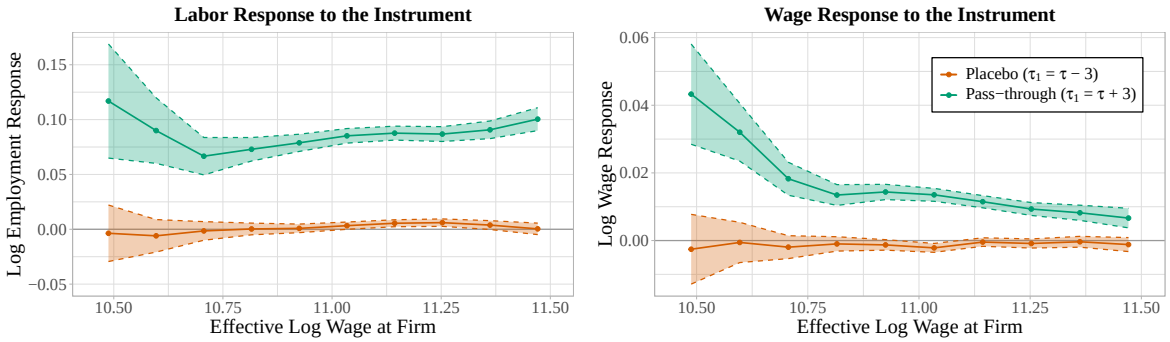
Description of Empirical Findings

Figure 4 presents reduced form estimates, averaged over worker skill types and plotted across the distribution of effective wages. These estimates correspond to the numerator and denominator of the estimator in equation (28). The green line shows estimates for $\tau_1 = \tau+3$, while the orange line plots differences in pre-trends between treated and untreated firms, with $\tau_1 = \tau-3$. The placebo effect is not significantly different from zero, lending credibility to the IV design. In the Appendix, I report reduced form estimates by skill type, and I present event study plots of the pass-through before and after the shock; these plots also show no evidence of pre-trends.

The reduced form estimates reveal two distinct patterns. First, the employment response to the instrument lies consistently between 7% and 11%, suggesting that employment growth is relatively independent of firm size. This observation aligns with Gibrat's law. Second, the earnings response ranges from 1% to 5%, and it declines with a firm's effective wage. Based on equation (17), this pattern could suggest that higher-paying firms tend to have lower returns to scale. Additionally, it could indicate that these firms face higher elasticities of labor supply.

[26] Specifically, I set $K_{z,j}(w) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{1}{h}(w^{\text{eff}}_{j\tau_0}(\chi)-w)\right)^2\right]$ where the parameter h governs the bandwidth.

Figure 4: Average Pass-Through by Workers' Effective Wages



Notes. This figure plots the reduced form estimates, averaged across skill types and years $\tau_1 \in \{2002, \ldots, 2017\}$. This specification includes market fixed effects. Standard errors are bootstrapped using 500 bootstrap samples.

Figure 5 presents IV estimates of the firm-specific labor supply elasticities, disaggregated by workers' college attainment and experience, along with the implied wage markdowns from equation (11). The estimated elasticities differ substantially across firms, ranging from about 1.5 to 10 for certain worker types. These differences strongly reject a model with constant β. Furthermore, they indicate that there are large differences in wage-setting power among firms.

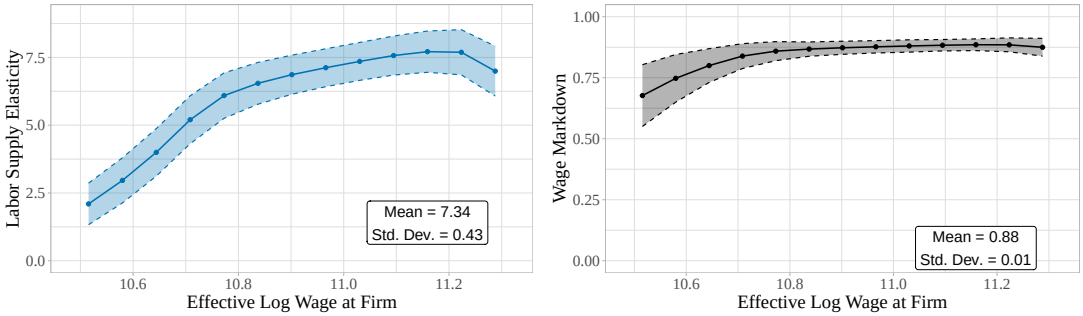
One key pattern that emerges from the estimates is that firms paying higher wages to a given skill group tend to face larger labor supply elasticities. This relationship holds consistently across all worker skill types, and it aligns with the model's prediction that (holding skills fixed) lower-paying firms tend to attract workers who are less responsive to earnings. This implies that market power is more concentrated at the lower end of the effective wage distribution.

The elasticity estimates also reveal notable differences by worker skill type. In particular, I find that the labor supplied to a firm is generally less elastic for more experienced workers, and it is especially inelastic for more experienced workers with a college degree. This finding implies that workers with more experience and more education tend to place greater value on non-wage amenities relative to wages when choosing a job. In addition, I find substantially more variation in labor supply elasticities among less experienced, college-educated workers, suggesting that there may be greater dispersion in wage-amenity trade-offs for this subgroup.

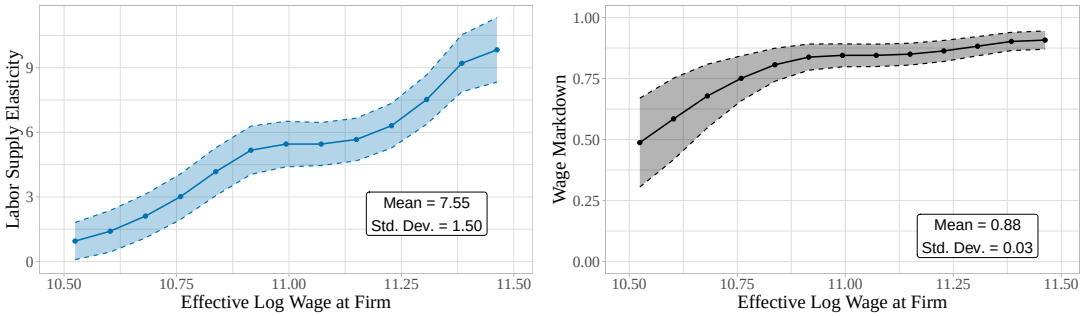
To further validate my approach, I conduct two robustness checks. First, I test whether the instrument Z_j affects labor on the intensive margin by changing work hours. This test addresses a potential concern that firms might respond to productivity shocks by adjusting workers' hours instead of workers' wages. If this were the case, it would threaten the validity of the IV design, which assumes that the pass-through of demand shocks operates only through wages. I find no significant effect on workers' hours, supporting the credibility of the IV assumptions.

Second, I test the assumption that firms are strategically small in the economy. A testable implication of this assumption is that, in my model, firms paying identical wages to workers

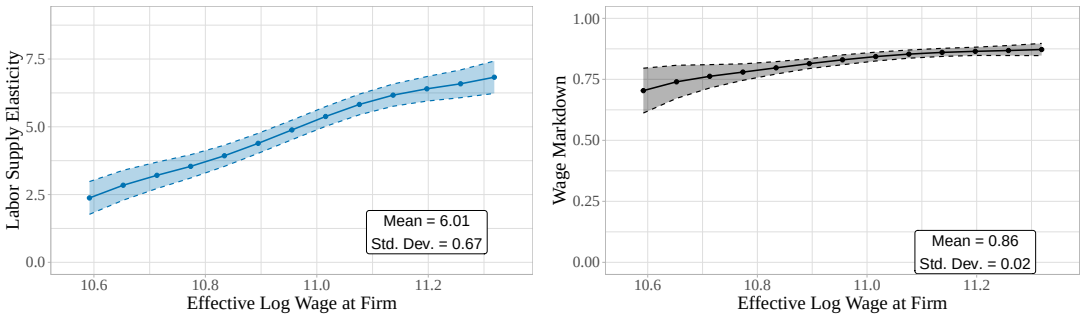
Figure 5: IV Estimates of Labor Supply Elasticities and Implied Wage Markdowns



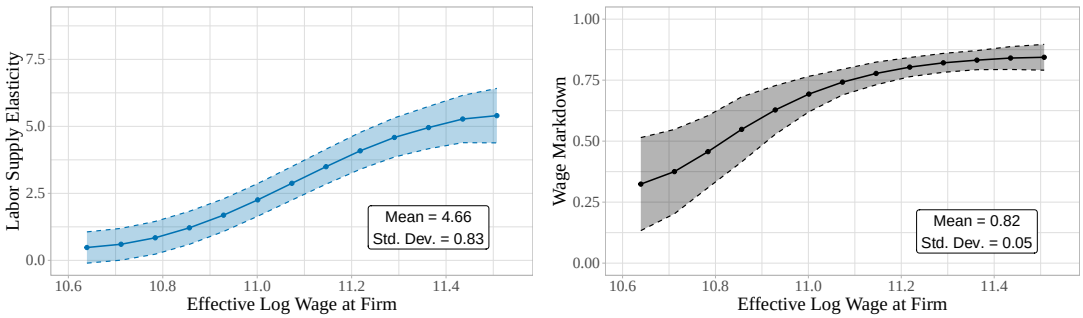
(a) No College Degree, < 10 Years Experience



(b) College Graduate, < 10 Years Experience



(c) No College Degree, ≥ 10 Years Experience



(d) College Graduate, ≥ 10 Years Experience

Notes. This figure plots IV estimates of firm-level labor supply elasticities and wage markdowns, averaged over all years of data, for $\tau_1 \in \{2002, \ldots, 2017\}$. The subgroup "No College Degree" is a worker-weighted average of two skill types: "Lower Secondary" and "Upper Secondary". Population-weighted moments are reported. This specification includes market fixed effects. Standard errors are bootstrapped using 500 bootstrap samples.

with the same skills face the same labor supply elasticities. By contrast, in a concentrated labor market with strategically large firms, two firms paying the same wages to the same types of workers can still face different elasticities if their market shares differ. Guided by this property, I test whether a firm's labor supply elasticity varies with its market share after controlling for effective wages. I find no evidence of a market share effect, lending support to the assumption.

V.B. Estimation of Structural Parameters in the Model

Next, I present estimates of the structural parameters in the model, including firm productivity and technology $(T_{j\tau}, \{\theta_{j\chi}\}_\chi, \rho_j, \alpha_j)$, non-wage amenities $a_{j\chi}$, and the densities $f_{\beta|\chi,\tau}$ governing workers' preferences. These estimates are summarized in Table 2 and in Figures 6 and 7.

Table 2: Structural Parameter Estimates—Means and Variances

	Notation	Mean	Variance
Panel A. Firm Production			
Log Total Factor Productivity	$t_{j\tau}$	12.940 (0.026)	1.187 (0.011)
Returns to Scale	$1-\alpha_j$	0.700 (0.006)	0.035 (0.001)
Substitution Parameter	ρ_j	0.746 (0.018)	0.463 (0.055)
Panel B. Firm Amenities	$a_{j\chi}$		
No College, < 10 Years Exp.		—	1.598 (0.124)
College, < 10 Years Exp.		—	2.120 (0.244)
No College, ≥ 10 Years Exp.		—	1.552 (0.074)
College, ≥ 10 Years Exp.		—	1.390 (0.079)
Panel C. Worker Skill Levels	φ_i		
No College, < 10 Years Exp.		—	0.062 (0.011)
College, < 10 Years Exp.		—	0.073 (0.021)
No College, ≥ 10 Years Exp.		—	0.064 (0.014)
College, ≥ 10 Years Exp.		—	0.076 (0.020)
Panel D. Worker Preferences	β_i		
No College, < 10 Years Exp.		7.270 (0.789)	5.204 (0.939)
College, < 10 Years Exp.		7.647 (1.120)	16.767 (4.911)
No College, ≥ 10 Years Exp.		6.075 (0.334)	7.340 (1.206)
College, ≥ 10 Years Exp.		4.898 (0.601)	9.269 (2.500)

Notes. This table reports cross-sectional means and variances of estimated model parameters, averaged across years. Means are omitted for parameters identified only up-to-location.

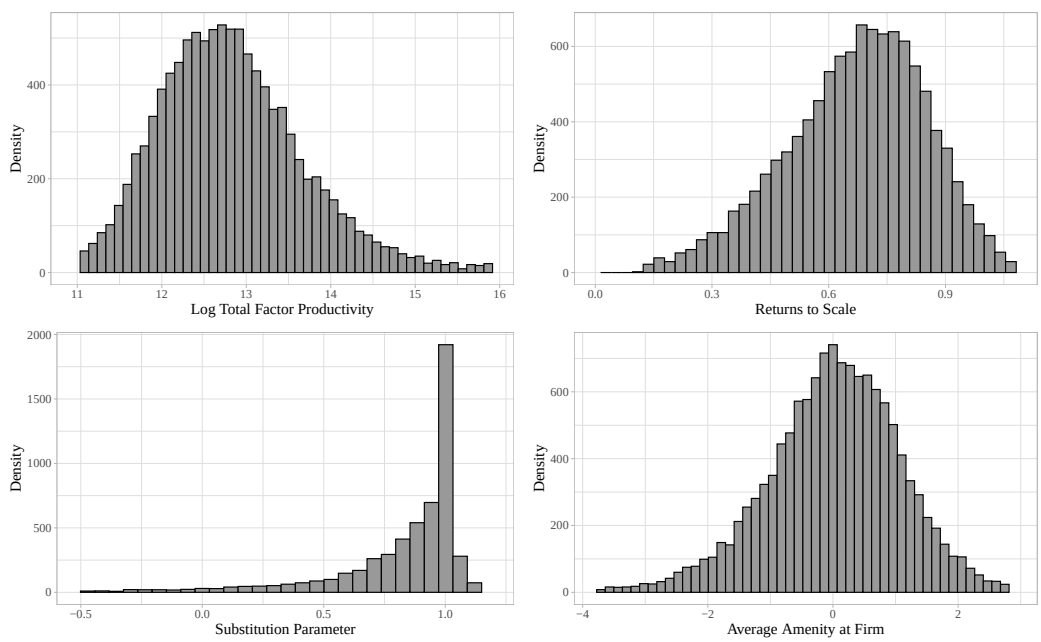
Firm Productivity, Technology, and Amenities

To estimate the production parameters, I apply the formulas in Propositions 3 and 4, which rely on firms' estimated elasticities, effective wages, and value added. I perform these calculations separately for each firm j and year τ, giving me a unique set of estimates for every combination

of (j, τ).[27] In Figure 6, I plot the marginal distributions of estimated TFP $T_{j\tau}$, returns to scale $1-\alpha_j$, and the substitution parameter ρ_j, averaged over all years. The means and variances are reported in Table 2. I find substantial dispersion in firms' returns to scale and substitution parameters, indicating that differences in firm technology play a key role in wage determination.

To estimate the value of firms' non-wage amenities, I follow Proposition 5 by integrating the estimated labor supply elasticities with respect to effective wages. This exercise allows me to compare the labor supplied to firms in a counterfactual where all firms pay the same wage.[28] Figure 6 plots the distribution of firms' average amenities, and Table 2 reports the variances by worker type. I find that the variances are broadly similar across worker types, with the exception of less experienced, college-educated workers, who face greater dispersion in amenities.

Figure 6: Histograms of Estimated Firm Productivity, Technology, and Amenities



Notes. This figure plots histograms of the estimated (log) total factor productivity $\log T_{j\tau}$, the returns to scale $1-\alpha_j$, the substitution parameter ρ_j, and the average amenities $\mathrm{E}(a_{j\chi}|j)$ across firms in the estimation sample.

Worker Preferences

To estimate the density $f_{\beta|\chi,\tau}$, I draw on Proposition 6, which shows that $f_{\beta|\chi,\tau}$ is uniquely determined by the labor supply curve to a firm. Therefore, it is identified as the unique solution to:

[27] Recall that recovering the substitution parameter ρ_j for a firm j requires me to compare estimates over two different time periods, τ and τ'. In my implementation, I set τ' to be $\tau - 1$, corresponding to the previous year.

[28] For each skill type χ, amenities are normalized relative to a "reference firm" j^*, where $a_{j^*\chi} = 0$. In my analysis, j^* is defined as the firm offering an average wage equal to the median of the distribution in the sample.

$$L_{j\tau}(\chi, W) = \int \frac{\exp\left(\beta \log W + a_{j\chi}\right)}{I_\tau^{\text{eff}}(\beta, \chi)} f_{\beta|\chi,\tau}(\beta|\chi, \tau) f_{\chi|\tau}(\chi|\tau) d\beta, \tag{29}$$

for all $W > 0$, where the labor supply curve $L_{j\tau}(\chi, W)$, the effective wage index $I_\tau^{\text{eff}}(\beta, \chi)$, the density of skill types $f_{\chi|\tau}$, and the non-wage amenities $a_{j\chi}$ can all be recovered from the data.

To make the estimation problem tractable, I introduce two important modifications. First, since the density function $f_{\beta|\chi,\tau}$ is an infinite dimensional parameter, I approximate it using a finite dimensional basis expansion. Specifically, I replace $f_{\beta|\chi,\tau}$ with a function $\hat{f}_{\beta|\chi,\tau}$, where:

$$\hat{f}_{\beta|\chi,\tau}(b) = \sum_{k=1}^{d_\phi} \phi_k \hat{f}^{(k)}_{\beta|\chi,\tau}(b), \quad \text{for } \phi = (\phi_1, \ldots, \phi_{d_\phi}) \in \Phi. \tag{30}$$

In this expression, ϕ is a d_ϕ-dimensional vector of unknown coefficients with support Φ, and $\{\hat{f}^{(k)}_{\beta|\chi,\tau}\}_k$ are basis functions known to the researcher. By defining the basis in this way, I can approximate $f_{\beta|\chi,\tau}$ with a function $\hat{f}_{\beta|\chi,\tau}$ that is parametrized by a finite dimensional vector ϕ.

Second, rather than imposing an infinite number of constraints, I evaluate equation (29) on a finite grid of effective wages $\mathbf{W} \subset \mathbb{R}_+$, noting that identification is achieved as $|\mathbf{W}| \to \infty$. This modification allows me to recast (29) as a convex optimization problem, minimizing the ℓ^2-distance between the estimated choice probabilities and the predicted probabilities implied by the density $\hat{f}_{\beta|\chi,\tau}$. Specifically, my estimation procedure involves choosing ϕ^* by solving:

$$\phi^* = \text{argmin}_\phi \sum_{W \in \mathbf{W}} \left[\phi' \hat{h}_\tau(\chi, W) - \hat{L}_{j\tau}(\chi, W)\right]^2. \tag{31}$$

where $\hat{L}_{j\tau}(\chi, W)$ is the estimated labor supply at $W^{\text{eff}}_{j\tau}(\chi) = W$ and $\hat{h}_\tau(\chi, W) \in \mathbb{R}^{d_\phi}$ denotes a vector with entries $\hat{h}_{\tau,k}(\chi, W) = \int \left[\exp\left(\beta \log W + \hat{a}_{j\chi}\right) / \hat{I}_\tau^{\text{eff}}(\beta, \chi)\right] \hat{f}^{(k)}_{\beta|\chi,\tau}(\beta|\chi, \tau) \hat{f}_{\chi|\tau}(\chi|\tau) d\beta$.[29]

In the baseline specification, I construct a basis using mixtures of $d_\phi = 18$ Gamma density functions. This type of density is shown to be particularly well suited for nonparametric estimation of functions that take support on the positive real line (e.g., Wiper et al., 2001). As a robustness check, I investigate the sensitivity of the estimates to the choice of basis function, the basis size d_ϕ, and the number of points $|\mathbf{W}|$ in the wage grid.[30] I find that the estimates are robust across alternative basis functions and that they stabilize once the basis size and the wage grid become sufficiently rich. Moreover, I show that the wage and employment distributions implied by the estimated densities $\hat{f}_{\beta|\chi,\tau}$ closely match those in the sample, indicating good fit.

Figure 7 plots the estimated density of β among workers in the population. I also compare

[29] Note that equation (31) can be written in matrix form as $\phi^* = \text{argmin}_\phi \left\{\phi^\top \hat{h}^\top \hat{h} \phi - 2\phi^\top \hat{h}^\top \hat{L} + \hat{L}^\top \hat{L}\right\}$, where $\hat{L} \in \mathbb{R}^{|\mathbf{W}|}$ and $\hat{h} \in \mathbb{R}^{|\mathbf{W}| \times d_\phi}$. This framing of the problem is more convenient for computational implementation.

[30] For example, I also approximate $f_{\beta|\chi,\tau}$ using a linear sieve constructed from Laguerre polynomials. These functions are capable of approximating any square-integrable function with positive support (see Chen, 2007).

moments of this distribution for different worker types in Table 2. I find that more educated and more experienced workers have lower average values of β, implying that they tend to have a higher willingness to pay for non-wage amenities. There is, however, substantial variation in β within each worker type. The full distributions by worker type are shown in the Appendix.

Figure 7: Density of Workers' Individual Labor Supply Elasticities

Notes. This figure plots the estimated density f_β, corresponding to the distribution of workers' marginal rates of substitution between wages and amenities. The confidence region is bootstrapped using 500 bootstrap samples.

VI. Evaluating the Incidence and Consequences of Labor Market Power

I now present the main findings from the empirical framework, quantifying imperfect competition in the labor market and evaluating its impacts on wage inequality, sorting, and welfare.

VI.A. Wage Markdowns and Rent Sharing

I begin by quantifying the incidence of labor market power, as measured by wage markdowns and the division of rents between workers and firms. Table 3 reports key moments of the estimated distributions—means, standard deviations, and a variance decomposition into within- and between-firm components—and compares them to the corresponding moments implied by a restricted model with constant β.[31] This comparison allows me to isolate the role of heterogeneous labor supply responses in shaping both the level and dispersion of market power.

[31] In the restricted model, I set β to the economy-wide mean from the full model, estimated as $\mathrm{E}(\beta) \approx 6.9$. I also consider an alternative specification that sets β equal to the IV estimand from a constant-effects model (e.g., Lamadon et al., 2022). I compute this estimand as an unconditional version of (21), yielding $\beta^{\mathrm{IV}} \approx 6.4$. The inequality $\beta^{\mathrm{IV}} < \mathrm{E}(\beta)$ is consistent with the fact that, under heterogeneous effects, linear IV recovers a weighted average of elasticities that is attenuated relative to the unweighted mean (Angrist et al., 2000; Gaarder et al., 2026).

On average, I find that workers are paid 86.5% of their marginal product. However, wage markdowns vary greatly both within and across firms, ranging from about 70% to 95% at the 1st and 99th percentiles. About 60% of the variation in markdowns is within firms, while the remaining 40% is between firms. In the model, within-firm variation reflects systematic differences in labor supply elasticities across worker skill types, while between-firm variation reflects selection: conditional on skills, firms attract workers with different labor supply responses.

Table 3: Estimates of Wage Markdowns and Rents

	Full Model: Heterogeneous β			Restricted Model: Homogeneous β		
	Mean	Std. Dev.	Variance Decomp. (Within Firm / Between Firm)	Mean	Std. Dev.	Variance Decomp. (Within Firm / Between Firm)
Wage Markdowns	0.865	0.036	62.46% / 37.54%	0.874	0.000	——
	(0.007)	(0.010)		(0.007)		
Worker Rents						
Dollars per Worker	\$12,976	\$3,946		\$10,305	\$1,818	
	(800)	(1,292)		(580)	(118)	
Share of Earnings	16.1%	6.2%	68.29% / 31.71%	12.6%	1.2%	100.00% / 0.00%
	(1.0)	(1.4)		(0.7)	(0.1)	
Firm Rents						
Dollars per Worker	\$8,375	\$3,031		\$7,990	\$2,853	
	(796)	(1,096)		(622)	(169)	
Share of Profits	23.0%	20.1%	0.00% / 100.00%	21.6%	19.1%	0.00% / 100.00%
	(1.2)	(0.4)		(1.1)	(0.5)	

Notes. This table shows estimated wage markdowns and rents under two model specifications. The main specification allows for unrestricted variation in workers' wage-amenity trade-offs (β), and the restricted specification sets β constant across workers, equal to the mean $\mathrm{E}(\beta)$ under the full model. Variance decompositions show the share of variation in each quantity attributable to within-firm and between-firm components, calculated via the Law of Total Variance: $\mathrm{Var}(X) = \mathrm{E}(\mathrm{Var}(X|j)) + \mathrm{Var}(\mathrm{E}(X|j))$. Standard errors are bootstrapped using 500 replications.

To further characterize the incidence of wage markdowns, Appendix Figure A.18 plots the markdown distribution by worker education and experience. Markdowns tend to be larger for more experienced workers, especially those with a college degree, and dispersion is also larger for college-educated workers. Table A.2 reports means and variances by industry and region. Overall, industry differences are modest, with average markdowns ranging from 85% to 88% across industries. Markdowns are somewhat smaller in high-wage sectors such as finance and larger in healthcare, education, and retail, though these patterns vary by skill type. By contrast, regional differences are sizable: the wage-to-marginal-product ratio averages 84.5% in rural areas versus 87.5% in urban areas. This gap is consistent with greater wage-setting power in rural labor markets, and it suggests that monopsony contributes to the urban-rural wage gap.[32]

[32] See Manning (2010), Hirsch et al. (2022), and Azkarate-Askasua & Zerecero (2025) for more evidence on the role of monopsony in generating regional wage differences. My framework builds on previous work by estimating an entire distribution of wage markdowns in each region, rather than a single region-specific markdown.

I find that many workers and firms capture substantial rents. The average worker is willing to pay \$12,976 (about 16% of earnings) to be at her current firm, while the average firm captures \$8,375 per worker (about 23% of profits) through wage-setting power. Appendix Figure A.19 plots the estimated distributions of worker and firm rents. This figure shows that rents are highly dispersed: worker rent shares range from less than 10% to above 25% of earnings, while firm rent shares range from below 5% to above 50% of profits. Also, Figures A.20 and A.21 show that worker rent shares tend to be larger for more educated and experienced workers, and that firm rent shares tend to be larger for firms with lower estimated productivity T_j.

Finally, I benchmark these estimates against those from the restricted model with constant β. This restriction rules out all heterogeneity in workers' wage markdowns, within and across firms. It also forces worker rent shares to be independent of one's skills X and firm j. Moreover, it implies that firm rent shares cannot depend on workforce composition, so any variation is driven by differences in technology (α_j, ρ_j). Figure A.19 compares the densities of rents in the restricted model to those in the full model. The marked differences between these densities show that allowing for heterogeneous labor supply responses is important for measuring rents.

VI.B. Consequences for Wage Inequality and the Gender Wage Gap

To evaluate the implications of labor market power for wage inequality, I compare the equilibrium outcomes in the monopsonistic economy to counterfactual outcomes that would arise in a competitive (Walrasian) economy, in which firms set wages equal to workers' marginal products. I solve for the counterfactual equilibrium, taking into account how workers reallocate across firms in response to wage changes. Table 4 reports these counterfactual estimates and compares them to the corresponding estimates implied by a restricted model with constant β.

I find that labor market power has sizable impacts on wage inequality, as measured by the variance of log earnings. Eliminating labor wedges would increase within-firm inequality by 1.3% and reduce between-firm inequality by 16%, leading to a 3.9% decline in total inequality. This finding reflects that, within firms, more educated and experienced workers tend to face larger wage markdowns, while, across firms, markdowns are largest among employers that pay lower wages to a given skill group. Removing labor wedges therefore widens within-firm skill premia while compressing wage differences between firms. A restricted model with constant β would not account for these effects, as it forces wage markdowns to be fixed across workers.

Next, I apply my framework to study the gender wage gap. Several recent papers, such as Card et al. (2016), Sharma (2023), and Caldwell & Oehlsen (2023), infer gender differences in monopsony power by estimating labor supply elasticities separately for men and women. I extend this analysis by allowing the elasticities to vary within gender and, for each gender, estimating the full distribution of workers' wage-amenity trade-offs. This distributional perspective delivers a richer characterization of gender wage gaps, allowing me to assess whether

gender differences in monopsony are widespread or concentrated in certain parts of the labor market. The framework assumes that firms cannot condition wages directly on gender. Thus, conditional on workers' skills, gender differences in wage markdowns arise from differences in firm composition: differential sorting of men and women to firms with different markdowns.

Table 4: Counterfactual Estimates—Measuring Impacts on Wage Inequality and Welfare

	Full Model: Heterogeneous β			Restricted Model: Homogeneous β		
	Monopsonistic Labor Market	Competitive Labor Market	% Difference in Dollars	Monopsonistic Labor Market	Competitive Labor Market	% Difference in Dollars
Panel A. Market Aggregates						
Mean Earnings (Log Dollars)	11.291	11.393	10.74%	11.291	11.400	11.52%
		(0.010)			(0.009)	
Mean Profit (Log Dollars)	14.687	14.334	-29.73%	14.687	14.439	-21.91%
		(0.014)			(0.008)	
Panel B. Earnings Inequality						
Variance in Log Earnings						
Total	0.102	0.098	-3.92%	0.102	0.102	0.00%
		(0.004)				
Between Firms	0.025	0.021	-16.00%	0.025	0.025	0.00%
		(0.003)				
Within Firms	0.076	0.077	1.32%	0.076	0.076	0.00%
		(0.001)				
Gender Wage Gap	14.3%	11.0%	-23.1%	14.3%	14.3%	0.00%
		(1.2)				
Panel C. Allocative Inefficiency						
Total Welfare (Log Dollars)	13.277	13.378	9.62%	12.864	12.864	0.00%
	(0.211)	(0.223)		(0.064)	(0.064)	
Labor-TFP Correlation	0.182	0.156	-14.26%	0.180	0.180	0.00%
	(0.006)	(0.014)		(0.005)	(0.005)	

Notes. This table considers how key economic quantities, such as measures of wage inequality and welfare, differ between the monopsonistic labor market and a counterfactual in which all firms are price-takers, so that the labor wedge in the first-order conditions is removed. Results are reported for two model specifications: the full model, which allows for unrestricted variation in workers' wage-amenity trade-offs (β), and a restricted model where β is constant, equal to the mean E(β) under the full model. Standard errors are bootstrapped using 500 replications.

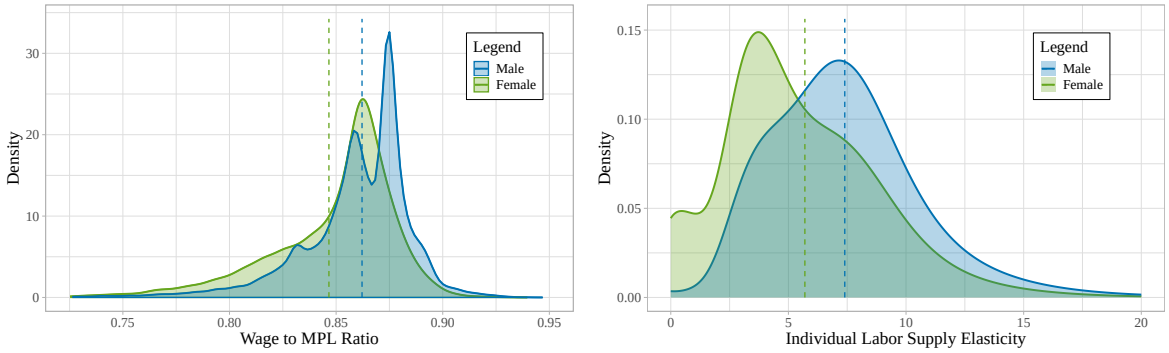
To analyze the gender wage gap, I take three steps. First, I examine whether women systematically face more wage-setting power than men. To do so, I compare the estimated distribution of wage markdowns across firms, summarized in Table 3, with the gender composition of each firm, evaluating how the incidence of markdowns varies by gender. The estimates, shown in Figure 8(a), indicate that, although there is substantial variation within each gender, women tend to face larger wage markdowns than men: on average, the wage-to-marginal-product ratio is 84% for women and 87% for men. This gap is primarily driven by between-firm rather than within-firm differences, with women disproportionately employed at high-markdown firms.

Second, I estimate the gender-specific densities of β, applying the estimator from equation

(30) separately for men and women. The resulting densities, plotted in Figure 8(b), reveal that there is substantial heterogeneity in wage-amenity trade-offs within each gender; however, on average, women place greater value on amenities than men. This finding is consistent with evidence from choice experiments (Mas & Pallais, 2017; Wiswall & Zafar, 2018) that many women exhibit a higher willingness to pay for non-wage job attributes like workplace flexibility.

Third, I evaluate how the gender wage gap would change if labor wedges were eliminated, taking into account how men and women reallocate differently in response to wage changes. As reported in Table 4, removing labor wedges reduces the gender wage gap by 3.3 percentage points, from 14.3% to 11%. In Appendix D.X, I decompose this effect into within- and between-firm components and show that it is primarily driven by changes in wages between firms.

Figure 8: Individual Labor Supply Elasticities and Wage Markdowns by Gender



(a) Density of Wage Markdowns (b) Density of Individual Labor Supply Elasticities

Notes. Figure 8(a) plots the estimated densities of wage markdowns by gender. Figure 8(b) plots the estimated densities f_β of workers' marginal rates of substitution between wages and amenities by gender. To compute f_β, I first estimate the firm-specific labor supply curves separately for men and women, and then I implement the estimator from equation (30). The dashed lines in each figure indicate the means of the estimated distribution.

VI.C. Consequences for Worker Sorting

I now examine how labor market power affects worker sorting. These effects depend critically on where wage markdowns are concentrated in the economy. If markdowns are identical across firms, then market power would not distort the allocation of labor. By contrast, if some firms set larger markdowns than others, this can distort workers' rankings over employers and lead them to choose firms they would not select in a competitive labor market. Quantifying misallocation therefore necessitates a rich characterization of how wage markdowns vary in the economy, which requires a framework that allows for heterogeneous labor supply responses.

To see how workers would reallocate in the absence of market power, recall that wage markdowns are largest at firms that pay lower wages to a given skill group. These firms tend to be those where labor is less-productive (low T_j). Indeed, Appendix Figure A.21 plots firms' aver-

age wage markdowns against TFP and finds that markdowns tend to be larger the bottom of the TFP distribution. Given this relationship, removing labor wedges would yield larger wage and employment gains for low-productivity firms. In line with this prediction, I find that the correlation between log employment ℓ_j and log TFP t_j is higher in the present allocation (0.182) than in the competitive allocation (0.156). Figure 9 illustrates this effect by plotting, across the firm productivity distribution, the predicted change in firms' labor shares from removing labor wedges. In the competitive allocation, more workers are employed at low-productivity firms.

To further decompose the allocative effects of market power, it is important to understand how the incidence of wage markdowns across firms varies by skill type. If the same firms are uniformly "high-markdown" or "low-markdown" for all workers, then removing labor wedges would shift each firm's employment in the same direction for all skill types. However, if the incidence of markdowns is skill-specific, then removing wedges can generate more complex reallocation patterns, with the same firm employing more of some workers and fewer of others.

I assess this question empirically by comparing how the ranking of wage markdowns across firms differs by skill type. Appendix Figure A.22 presents rank-rank plots of firm-level markdowns for each pair of skill types and reports Spearman rank correlations as a summary measure of the similarity of these rankings. I find that rankings are positively correlated across skill types, but the relationship is not one-to-one. Especially for college versus non-college workers, many firms rank as high-markdown for one group and low-markdown for the other.

I then examine, by skill type, the implied reallocation of labor when wedges are removed. Appendix Figure A.23 plots these reallocations by industry and region. For some sectors, the direction of reallocation is uniform across skill types: finance and business services contract, while retail, education, and healthcare expand; employment also rises in rural areas. In other sectors, such as trade, manufacturing, and construction, the direction of reallocation is skill-specific, with the same sector expanding for some groups while contracting for others. These results highlight the importance of allowing wage markdowns to vary within and across firms. Imposing a common markdown distribution would misrepresent the nature of misallocation.

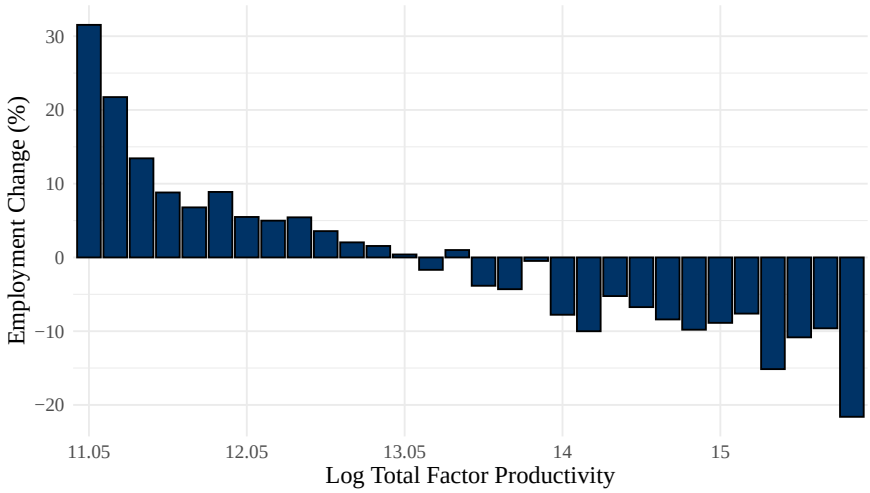
VI.D. Welfare Costs and the Role of Policy

Lastly, I assess the welfare cost of labor market power, taking into account firm profits and the value of non-wage amenities to workers. In the model, welfare losses arise from misallocation of workers to firms, driven by differences in firms' wage markdowns. In a version of the model with constant β, markdowns do not vary, leaving no scope for market power to affect welfare.

I estimate large efficiency gains from eliminating labor wedges, with total welfare increas-

ing by 9.62% in dollar terms (Table 4, Panel C).[33] In principle, a planner could implement the first-best allocation through a redistributive tax system that offers wage-specific subsidies to workers, with the shape of the subsidy schedule determined by the distribution of preferences. In practice, however, such a finely targeted policy is hard to envision. Nevertheless, because labor wedges are not purely idiosyncratic but instead differ systematically based on worker, firm, and regional characteristics, second-best policies that target earnings along observable dimensions may compress the dispersion in labor wedges and help to reduce misallocation.

Figure 9: Estimated Reallocation of Labor by Eliminating Labor Wedges



Notes. This figure depicts the implied reallocation of labor resulting from eliminating labor wedges. It plots the estimated percent change in employment across different groups of firms, grouped by their estimated log TFPs.

One policy tool that is often debated as a response to labor market power is the minimum wage.[34] The welfare effect of this policy depends not just on the average level of market power but also on where it is located; namely, how pervasive it is among workers who would be affected by a minimum wage. If wage-setting power only affects high-wage workers, then a minimum wage is unlikely to target distortions and may reduce welfare. If instead it is concentrated among low-wage workers, then a minimum wage may have more scope for efficiency gains.

Since the model abstracts from nonemployment and firm entry and exit, I do not run a full counterfactual to study the equilibrium effects of a minimum wage, as in Berger et al. (2025) and Haanwinckel (2025). Instead, I examine the estimated incidence of labor wedges across the wage distribution to provide suggestive evidence on the policy's potential welfare effects.

[33] In interpreting these results, it is important to recall that firms may choose amenities $a_j(X)$ initially, those $a_j(X)$ is held fixed in the counterfactuals. With richer data on amenities and a credible instrument, an interesting extension would allow firms to adjust $a_j(X)$ in response to the counterfactual changes. This task necessitates further information—or assumptions—about how firms choose and pay for the amenities provided to workers.

[34] In Norway, there is no government-mandated minimum wage. Instead, minimum wages are bargained at the sector level. In the administrative data, I find fewer than 4% of workers are bound to their minimum wages.

The results are a priori ambiguous: more educated and experienced workers face larger markdowns on average, yet conditional on skill, lower-paying firms set larger markdowns. Appendix Figure A.24 reconciles these patterns by plotting workers' average markdowns against their realized wages. While labor wedges are highly dispersed throughout the wage distribution, I find they are larger on average among lower-paid workers, indicating that wage-setting power is concentrated among workers most likely to be bound by the minimum wage.

VII. Conclusions

In this paper, I examined how an employer's wage setting power depends on the characteristics of its workforce, and I assessed the consequences of this relationship for welfare and inequality. To this end, I developed an equilibrium model of a frictional labor market with two-sided heterogeneity, in which workers make different trade-offs between wages and non-wage aspects of jobs. Using this framework, I showed that a firm's wage setting power depends directly on the types of workers it employs, which is influenced by the wages and amenities the firm provides. As a result, wage markdowns and rents can vary both within and across firms, and are also endogenous to wage changes in the labor market. I then showed that this framework is empirically tractable and can be used to draw inferences about sorting, imperfect competition, and rent sharing. I estimated the model using a matched worker-firm panel dataset that covers the universe of workers and firms in Norway between 1995 and 2018. My estimates reveal that wage setting power varies greatly in the economy, and that this variation has key implications for how imperfect competition impacts wage inequality and welfare.

This paper provides an initial examination into the link between firm workforce composition and wage setting power; however, there is much room for future work. For example, my framework abstracts from strategic competition where firms internalize and leverage their market shares. In the Norwegian setting, I validate this assumption by testing whether a firm's market share affects its labor supply elasticities, and I find no significant effect. However, this test does not definitively rule out the existence of strategic interactions among firms. A meaningful extension of the framework would be to examine how a firm's wage setting power is influenced not only by the composition of its workforce but also by its size in the market.

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Appendix

This appendix accompanies the paper "Job Preferences, Labor Market Power, and Inequality". It contains derivations of key economic quantities, model extensions, proofs of equilibrium properties and identification results, details about data preparation and estimation, and robustness analyses. All notation is consistent with the rest of the paper.

A. *Derivation of Equilibrium Quantities*

A.1. *Firm Labor Supply*

Given any set of wage offers $\mathbf{W}(X) = \{W_k(X)\}_k$, a worker with skills X and marginal utility of (log) earnings β would choose to work at firm j with probability $\mathrm{P}(j(i) = j|\beta, X)$, where:

$$\begin{aligned}
\mathrm{P}(j(i) = j|\beta, X) &= \mathrm{P}\left(u_{ij}\big(W_j(X_i), a_j(X_i)\big) > \max_{k\neq j}\left\{u_{ik}\big(W_k(X_i), a_k(X_i)\big)\right\} \,\middle|\, \beta_i = \beta, X_i = X\right) \\
&= \mathrm{P}\left(\beta \log W_j(X) + a_j(X) + \epsilon_{ij} > \beta \log W_k(X) + a_k(X) + \epsilon_{ik}, \forall k \neq j\right) \\
&= \mathrm{P}\left(\epsilon_{ik} < \beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \epsilon_{ij}, \forall k \neq j, \forall k \neq j\right) \\
&= \int_{-\infty}^{\infty} \mathrm{P}\left(\epsilon_{ik} < \beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \epsilon_{ij}, \forall k \neq j \,\middle|\, \epsilon_{ij} = \tilde{\epsilon}\right) f_{\epsilon_{ij}}(\tilde{\epsilon}) d\tilde{\epsilon} \\
&= \int_{-\infty}^{\infty} \prod_{k\neq j} \mathrm{P}\left(\epsilon_{ik} < \beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \epsilon_{ij} \,\middle|\, \epsilon_{ij} = \tilde{\epsilon}\right) f_{\epsilon_{ij}}(\tilde{\epsilon}) d\tilde{\epsilon} \\
&= \int_{-\infty}^{\infty} \prod_{k\neq j} F_\epsilon\left(\beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \tilde{\epsilon}\right) f_\epsilon(\tilde{\epsilon}) d\tilde{\epsilon}.
\end{aligned}$$

Here, the second-to-last equality holds because $\{\epsilon_{ij}\}_{i,j}$ are independent, and the final equality holds because $\{\epsilon_{ij}\}_{i,j}$ are identically distributed. The mass of workers with skills X at firm j is:

$$S_j(X) = \int \left(\int_{-\infty}^{\infty} \prod_{k\neq j} F_\epsilon\left(\beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \tilde{\epsilon}\right) f_\epsilon(\tilde{\epsilon}) d\tilde{\epsilon}\right) f_{\beta,X}(\beta, X) d\beta.$$

Under a logit error structure, where $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$, the choice probability becomes:

$$\mathrm{P}(j(i) = j|\beta, X) = \frac{\exp\big(\beta \log W_j(X) + a_j(X)\big)}{\sum_{k=1}^{J} \exp\big(\beta \log W_k(X) + a_k(X)\big)}.$$

Defining $I(\beta, X) = \sum_{k=1}^{J} \exp\big(\beta \log W_k(X) + a_k(X)\big)$ to be the worker's wage index, I obtain:

$$S_j(X) = \int \frac{1}{I(\beta, X)} \exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta,X}(\beta, X) d\beta.$$

Using this parameterization for F_ε, I will now derive the firm-specific labor supply elasticity $\frac{\partial \log S_j(X)}{\partial \log W_j(X)}$, as well as the higher-order derivatives of $\log S_j(X)$. Consider the following property.

Property A.1. Assume $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$. From the perspective of a strategically small firm, the first five derivatives of (log) labor supply curve with respect to the (log) wage are:

$$\begin{aligned}
\frac{\partial \log S_j(X)}{\partial \log W_j(X)} &= \mathrm{E}\big(\beta | X, j(i) = j\big) \\
\frac{\partial^2 \log S_j(X)}{\partial \left[\log W_j(X)\right]^2} &= \mathrm{Var}\big(\beta | X, j(i) = j\big) \\
\frac{\partial^3 \log S_j(X)}{\partial \left[\log W_j(X)\right]^3} &= \mathrm{E}\big([\beta - \mathrm{E}(\beta | X, j(i) = j)]^3 \big| X, j(i) = j\big) \\
\frac{\partial^4 \log S_j(X)}{\partial \left[\log W_j(X)\right]^4} &= \mathrm{E}\big([\beta - \mathrm{E}(\beta | X, j(i) = j)]^4 \big| X, j(i) = j\big) - 3\,\mathrm{Var}\big(\beta | X, j(i) = j\big)^2 \\
\frac{\partial^5 \log S_j(X)}{\partial \left[\log W_j(X)\right]^5} &= \mathrm{E}\big([\beta - \mathrm{E}(\beta | X, j(i) = j)]^5 \big| X, j(i) = j\big) \\
&\quad - 10\,\mathrm{Var}\big(\beta | X, j(i) = j\big)\,\mathrm{E}\big([\beta - \mathrm{E}(\beta | X, j(i) = j)]^3 \big| X, j(i) = j\big).
\end{aligned}$$

Proof. If the firm views itself as strategically small, then it does not internalize the impact of its own wage on each worker's wage index. Specifically, it sets $\partial I(\beta, X)/\partial W_j(X) = 0$ for all β and X. Under this assumption, the following relationships hold for any function $g : \mathbb{R} \to \mathbb{R}$:

$$\begin{aligned}
\frac{\partial\, \mathrm{E}(g(\beta)\, \mathrm{P}(j(i) = j | \beta, X) \mid X)}{\partial \log W_j(X)} &= \int \frac{\partial}{\partial \log W_j(X)} \left[\frac{g(\beta) \exp\big(\beta \log W_j(X) + a_j(X)\big)}{I(\beta, X)}\right] f_{\beta|X}(\beta|X) d\beta \\
&= \int \frac{\beta g(\beta) \exp\big(\beta \log W_j(X) + a_j(X)\big)}{I(\beta, X)} f_{\beta|X}(\beta|X) d\beta \\
&= \mathrm{E}\left(\beta g(\beta)\, \mathrm{P}(j(i) = j | \beta, X) \mid X\right). \qquad \text{(A.1)}
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{E}(g(\beta)\, \mathrm{P}(j(i) = j | \beta, X) \mid X)}{\mathrm{P}(j(i) = j | X)} &= \frac{\sum_{k=1}^{J} \mathrm{E}(g(\beta)\, \mathrm{P}(j(i) = j | \beta, X) \mid X, j(i) = k) \times \mathrm{P}(j(i) = k | X)}{\mathrm{P}(j(i) = j | X)} \\
&= \frac{\mathrm{E}(g(\beta) \mid X, j(i) = j) \times \mathrm{P}(j(i) = j | X) + \sum_{k \neq j} 0 \times \mathrm{P}(j(i) = k | X)}{\mathrm{P}(j(i) = j | X)} \\
&= \mathrm{E}(g(\beta) \mid X, j(i) = j)\,. \qquad \text{(A.2)}
\end{aligned}$$

To ease notation, let $s_j(X) = \log S_j(X)$ and $w_j(X) = \log W_j(X)$. Also, define $\tau_{jX,s}$ such that:

$$\tau_{jX,s}(w) = \mathrm{E}\big(\beta^s \exp\big(\beta w + a_j(X)\big)/I(\beta, X)|X\big).$$

Equation (A.1) implies $\partial\tau_{jX,s}(w_j(X))/\partial w_j(X) = \tau_{jX,s+1}(w_j(X))$, while equation (A.2) implies $\tau_{jX,s}(w_j(X))/\tau_{jX,0}(w_j(X)) = \mathrm{E}(\beta^s|X, j(i) = j)$. Using these properties, I derive the following:

<u>*First Derivative*</u>. The first derivative of the labor supply curve, $\partial s_j(X)/\partial w_j(X)$, is:

$$\frac{\partial s_j(X)}{\partial w_j(X)} = \frac{\partial \log\big(\tau_{jX,0}(w_j(X))\big)}{\partial w_j(X)} = \frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))} = \mathrm{E}\big(\beta|X, j(i) = j\big).$$

<u>*Second Derivative*</u>. The second derivative of the labor supply curve, $\partial^2 s_j(X)/\partial w_j^2(X)$, is:

$$\begin{aligned}
\frac{\partial^2 s_j(X)}{\partial w_j^2(X)} &= \frac{\partial\big(\tau_{jX,1}(w_j(X))/\tau_{jX,0}(w_j(X))\big)}{\partial w_j(X)} \\
&= \frac{\tau_{jX,2}(w_j(X))\tau_{jX,0}(w_j(X)) - \tau_{jX,1}^2(w_j(X))}{\tau_{jX,0}^2(w_j(X))} = \frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))} - \left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2 \\
&= \mathrm{E}\big(\beta^2|X, j(i) = j\big) - \mathrm{E}\big(\beta|X, j(i) = j\big)^2 = \mathrm{Var}\big(\beta|X, j(i) = j\big).
\end{aligned}$$

<u>*Third Derivative*</u>. The third derivative of the labor supply curve, $\partial^3 s_j(X)/\partial w_j^3(X)$, is:

$$\begin{aligned}
\frac{\partial^3 s_j(X)}{\partial w_j^3(X)} &= \frac{\partial}{\partial w_j(X)}\left(\frac{\tau_{jX,2}(w_j(X))\tau_{jX,0}(w_j(X)) - \tau_{jX,1}^2(w_j(X))}{\tau_{jX,0}^2(w_j(X))}\right) \\
&= \frac{\tau_{jX,3}(w_j(X))\tau_{jX,0}^3(w_j(X)) - 3\tau_{jX,2}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^2(w_j(X)) + 2\tau_{jX,1}^3(w_j(X))\tau_{jX,0}(w_j(X))}{\tau_{jX,0}^4(w_j(X))} \\
&= \frac{\tau_{jX,3}(w_j(X))}{\tau_{jX,0}(w_j(X))} - 3\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) + 2\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^3 \\
&= \mathrm{E}\big(\beta^3|X, j(i) = j\big) - 3\,\mathrm{E}\big(\beta|X, j(i) = j\big)\,\mathrm{E}\big(\beta^2|X, j(i) = j\big) + 2\,\mathrm{E}\big(\beta|X, j(i) = j\big)^3 \\
&= \mathrm{E}\big([\beta - \mathrm{E}(\beta|X, j(i) = j)]^3\big|X, j(i) = j\big).
\end{aligned}$$

Fourth Derivative. The fourth derivative of the labor supply curve, $\partial^4 s_j(X)/\partial w_j^4(X)$, is:

$$
\begin{aligned}
\frac{\partial^4 s_j(X)}{\partial w_j^4(X)} &= \frac{\partial}{\partial w_j(X)}\left(\frac{\tau_{jX,3}(w_j(X))\tau_{jX,0}^3(w_j(X)) - 3\tau_{jX,2}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^2(w_j(X)) + 2\tau_{jX,1}^3(w_j(X))\tau_{jX,0}(w_j(X))}{\tau_{jX,0}^4(w_j(X))}\right) \\
&= \frac{\tau_{jX,4}(w_j(X))\tau_{jX,0}^7(w_j(X)) - 4\tau_{jX,3}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^6(w_j(X)) - 3\tau_{jX,2}^2(w_j(X))\tau_{jX,0}^6(w_j(X))}{\tau_{jX,0}^8(w_j(X))} \\
&\quad + \frac{12\tau_{jX,2}(w_j(X))\tau_{jX,1}^2(w_j(X))\tau_{jX,0}^5(w_j(X)) - 6\tau_{jX,1}^4(w_j(X))\tau_{jX,0}^4(w_j(X))}{\tau_{jX,0}^8(w_j(X))} \\
&= \frac{\tau_{jX,4}(w_j(X))}{\tau_{jX,0}(w_j(X))} - 4\left(\frac{\tau_{jX,3}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) - 3\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2 \\
&\quad + 12\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2 - 6\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^4 \\
&= \mathrm{E}\big(\beta^4|X, j(i) = j\big) - 4\,\mathrm{E}\big(\beta|X, j(i) = j\big)\,\mathrm{E}\big(\beta^3|X, j(i) = j\big) - 3\,\mathrm{E}\big(\beta^2|X, j(i) = j\big)^2 \\
&\quad + 12\,\mathrm{E}\big(\beta|X, j(i) = j\big)^2\,\mathrm{E}\big(\beta^2|X, j(i) = j\big) - 6\,\mathrm{E}\big(\beta|X, j(i) = j\big)^4 \\
&= \mathrm{E}\big([\beta - \mathrm{E}(\beta|X, j(i) = j)]^4\big|X, j(i) = j\big) - 3\,\mathrm{Var}\big(\beta|X, j(i) = j\big)^2.
\end{aligned}
$$

Fifth Derivative. The fifth derivative of the labor supply curve, $\partial^5 s_j(X)/\partial w_j^5(X)$, is:

$$
\begin{aligned}
\frac{\partial^5 s_j(X)}{\partial w_j^5(X)} &= \frac{\partial}{\partial w_j(X)}\left(\frac{\tau_{jX,4}(w_j(X))\tau_{jX,0}^7(w_j(X)) - 4\tau_{jX,3}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^6(w_j(X)) - 3\tau_{jX,2}^2(w_j(X))\tau_{jX,0}^6(w_j(X))}{\tau_{jX,0}^8(w_j(X))}\right. \\
&\qquad \left. + \frac{12\tau_{jX,2}(w_j(X))\tau_{jX,1}^2(w_j(X))\tau_{jX,0}^5(w_j(X)) - 6\tau_{jX,1}^4(w_j(X))\tau_{jX,0}^4(w_j(X))}{\tau_{jX,0}^8(w_j(X))}\right) \\
&= \frac{\tau_{jX,5}(w_j(X))\tau_{jX,0}^{15}(w_j(X))}{\tau_{jX,0}^{16}(w_j(X))} - \frac{60\tau_{jX,2}(w_j(X))\tau_{jX,1}^3(w_j(X))\tau_{jX,0}^{12}(w_j(X)) - 24\tau_{jX,1}^5(w_j(X))\tau_{jX,0}^{11}(w_j(X))}{\tau_{jX,0}^{16}(w_j(X))} \\
&\quad - \frac{5\tau_{jX,4}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^{14}(w_j(X)) - 20\tau_{jX,3}(w_j(X))\tau_{jX,1}^2(w_j(X))\tau_{jX,0}^{13}(w_j(X))}{\tau_{jX,0}^{16}(w_j(X))} \\
&\quad - \frac{10\tau_{jX,3}(w_j(X))\tau_{jX,2}(w_j(X))\tau_{jX,0}^{14}(w_j(X)) - 30\tau_{jX,2}^2(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^{13}(w_j(X))}{\tau_{jX,0}^{16}(w_j(X))} \\
&= \frac{\tau_{jX,5}(w_j(X))}{\tau_{jX,0}(w_j(X))} - 5\left(\frac{\tau_{jX,4}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) + 20\left(\frac{\tau_{jX,3}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2 + 24\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^5 \\
&\quad - 10\left(\frac{\tau_{jX,3}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) + 30\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) - 60\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^3 \\
&= \mathrm{E}\big(\beta^5|X, j(i) = j\big) - 5\,\mathrm{E}\big(\beta|X, j(i) = j\big)\,\mathrm{E}\big(\beta^4|X, j(i) = j\big) + 20\,\mathrm{E}\big(\beta|X, j(i) = j\big)^2\,\mathrm{E}\big(\beta^3|X, j(i) = j\big) \\
&\quad - 10\,\mathrm{E}\big(\beta^2|X, j(i) = j\big)\,\mathrm{E}\big(\beta^3|X, j(i) = j\big) + 30\,\mathrm{E}\big(\beta|X, j(i) = j\big)\,\mathrm{E}\big(\beta^2|X, j(i) = j\big)^2 \\
&\quad - 60\,\mathrm{E}\big(\beta|X, j(i) = j\big)^3\,\mathrm{E}\big(\beta^2|X, j(i) = j\big) + 24\,\mathrm{E}\big(\beta|X, j(i) = j\big)^5 \\
&= \mathrm{E}\big([\beta - \mathrm{E}(\beta|X, j(i) = j)]^5\big|X, j(i) = j\big) - 10\,\mathrm{Var}\big(\beta|X, j(i) = j\big)\,\mathrm{E}\big([\beta - \mathrm{E}(\beta|X, j(i) = j)]^3\big|X, j(i) = j\big).
\end{aligned}
$$

□

A.2. Equilibrium Wage Equation

In equilibrium, each firm chooses a set of wages $\{W_j(\chi,\varphi)\}_{\chi,\varphi}$ to maximize profit Π_j, where:

$$\Pi_j = T_j\left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(\int \varphi D_j(\chi,\varphi)d\varphi\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}} - \sum_{\chi\in\mathcal{X}}\left(\int W_j(\chi,\varphi)D_j(\chi,\varphi)d\varphi\right).$$

Let $L_j(\chi,\varphi)$ denote the equilibrium employment of skill group $X=(\chi,\varphi)$ at firm j, defined as the amount of labor hired when $D_j(\chi,\varphi)=S_j(\chi,\varphi)$. The firm's first order condition is:

$$\begin{aligned}\frac{\partial\Pi_j}{\partial W_j(\chi,\varphi)} =& \varphi T_j(1-\alpha_j)\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j-1}\left(\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-1}\frac{\partial L_j(\chi,\varphi)}{\partial W_j(\varphi,\chi)}\\ &- L_j(\chi,\varphi) - W_j(\chi,\varphi)\frac{\partial L_j(\chi,\varphi)}{\partial W_j(\varphi,\chi)} = 0, \quad \text{for all } (\chi,\varphi)\in\mathcal{X}\times\mathbb{R},\end{aligned}$$

where $L_j^{\text{eff}}(\chi)=\int\varphi L_j(\chi,\varphi)d\varphi$ denotes the efficiency units of labor for a given skill type χ. Defining $\boldsymbol{\varepsilon}_j(X)$ as the labor supply elasticity $\partial\log L_j(X)/\partial\log W_j(X)$, this condition becomes:

$$W_j(\chi,\varphi) = \frac{\boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)}\times\varphi T_j(1-\alpha_j)\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j-1}\left(\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-1}.$$

The wage markdown at firm j for workers with skills X is: $\frac{W_j(X)}{\partial Y_j/\partial L_j(X)} = \boldsymbol{\varepsilon}_j(X)/(1+\boldsymbol{\varepsilon}_j(X))$.

Lemmas 1 and 2 establish the existence of an equilibrium corresponding to a unique set of profit-maximizing wages $[W_j(X)]_X$ for each firm j. In this equilibrium, the firm's problem has an interior solution, which satisfies the first-order condition. Taking logs, the condition is:

$$\begin{aligned}w_j(\chi,\varphi) =& \log\varphi + \log T_j + \log(1-\alpha_j) + \log\theta_{j\chi} - \log\left(1+\boldsymbol{\varepsilon}_j^{-1}(\chi,\varphi)\right)\\ &- (1-\rho_j)\log L_j^{\text{eff}}(\chi) + \frac{1-\alpha_j-\rho_j}{\rho_j}\log\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}.\end{aligned}$$

A.3. Firm Production

After plugging in the labor supply constraint, a firm j's value added from hiring workers is:

$$Y_j = T_j\left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(\int\varphi L_j(\chi,\varphi)d\varphi\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}}.$$

For workers with skills $X = (\chi, \varphi)$, the marginal product of labor at firm j is $\frac{\partial Y_j}{\partial L_j(\chi,\varphi)}$, where:

$$\frac{\partial Y_j}{\partial L_j(\chi,\varphi)} = \varphi T_j(1-\alpha_j)\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j-1}\left(\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-1}.$$

To better understand the relationship between skill types χ and skill levels φ in the production function, it will be useful to define the standardized unit of labor at the firm to be N_j, where:

$$N_j = \left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(\int \varphi L_j(\chi,\varphi)d\varphi\right)^{\rho_j}\right)^{1/\rho_j}.$$

For any skill profile $X = (\chi, \varphi)$, the elasticity of N_j with respect to $L_j(\chi, \varphi)$ is defined by:

$$\frac{\partial \log N_j}{\partial \log L_j(\chi,\varphi)} = \underbrace{\frac{\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}}{\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}}}_{\partial\log N_j/\partial\log L_j^{\text{eff}}(\chi)} \times \underbrace{\frac{\varphi L_j(\chi,\varphi)}{\int \varphi' L_j(\chi,\varphi')d\varphi'}}_{\partial\log L_j^{\text{eff}}(\chi)/\partial\log L_j(\chi,\varphi)}.$$

Observe that these elasticities sum up to one: $\sum_{\chi\in\mathcal{X}}\int \frac{\partial \log N_j}{\partial \log L_j(\chi,\varphi)}d\varphi = 1$. They can therefore be interpreted as the share of effective labor contributed to firm j by workers with skills (χ, φ).

Property A.2. Suppose that $\alpha_j \in (0, 1)$ and $\rho_j < 1$. Then a firm's value added Y_j is concave in labor inputs. Additionally, Y_j exhibits decreasing marginal returns within each skill type:

$$\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} < 0, \quad \text{for all } \chi \in \mathcal{X} \text{ and } \varphi, \varphi' \in \mathbb{R}_{++},$$

and complementarity (or substitutability) between skill types depending on the value of $1 - \rho_j$:

$$\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi',\varphi')}\begin{cases} > 0 & \text{if } \alpha_j < 1-\rho_j \\ < 0 & \text{if } \alpha_j > 1-\rho_j, \end{cases} \quad \text{for all } \chi \neq \chi' \text{ and } \varphi, \varphi' \in \mathbb{R}_{++}.$$

Proof. To show that Y_j is concave, first note that it is continuous, strictly increasing, and homogeneous of degree $1 - \alpha_j \in (0, 1]$. Therefore, by Jehle & Reny (2011), Theorem 3.1, it is sufficient to show that Y_j is strictly quasiconcave. Note that Y_j can be written as a composition of two functions, $Y_j = h \circ g(\{L_j^{\text{eff}}(\chi)\}_\chi)$, where $g : \mathbb{R}_{++}^{|\mathcal{X}|} \to \mathbb{R}$ and $h : \mathbb{R} \to \mathbb{R}$ are defined by:

$$g(\{L_j^{\text{eff}}(\chi)\}_\chi) = \sum_{\chi\in\mathcal{X}}\theta_{j\chi}\big(L_j^{\text{eff}}(\chi)\big)^{\rho_j} \quad \text{and} \quad h(x) = T_j x^{(1-\alpha_j)/\rho_j}.$$

The function g has a diagonal Hessian matrix with entries: $\frac{\partial^2 g}{\partial[L_j^{\text{eff}}(\chi)]^2} = -\rho_j(1-\rho_j)\theta_{j\chi}\big(L_j^{\text{eff}}(\chi)\big)^{\rho_j-2}$.

I now analyze three cases. First, if $\rho_j \in (0,1)$, then h is strictly increasing and g is strictly concave, because its Hessian is negative definite. It follows that Y_j is strictly quasiconcave in $\{L_j^{\text{eff}}(\chi)\}_\chi$, as it is a strictly increasing transformation of a strictly concave function. Second, if $\rho_j < 0$, then h is strictly decreasing and g is strictly convex, because its Hessian is positive definite. Therefore, Y_j is still strictly quasiconcave in $\{L_j^{\text{eff}}(\chi)\}_\chi$, as it is a strictly decreasing transformation of a strictly convex function. Third, if ρ_j approaches zero, then Y_j reduces to a Cobb-Douglas production function that is strictly concave in $\{L_j^{\text{eff}}(\chi)\}_\chi$. In all three cases, Y_j is shown to be strictly quasiconcave in $\{L_j^{\text{eff}}(\chi)\}_\chi$. By Jehle & Reny (2011), Theorem 3.1, this implies that Y_j is concave in $\{L_j^{\text{eff}}(\chi)\}_\chi$. Furthermore, since $L_j^{\text{eff}}(\chi)$ is a linear function of $\{L_j(\chi,\varphi)\}_\varphi$, it follows that a firm's value added Y_j is a concave function of all labor inputs.

To show that $\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} < 0$, fix some $\chi \in \mathcal{X}$. For any skill levels φ and φ', I write:

$$\begin{aligned}
\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} &= \varphi T_j(1-\alpha_j)\theta_{j\chi}\Bigg[\varphi'(\rho_j-1)\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j-2}\left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-1} \\
&\quad + \varphi'(1-\alpha_j-\rho_j)\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{2\rho_j-2}\left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-2}\Bigg] \\
&= \varphi'\left(L_j^{\text{eff}}(\chi)\right)^{-1}\frac{\partial Y_j}{\partial L_j(\chi,\varphi)}\left[(\rho_j-1)+(1-\alpha_j-\rho_j)\frac{\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}}{\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}}\right].
\end{aligned}$$

With positive employment, $\varphi'[L_j^{\text{eff}}(\chi)]^{-1}\frac{\partial Y_j}{\partial L_j(\chi,\varphi)} > 0$. So, $\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} < 0$ if and only if:

$$\begin{aligned}
0 &> (\rho_j-1)+(1-\alpha_j-\rho_j)\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)} \\
&= (\rho_j-1)\left[1-\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)}\right]-\alpha_j\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)} > (\rho_j-1)-\alpha_j\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)},
\end{aligned}$$

where the final inequality follows because $\partial \log N_j/\partial \log L_j^{\text{eff}}(\chi)$ is bounded between 0 and 1, and $\rho_j < 1$ by assumption. Re-arranging terms, this inequality may be re-written as follows:

$$\rho_j < 1+\alpha_j\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)}.$$

Since this holds trivially when $\alpha_j \in (0,1)$ and $\rho_j < 1$, I conclude that $\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} < 0$.[35]

[35] Note that values of ρ_j above one can also satisfy this inequality, particularly if the returns to scale are small.

Finally, fix $\chi, \chi' \in \mathcal{X}$, where $\chi \neq \chi'$, and $\varphi, \varphi' \in \mathbb{R}$. The derivative $\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi',\varphi')}$ is:

$$\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi',\varphi')} = (1-\alpha_j-\rho_j)\underbrace{\varphi\varphi' T_j(1-\alpha_j)\theta_{j\chi}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi)L_j^{\text{eff}}(\chi')\right)^{\rho_j-1}\left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-2}}_{(*)}.$$

Observe that the term $(*)$ is strictly positive. Thus, the sign of the derivative is pinned down by $1-\alpha_j-\rho_j$. It is positive whenever $\rho_j < 1-\alpha_j$, and it is negative whenever $\rho_j > 1-\alpha_j$. □

A.4. Firm Profit

After plugging in the labor supply constraint, the profit function Π_j for a firm j is defined by:

$$\Pi_j = Y_j - \int W_j(X)L_j(X)dX.$$

In equilibrium, wages are determined by setting the derivatives $\partial\Pi_j/\partial W_j(X)$ to zero, where:

$$\frac{\partial \Pi_j}{\partial W_j(X)} = \left(\frac{\partial Y_j}{\partial L_j(X)} - W_j(X)\right)\frac{\partial L_j(X)}{\partial W_j(X)} - L_j(X).$$

To understand when profit is a concave function of wages, I evaluate the Hessian matrix of Π_j.

Property A.3. The Hessian of the profit function Π_j is given by $\mathrm{H}_{\Pi_j} = \mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A}+\mathbf{B}$, where $\mathrm{H}_{Y_j} = [\frac{\partial^2 Y_j}{\partial L_j(X)\partial L_j(X')}]_{X,X'}$ is the Hessian of Y_j, and $\mathbf{A}$ and $\mathbf{B}$ are diagonal matrices with entries:

$$\mathbf{A}[X,X] = \frac{\partial L_j(X)}{\partial W_j(X)} \quad \text{and} \quad \mathbf{B}[X,X] = \frac{L_j(X)}{W_j(X)}\left(\frac{\partial^2 \log L_j(X)}{\partial\left[\log W_j(X)\right]^2}\boldsymbol{\varepsilon}_j^{-1}(X) - (\boldsymbol{\varepsilon}_j(X)+1)\right).$$

Proof. The diagonal entries of the matrix H_{Π_j}, denoted by $\mathrm{H}_{\Pi_j}[X,X] = \frac{\partial^2\Pi_j}{\partial W_j^2(X)}$, are given by:

$$\begin{aligned}\frac{\partial^2\Pi_j}{\partial W_j^2(X)} &= \frac{\partial^2 Y_j}{\partial L_j^2(X)}\times\left(\frac{\partial L_j(X)}{\partial W_j(X)}\right)^2 + \frac{\partial Y_j}{\partial L_j(X)}\times\frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - \frac{\partial L_j(X)}{\partial W_j(X)} - \frac{\partial L_j(X)}{\partial W_j(X)} - W_j(X)\frac{\partial^2 L_j(X)}{\partial W_j^2(X)}\\ &= \frac{\partial^2 Y_j}{\partial L_j^2(X)}\times\left(\frac{\partial L_j(X)}{\partial W_j(X)}\right)^2 + \left(\frac{\partial Y_j}{\partial L_j(X)} - W_j(X)\right)\frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - 2\frac{\partial L_j(X)}{\partial W_j(X)}.\end{aligned}$$

If the first-order condition binds, then $\frac{\partial Y_j}{\partial L_j(X)} - W_j(X) = W_j(X)/\boldsymbol{\varepsilon}_j(X)$, which implies that:

$$\begin{aligned}
\frac{\partial^2 \Pi_j}{\partial W_j^2(X)} &= \frac{\partial^2 Y_j}{\partial L_j^2(X)} \times \left(\frac{\partial L_j(X)}{\partial W_j(X)}\right)^2 + \frac{W_j(X)}{\boldsymbol{\varepsilon}_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - 2\frac{\partial L_j(X)}{\partial W_j(X)} \\
&= \frac{\partial^2 Y_j}{\partial L_j^2(X)} \times \left(\frac{\partial L_j(X)}{\partial W_j(X)}\right)^2 + \frac{L_j(X)}{W_j(X)}\left(\frac{W_j^2(X)}{\boldsymbol{\varepsilon}_j(X) L_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - 2\,\boldsymbol{\varepsilon}_j(X)\right) \\
&= \frac{\partial^2 Y_j}{\partial L_j^2(X)} \times \left(\frac{\partial L_j(X)}{\partial W_j(X)}\right)^2 + \frac{L_j(X)}{W_j(X)}\left(\frac{\partial^2 \log L_j(X)}{\partial \left[\log W_j(X)\right]^2}\,\boldsymbol{\varepsilon}_j^{-1}(X) - (\boldsymbol{\varepsilon}_j(X) + 1)\right),
\end{aligned}$$

where the final equality relies on the observation that $\partial^2 \log L_j(X)/\partial[\log W_j(X)]^2$ equals:

$$\begin{aligned}
\frac{\partial^2 \log L_j(X)}{\partial \left[\log W_j(X)\right]^2} &= \frac{W_j^2(X)}{L_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - \left(\frac{\partial L_j(X)}{\partial W_j(X)}\right)^2 \times \frac{W_j^2(X)}{L_j^2(X)} + \frac{\partial L_j(X)}{\partial W_j(X)} \times \frac{W_j(X)}{L_j(X)} \\
&= \frac{W_j^2(X)}{L_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - \boldsymbol{\varepsilon}_j^2(X) + \boldsymbol{\varepsilon}_j(X).
\end{aligned}$$

The off-diagonal entries of H_{Π_j}, denoted by $\mathrm{H}_{\Pi_j}[X, X'] = \frac{\partial^2 \Pi_j}{\partial W_j(X) \partial W_j(X')}$ for $X \neq X'$, equal:

$$\frac{\partial^2 \Pi_j}{\partial W_j(X) \partial W_j(X')} = \frac{\partial^2 Y_j}{\partial L_j(X) \partial L_j(X')} \times \frac{\partial L_j(X)}{\partial W_j(X)} \times \frac{\partial L_j(X')}{\partial W_j(X')}.$$

Taken together, I can write $\mathrm{H}_{\Pi_j} = \mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A} + \mathbf{B}$, where H_{Y_j}, $\mathbf{A}$, and $\mathbf{B}$ are all defined above. □

I now highlight two notable special cases where Π_j is a concave function of wages $\{W_j(X)\}_X$.

Example (Homogeneous β). Suppose β is constant within each skill group: $\mathrm{Var}(\beta_i | X_i = X) = 0$. Then by Property A.1, $\partial^2 \log L_j(X)/\partial[\log W_j(X)]^2 = \mathrm{Var}(\beta_i | X_i = X, j(i) = j) = 0$. In this case, $\mathbf{B}$ is a negative definite matrix with entries $\mathbf{B}[X, X] = -[L_j(X)/W_j(X)] \times [\boldsymbol{\varepsilon}_j(X) + 1]$. Since H_{Y_j} is negative semidefinite by Property A.2, H_{Π_j} is also negative semidefinite, because:

$$\begin{aligned}
v'\mathrm{H}_{\Pi_j} v &= v'\big(\mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A} + \mathbf{B}\big)v \\
&= v'\mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A}v + v'\mathbf{B}v \\
&= (\mathbf{A}v)'(\mathrm{H}_{Y_j})(\mathbf{A}v) + v'\mathbf{B}v \leq 0,
\end{aligned}$$

for any vector v. Therefore, the firm's profit function Π_j must be concave in wages $\{W_j(X)\}_X$.

Example (Firm is a Price-taker). Suppose that the firm does not exercise any wage-setting power, setting $W_j(X) = \frac{\partial Y_j}{\partial L_j(X)}$ for all skill groups X. In this case, $\mathbf{B}$ will reduce to a negative definite matrix with entries $\mathbf{B}[X, X] = -2[\partial L_j(X)/\partial W_j(X)]$. Just as in the previous case, this implies that H_{Π_j} is negative semidefinite. So, the firm's profit function must be concave.

A.5. Worker Rents

Worker rents R_i^w are defined so that $u_{ij}\big(W_j(X_i) - R_i^w, a_j(X_i)\big) = \max_{k\neq j(i)} u_{ik}\big(W_k(X_i), a_k(X_i)\big)$. As a share of a worker's earnings, these rents can be expressed as a strictly increasing function of the utility difference between a worker i's chosen firm $j(i)$ and the next-best alternative:

$$\frac{R_i^w}{W_{j(i)}(X_i)} = 1 - \exp\left(-\frac{u_{ij(i)}\big(W_{j(i)}(X_i), a_{j(i)}(X_i)\big) - \max_{k\neq j(i)} u_{ik}\big(W_k(X_i), a_k(X_i)\big)}{\beta}\right).$$

I now derive the distribution of these rent shares among workers with skills $X_i = X$ at firm j. I show that, if β is homogeneous, then these shares are independent of a worker's skills and firm.

Property A.4. For workers with skills X at firm j, the density of $\omega_i = R_i^w / W_{j(i)}(X_i)$ equals:

$$f_{\omega|X,j}(\omega|X_i = X, j(i) = j) = \mathrm{E}\Big(\beta_i(1-\omega)^{\beta_i - 1} \;\Big|\; X_i = X, j(i) = j\Big).$$

Whenever β is homogeneous, ω_i will be independent of both a worker's skills X_i and firm $j(i)$.

Proof. Let $W_j(X)$ be the wage that a firm j pays workers with skills X. For any $W \leq W_j(X)$, the density of these workers who would be willing to accept their current firm at wage W is:

$$\frac{\partial L_j(X, W)}{\partial W} \times \frac{1}{L_j(X, W_j(X))}.$$

By the change-of-variables formula, the density of ω_i given $X_i = X$ and $j(i) = j$ is equal to:

$$\begin{aligned}
f_{\omega|X,j}(\omega|X_i = X, j(i) = j) &= -\frac{\partial L_j(X, (1-\omega)W_j(X))}{\partial \omega} \times \frac{1}{L_j(X, W_j(X))} \\
&= -\frac{\partial}{\partial \omega}\left(\int \frac{\exp\big(\beta \log\big((1-\omega)W_j(X)\big) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X)}{I(\beta, X)} d\beta\right) \times \frac{1}{L_j(X, W_j(X))} \\
&= \left(\int \beta(1-\omega)^{\beta-1} \frac{\exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X)}{I(\beta, X)} d\beta\right) \times \frac{1}{L_j(X, W_j(X))} \\
&= \frac{\mathrm{E}\Big(\beta(1-\omega)^{\beta-1} \times \exp\big(\beta \log W_j(X) + a_j(X)\big)/I(\beta, X) \;\Big|\; X\Big)}{\mathrm{E}\Big(\exp\big(\beta \log W_j(X) + a_j(X)\big)/I(\beta, X) \;\Big|\; X\Big)} \\
&= \mathrm{E}\Big(\beta_i(1-\omega)^{\beta_i - 1} \;\Big|\; X_i = X, j(i) = j\Big).
\end{aligned}$$

If β is homogeneous, then this density simplifies to $f_{\omega|X,j}(\omega|X_i = X, j(i) = j) = \beta(1-\omega)^{\beta-1}$. Since this function does not depend on a worker's skills $X_i = X$ or firm $j(i) = j$, I can write:

$$f_{\omega|X,j}(\omega|X_i = X, j(i) = j) = f_\omega(\omega), \quad \text{for all } X \text{ and } j.$$

□

When β is heterogeneous, the share of wages that workers earn as rents systematically varies

across firms and skill groups. To characterize this variation, I compute the average rent share, $\mathrm{E}(R_i^w/W_{j(i)}(X_i)|X_i = X, j(i) = j)$, for workers with skills X at firm j. I also show that (fixing skills) workers who sort into higher paying firms receive a smaller share of their wage as rents.

Property A.5. Among workers with skills X employed at firm j, the expected rent share is:

$$\mathrm{E}\left(\frac{R_i^w}{W_{j(i)}(X_i)} \;\middle|\; X_i = X, j(i) = j\right) = \mathrm{E}\left(\frac{1}{1+\beta_i} \;\middle|\; X_i = X, j(i) = j\right).$$

If $\mathrm{Var}(\beta_i|X_i = X) > 0$, then $\mathrm{E}(R_i^w/W_{j(i)}(X_i)|X_i = X, j(i) = j)$ is strictly decreasing in $W_j(X)$.

Proof. Average worker rents $\mathrm{E}(R_i^w|X, j(i) = j)$ are derived by integrating $W_j(X) - W$ with respect to the density of workers who would be willing to accept their current job at wage W.

$$\begin{aligned}
\mathrm{E}(R_i^w|X, j(i) = j) &= \int_0^{W_j(X)} \big(W_j(X) - W\big)\left(\frac{\partial L_j(X,W)}{\partial W} \times \frac{1}{L_j(X, W_j(X))}\right) dW \\
&= -\frac{W_j(X)}{L_j(X, W_j(X))} \times \int_0^1 \omega \left(\frac{\partial}{\partial \omega} \int \frac{\exp\big(\beta \log\big((1-\omega)W_j(X)\big) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X)}{I(\beta, X)} d\beta\right) d\omega \\
&= \frac{W_j(X)}{L_j(X, W_j(X))} \times \int_0^1 \omega \left(\int \frac{\beta(1-\omega)^{\beta-1} \exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X)}{I(\beta, X)} d\beta\right) d\omega.
\end{aligned}$$

The final equality relies on the assumption that firms view themselves to be strategically small in the market. Changing the order of integration, the average worker rents can be re-written as:

$$\begin{aligned}
\mathrm{E}(R_i^w|X, j(i) = j) &= \frac{W_j(X)}{L_j(X, W_j(X))} \times \int \frac{1}{I(\beta, X)} \exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X) \left(\int_0^1 \omega\beta(1-\omega)^{\beta-1} d\omega\right) d\beta \\
&= \frac{W_j(X)}{L_j(X, W_j(X))} \times \int \frac{1}{I(\beta, X)} \left(\frac{1}{1+\beta}\right) \exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X) d\beta \\
&= W_j(X) \times \frac{\mathrm{E}\Big(\frac{1}{1+\beta} \times \exp\big(\beta \log W_j(X) + a_j(X)\big)/I(\beta, X) \mid X\Big)}{\mathrm{E}\big(\exp\big(\beta \log W_j(X) + a_j(X)\big)/I(\beta, X) \mid X\big)} \\
&= W_j(X) \times \mathrm{E}\left(\frac{1}{1+\beta_i} \;\middle|\; X_i = X, j(i) = j\right).
\end{aligned}$$

To see how $\mathrm{E}(R_i^w/W_{j(i)}(X_i)|X_i = X, j(i) = j)$ depends on $W_j(X)$, I evaluate the elasticity:

$$\frac{\partial \log \mathrm{E}(R_i^w/W_{j(i)}(X_i)|X_i = X, j(i) = j)}{\partial \log W_j(X)} = \frac{\frac{\partial \mathrm{E}\left(\frac{1}{1+\beta_i}\middle|X_i = X, j(i) = j\right)}{\partial \log W_j(X)}}{\mathrm{E}\left(\frac{1}{1+\beta_i} \mid X_i = X, j(i) = j\right)} = \frac{\mathrm{Cov}\left(\beta_i, \frac{1}{1+\beta_i} \mid X_i = X, j(i) = j\right)}{\mathrm{E}\left(\frac{1}{1+\beta_i} \mid X_i = X, j(i) = j\right)}.$$

As long as $\mathrm{Var}(\beta_i|X_i = X) > 0$, the covariance term is strictly negative. Hence, fixing skills, workers at higher paying firms tend to receive a smaller share of their wage in the form of rents.

□

A.6. Employer Rents

Employer rents come in the form of excess profits that firms obtain by exploiting their wage-setting power. To calculate these rents, I consider a counterfactual setting where firms act as price-takers in the market, facing perfectly-elastic labor supply curves. I then define the rents at firm j to be the difference between the realized and counterfactual profits $R_j^e = \Pi_j - \Pi_j^{\text{pt}}$.

Property A.6. Let Y_j^{pt} be a firm j's value added when it is a price taker. The firm's rents are:

$$R_j^e = Y_j \times \left[\sum_{\chi \in \mathcal{X}} \int \left(\frac{1 + \alpha_j \boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \right) \frac{\partial \log N_j}{\partial \log L_j(\chi, \varphi)} d\varphi - \alpha_j \left(\frac{Y_j^{\text{pt}}}{Y_j} \right) \right].$$

If β is homogeneous, then all firms receive the same rents as a share of their profits, provided that they have the same returns to scale, α_j, and that labor is perfectly substitutable, $\rho_j = 1$.

Proof. For any firm j, the profit Π_j that the firm obtains in the monopsonistic equilibrium is:

$$\begin{aligned}
\Pi_j &= Y_j - \sum_{\chi \in \mathcal{X}} \left(\int \left(\frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \right) \left(\frac{\partial Y_j}{\partial L_j(\chi, \varphi)} \right) L_j(\chi, \varphi) d\varphi \right) \\
&= Y_j - \sum_{\chi \in \mathcal{X}} \left(\int \left(\frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \right) \varphi T_j (1 - \alpha_j) \theta_{j\chi} \left(L_j^{\text{eff}}(\chi) \right)^{\rho_j - 1} \left(\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi') \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j} - 1} L_j(\chi, \varphi) d\varphi \right) \\
&= Y_j \times \left(1 - (1 - \alpha_j) \sum_{\chi \in \mathcal{X}} \left(\frac{\theta_{j\chi} \left(L_j^{\text{eff}}(\chi) \right)^{\rho_j}}{\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi') \right)^{\rho_j}} \right) \left(\int \left(\frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \right) \left(\frac{\varphi L_j(\chi, \varphi)}{\int \varphi' L_j(\chi, \varphi') d\varphi'} \right) d\varphi \right) \right). \\
&= Y_j \times \left(1 - (1 - \alpha_j) \sum_{\chi \in \mathcal{X}} \int \left(\frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \right) \frac{\partial \log N_j}{\partial \log L_j(\chi, \varphi)} d\varphi \right) \\
&= Y_j \times \sum_{\chi \in \mathcal{X}} \int \left(\frac{1 + \alpha_j \boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \right) \frac{\partial \log N_j}{\partial \log L_j(\chi, \varphi)} d\varphi.
\end{aligned}$$

If the firm is a price-taker in the market, then its profit Π_j^{pt} is defined in the following way:

$$\Pi_j^{\text{pt}} = \max_{\{D_j^{\text{pt}}(\chi, \varphi)\}_{\chi, \varphi}} T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j^{\text{pt}}(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} - \sum_{\chi \in \mathcal{X}} \left(\int W_j^{\text{pt}}(\chi, \varphi) D_j^{\text{pt}}(\chi, \varphi) d\varphi \right).$$

Taking first-order conditions with respect to labor demand yields the following wage equation:

$$W_j^{\text{pt}}(\chi, \varphi) = \varphi T_j (1 - \alpha_j) \theta_{j\chi} \left(\int \varphi D_j^{\text{pt}}(\chi, \varphi) d\varphi \right)^{\rho_j - 1} \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j^{\text{pt}}(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j} - 1}.$$

In equilibrium, the firm's labor demand equals its labor supply evaluated at the wage $W_j^{\text{pt}}(X)$. Let $L_j^{\text{pt}}(X)$ be the market-clearing quantity of labor, so that $D_j^{\text{pt}}(X) = S_j\big(X, W_j^{\text{pt}}(X)\big)$. I write:

$$\begin{aligned}
\Pi_j^{\text{pt}} &= T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j^{\text{pt}}(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} \times \left(1 - (1-\alpha_j) \sum_{\chi \in \mathcal{X}} \int \frac{\partial \log N_j}{\partial \log L_j(\chi, \varphi)} d\varphi \right) \\
&= \alpha_j T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j^{\text{pt}}(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} \\
&= \alpha_j Y_j^{\text{pt}},
\end{aligned}$$

where the second equality above follows because the elasticities $\frac{\partial \log N_j}{\partial \log L_j(\chi,\varphi)}$ aggregate to one. Given these expressions for the firm's realized and counterfactual profits, Π_j and Π_j^{pt}, I obtain:

$$R_j^e = Y_j \times \left[\sum_{\chi \in \mathcal{X}} \int \left(\frac{1 + \alpha_j \boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \right) \frac{\partial \log N_j}{\partial \log L_j(\chi, \varphi)} d\varphi - \alpha_j \left(\frac{Y_j^{\text{pt}}}{Y_j} \right) \right].$$

If β is homogeneous across workers, then the labor supply elasticity $\boldsymbol{\varepsilon}_j(X)$ is constant across all firms j and skill groups X. In this case, a firm j's rents as a share of its profits are given by:

$$\begin{aligned}
\frac{R_j^e}{\Pi_j} = 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j \boldsymbol{\varepsilon}} \times \frac{Y_j^{\text{pt}}}{Y_j} &= 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j \boldsymbol{\varepsilon}} \times \left(\frac{T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j^{\text{pt}}(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}}}{T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}}} \right) \\
&= 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j \boldsymbol{\varepsilon}} \times \left(\frac{\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi \left(\frac{W_j^{\text{pt}}(\chi,\varphi)}{W_j(\chi,\varphi)} \right)^{\beta} L_j(\chi, \varphi) d\varphi \right)^{\rho_j}}{\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j(\chi, \varphi) d\varphi \right)^{\rho_j}} \right)^{\frac{1-\alpha_j}{\rho_j}}.
\end{aligned}$$

If labor is perfectly substitutable, so that $\rho_j = 1$, then the wage ratio $W_j^{\text{pt}}(\chi, \varphi)/W_j(\chi, \varphi)$ is:

$$\begin{aligned}
\frac{W_j^{\text{pt}}(\chi, \varphi)}{W_j(\chi, \varphi)} &= \frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \left(\frac{\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \int \varphi L_j^{\text{pt}}(\chi, \varphi) d\varphi}{\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi, \varphi) d\varphi} \right)^{-\alpha_j} \\
&= \frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \left(\frac{\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \int \varphi \left(\frac{W_j^{\text{pt}}(\chi,\varphi)}{W_j(\chi,\varphi)} \right)^{\beta} L_j(\chi, \varphi) d\varphi}{\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi, \varphi) d\varphi} \right)^{-\alpha_j} \\
&= \frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \left(\frac{W_j^{\text{pt}}(\chi, \varphi)}{W_j(\chi, \varphi)} \right)^{\beta} \left(\frac{\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi, \varphi) d\varphi}{\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi, \varphi) d\varphi} \right)^{-\alpha_j} = \frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \left(\frac{W_j^{\text{pt}}(\chi, \varphi)}{W_j(\chi, \varphi)} \right)^{\beta} = \left(\frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \right)^{\frac{1}{1-\beta}},
\end{aligned}$$

where the third equality holds since $W_j^{\text{pt}}(\chi, \varphi)/W_j(\chi, \varphi)$ is constant within each firm j. Subs-

tituting this expression, I can write R_j^e/Π_j as a deterministic function of the returns to scale α_j.

$$\frac{R_j^e}{\Pi_j} = 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j\,\boldsymbol{\varepsilon}} \times \left(\frac{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi \left(\frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}}\right)^{\frac{1}{1-\beta}} L_j(\beta,\chi) d\varphi}{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi,\varphi) d\varphi} \right)^{1-\alpha_j}$$

$$= 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j\,\boldsymbol{\varepsilon}} \times \left(\frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \right)^{\frac{1-\alpha_j}{1-\beta}}.$$

□

A.7. Pass-through of TFP Shocks to Wages

In the next property, I derive an expression for the pass-through of a hypothetical shock to firm productivity (specifically, the TFP parameter T_j) on the wages $\{W_j(X)\}_X$ of different workers.

Property A.7. The vector of elasticities of firm j's wages $\{W_j(X)\}_X$ with respect its TFP T_j is $\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X = (I - \boldsymbol{\Gamma}_j - \mathbf{K}_j)^{-1}\mathbf{1}$, where $\boldsymbol{\Gamma}_j$ and $\mathbf{K}_j$ are both matrices in $\mathbb{R}^{|X|\times|X|}$, with entries:

$$\boldsymbol{\Gamma}_j[X,X'] = \mathbb{1}\{X = X'\} \frac{\frac{\partial^2 \log L_j(X)}{\partial[\log W_j(X)]^2}}{\boldsymbol{\varepsilon}_j(X)[1+\boldsymbol{\varepsilon}_j(X)]}$$

$$\mathbf{K}_j[X,X'] = \left((1-\alpha_j-\rho_j)\frac{\partial \log N_j}{\log L_j(\chi',\varphi')} - \mathbb{1}\{\chi=\chi'\}(1-\rho_j)\frac{\partial \log L_j^{\text{eff}}(\chi')}{\partial \log L_j(\chi',\varphi')} \right) \boldsymbol{\varepsilon}_j(\chi',\varphi').$$

If β is homogeneous, then $\frac{\partial \log W_j(X)}{\partial \log T_j}$ is constant across all firms j and skill groups X, provided that firms have the same returns to scale, α_j, and that labor is perfectly substitutable, $\rho_j = 1$.

Proof. For any skill group X, the elasticity of $W_j(X)$ with respect to firm TFP is given by:

$$\begin{aligned}
\frac{\partial \log W_j(\chi,\varphi)}{\partial \log T_j} &= 1 + \frac{\partial\left(\frac{\boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)}\right)}{\partial \log T_j} - (1-\rho_j)\frac{\partial \log L_j^{\text{eff}}(\chi)}{\partial \log T_j} + (1-\alpha_j-\rho_j)\frac{\partial \log N_j}{\partial \log T_j} \\
&= 1 + \frac{\partial\left(\frac{\boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)}\right)}{\partial \log W_j(\chi,\varphi)} \frac{\partial \log W_j(\chi,\varphi)}{\partial \log T_j} - (1-\rho_j)\int \frac{\partial \log L_j^{\text{eff}}(\chi)}{\partial \log L_j(\chi,\varphi')} \boldsymbol{\varepsilon}_j(\chi,\varphi') \frac{\partial \log W_j(\chi,\varphi')}{\partial \log T_j} d\varphi' \\
&\quad + (1-\alpha_j-\rho_j)\sum_{\chi'\in\mathcal{X}} \left(\int \frac{\partial \log N_j}{\partial \log L_j(\chi',\varphi')} \boldsymbol{\varepsilon}_j(\chi',\varphi') \frac{\partial \log W_j(\chi',\varphi')}{\partial \log T_j} d\varphi' \right) \\
&= 1 + \left(\frac{1}{\boldsymbol{\varepsilon}_j(\chi,\varphi)[1+\boldsymbol{\varepsilon}_j(\chi,\varphi)]} \right) \frac{\partial^2 \log L_j(\chi,\varphi)}{\partial[\log W_j(\chi,\varphi)]^2} \frac{\partial \log W_j(\chi,\varphi)}{\partial \log T_j} \\
&\quad + \sum_{\chi'\in\mathcal{X}} \left[(1-\alpha_j-\rho_j)\frac{\partial \log N_j}{\log L_j^{\text{eff}}(\chi')} - (1-\rho_j)\mathbb{1}\{\chi=\chi'\} \right] \int \frac{\partial \log L_j^{\text{eff}}(\chi')}{\partial \log L_j(\chi',\varphi')} \boldsymbol{\varepsilon}_j(\chi',\varphi') \frac{\partial \log W_j(\chi',\varphi')}{\partial \log T_j} d\varphi'.
\end{aligned}$$

So, in each firm j, the elasticities solve a linear system, which can be written in matrix form as:

$$\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X = \mathbf{1} + \mathbf{\Gamma}_j \left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X + \mathbf{K}_j \left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X,$$

with $\mathbf{\Gamma}_j$ and $\mathbf{K}_j$ defined above. If $(I - \mathbf{\Gamma}_j - \mathbf{K}_j)^{-1}$ exists, then $\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X = (I - \mathbf{\Gamma}_j - \mathbf{K}_j)^{-1}\mathbf{1}$.

If β is homogeneous across workers, the labor supply elasticity $\boldsymbol{\varepsilon}_j(X)$ is constant across firms j and skill groups X, and the second derivative $\partial^2 \log L_j(X)/\partial[\log W_j(X)]^2$ equals zero. So, if workers are perfectly substitutable in a firm's production function, so that $\rho_j = 1$, then:

$$\begin{aligned}\frac{\partial \log W_j(\chi, \varphi)}{\partial \log T_j} &= 1 - \alpha_j \boldsymbol{\varepsilon} \sum_{\chi' \in \mathcal{X}} \int \frac{\partial \log N_j}{\log L_j(\chi', \varphi')} \frac{\partial \log W_j(\chi', \varphi')}{\partial \log T_j} d\varphi' \\ &= 1 - \alpha_j \boldsymbol{\varepsilon} \frac{\partial \log W_j(\chi, \varphi)}{\partial \log T_j} = \frac{1}{1 - \alpha_j \boldsymbol{\varepsilon}}.\end{aligned}$$

So, $\frac{\partial \log W_j(X)}{\partial \log T_j}$ is constant across firms and skill groups if firms have the same returns to scale. □

A.8. Allocative Inefficiency

Social welfare is defined as the sum of workers' utilities (net of amenities) and firms' profits:

$$\mathcal{W} = \mathrm{E}\left(\max_j \left\{u_{ij}\big(W_j(X_i), a_j(X_i)\big)\right\}\right) + \log \sum_{j=1}^{J} \Pi_j.$$

Using the formula for the expectation of a maximum over T1EV random variables, I write:

$$\begin{aligned}\mathcal{W} &= \mathrm{E}\left(\max_j \left\{\beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij}\right\}\right) + \log \sum_{j=1}^{J} \Pi_j \\ &= \int \mathrm{E}\left[\max_j \left\{\beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij}\right\} \middle| \beta_i = \beta, X_i = X\right] f_{\beta,X}(\beta_i, X) d(\beta_i, X) + \log \sum_{j=1}^{J} \Pi_j \\ &= \int \left(\log \sum_{j=1}^{J} \exp\left[\beta \log W_j(X) + a_j(X)\right] + \gamma\right) f_{\beta,X}(\beta, X) d(\beta, X) + \log \sum_{j=1}^{J} \Pi_j \\ &= \int \log I(\beta, X) f_{\beta,X}(\beta, X) d(\beta, X) + \gamma + \log \sum_{j=1}^{J} \Pi_j,\end{aligned}$$

where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant.[36] The social planner seeks to maximize welfare by solving $\mathcal{W}^* = \max_{\{j(i)\}_i} \mathcal{W}$. The first order condition of the planner's problem is:

$$\frac{\partial \mathcal{W}}{\partial L_j(X)} = 0,$$

for all X and j. The following property characterizes when the optimal allocation is achieved.

Property A.8. The allocation of labor that solves the social planner's problem, denoted by $\{L_j^*(X)\}_{j,X}$, coincides with the equilibrium allocation in a competitive (Walrasian) economy.

Proof. The planner's solution is characterized by setting $\frac{\partial \mathcal{W}}{\partial L_j(X)}$ to zero for all X and j, where:

$$\begin{aligned}
\frac{\partial \mathcal{W}}{\partial L_j(X)} &= \frac{\partial \left[\int \log I(\beta, X) f_{\beta|X}(\beta|X) f_X(X) d\beta \right]}{\partial L_j(X)} + \frac{\partial \log \sum_{j=1}^J \Pi_j}{\partial L_j(X)} \\
&= \int \frac{\partial \log I(\beta, X))}{\partial L_j(X)} f_{\beta|X}(\beta|X) f_X(X) d\beta + \frac{\partial \Pi_j}{\partial L_j(X)} \left(\sum_{j=1}^J \Pi_j \right)^{-1} \\
&= \int \frac{\beta \exp\left(\beta \log W_j(X) + a_j(X)\right)}{I(\beta, X)} \left(\frac{\partial \log W_j(X)}{\partial L_j(X)} \right) f_{\beta|X}(\beta|X) f_X(X) d\beta + \frac{\partial \Pi_j}{\partial L_j(X)} \left(\sum_{j=1}^J \Pi_j \right)^{-1}. \\
&= \left(\frac{\partial \log W_j(X)}{\partial \log L_j(X)} \right) \mathrm{E}(\beta|X, j(i) = j) + \frac{\partial \Pi_j}{\partial L_j(X)} \left(\sum_{j=1}^J \Pi_j \right)^{-1} \\
&= \left(\frac{\partial \log W_j(X)}{\partial \log L_j(X)} \right) \mathrm{E}(\beta|X, j(i) = j) + \frac{\partial \left[Y_j - \int W_j(X') L_j(X') dX' \right]}{\partial L_j(X)} \left(\sum_{j=1}^J \Pi_j \right)^{-1} \\
&= \left(\frac{\partial \log W_j(X)}{\partial \log L_j(X)} \right) \mathrm{E}(\beta|X, j(i) = j) + \frac{\frac{\partial Y_j}{\partial L_j(X)} - W_j(X)}{\sum_{j=1}^J \Pi_j} - \frac{\int \frac{\partial \log W_j(X')}{\partial \log L_j(X)} W_j(X') dX'}{\sum_{j=1}^J \Pi_j}.
\end{aligned}$$

If all firms are price-takers, then they take wages as given, so: $\partial \log W_j(X') / \partial \log L_j(X) = 0$ for all $X, X' \in \mathcal{X} \times \mathbb{R}_+$. In this case, the solution to the social planner's problem reduces to:

[36] This property is proven in Small & Rosen (1981). Even without T1EV errors, expected maximal utility is:

$$\begin{aligned}
\mathrm{E}\left(\max_j \left\{ u_{ij}\left(W_j(X_i), a_j(X_i)\right) \right\} \right) &= \sum_{j=1}^J \left(\int \mathrm{E}\left[\beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij} \middle| j(i) = j, X_i = X \right] L_j(X) dX \right) \\
&= \sum_{j=1}^J \left(\int \left(\boldsymbol{\varepsilon}_j(X) \log W_j(X) + a_j(X) \right) L_j(X) dX \right) + \mathrm{E}\left[\epsilon_{ij} \middle| j(i) = j \right].
\end{aligned}$$

$$\frac{\partial \mathcal{W}}{\partial L_j(X)} = 0 \quad \Longleftrightarrow \quad W_j(X) = \frac{\partial Y_j}{\partial L_j(X)},$$

for all X and j, which coincides with the equilibrium wage condition in a Walrasian economy. □

Property A.5 implies that the first-best allocation is implemented in a competitive (Walrasian) economy. To compute the optimal welfare $\mathcal{W}^*$, I therefore consider a counterfactual setting where all firms are price takers in the labor market. In this setting, social welfare is given by:

$$\mathcal{W}^* = \int \log I^*(\beta, X) f_{\beta,X}(\beta, X) d(\beta, X) + \gamma + \log \sum_{j=1}^{J} \alpha_j Y_j^*.$$

I measure allocative inefficiency by comparing $\mathcal{W}^*$ to welfare in a monopsonistic equilibrium:

$$\mathcal{W} = \int \log I(\beta, X) f_{\beta,X}(\beta, X) d(\beta, X) + \gamma + \log \sum_{j=1}^{J} Y_j \left(\int \left(\frac{1 + \alpha_j \boldsymbol{\varepsilon}_j(X)}{1 + \boldsymbol{\varepsilon}_j(X)} \right) \omega_j(X) dX \right).$$

To implement the first-best optimal allocation, a planner can give wage-specific wage subsidies to workers, where the shape of the subsidy curve depends on the distribution of preferences.

B. Model Extensions

B.1. Micro-foundation for the Worker's Indirect Utility Function

I now present a simple micro-foundation for the indirect utility function of the worker, where each worker i chooses a firm j to maximize utility from consuming goods and leisure. Let C_{ij} denote the worker's expected consumption from working at the firm and let H_{ij} denote the worker's expected time spent working. A worker i's utility from choosing a firm j equals:

$$u_{ij} = f_i(C_{ij}, H_{ij}).$$

Let $\bar{H}_j(X_i)$ denote the scheduled work hours for employees with skills X_i at firm j, accounting for paid time off, overtime, vacation leave, and other benefits. Let $\tilde{H}_{ij}$ denote the idiosyncratic component of hours worked, accounting for commuting time and worker-firm match factors. Assume that $f_i : \mathbb{R}_+^2 \to \mathbb{R}$ is log-additive in consumption and scheduled hours, such that:

$$f_i(C_{ij}, H_{ij}) = a_i + \kappa_i \log C_{ij} - \eta \log \bar{H}_j(X_i) - \tilde{f}(\tilde{H}_{ij}).$$

The marginal rate of substitution between log consumption and log work hours equals $-\kappa_i/\eta$. Define the worker's budget constraint as $C_{ij} = W_j(X_i)$, where $W_j(X_i)$ denotes total earnings. Additionally, define $a_j(X) = -\log \bar{H}_j(X_i)$, $\beta_i = \kappa_i/\eta$, and $\epsilon_{ij} = -\eta^{-1} \tilde{f}(\tilde{H}_{ij})$. It follows that:

$$u_{ij} = \frac{a_i}{\eta} + \beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij},$$

which corresponds to utility specification (1) in the paper.[37] Note that this utility function can be extended to include non-labor income. For example, let V_i denote the worker's other forms of income. The budget constraint becomes $C_{ij} = W_j(X_i) + V_i$. If V_i is observed in data, then I can re-define earnings to be $\tilde{W}_{ij}(X_i) = W_j(X_i) + V_i$, and the same utility specification applies.

B.2. Capital and Monopolistic Competition in the Product Market

I now give an extension of the model that includes capital and monopolistic competition in the product market. Consider a monopolistic firm j with the following production function:

$$Q_j = T_j K_j^{\eta_j} N_j^{1-\alpha_j},$$

where K_j denotes capital. In a monopolistic product market, the revenue curve is $Y_j = Q_j^{1-\kappa_j}$. For each skill vector X, labor is hired according to the labor supply curve $L_j(X)$ and capital is rented at some fixed price r_j. The firm's profit function has the following representation:

$$\begin{aligned}\Pi_j &= Q_j^{1-\kappa_j} - \sum_{\chi\in\mathcal{X}} \left(\int W_j(\chi,\varphi) L_j(\chi,\varphi) d\varphi \right) - r_j K_j \\ &= \tilde{T}_j K_j^{\tilde{\eta}_j} N_j^{1-\tilde{\alpha}_j} - \sum_{\chi\in\mathcal{X}} \left(\int W_j(\chi,\varphi) L_j(\chi,\varphi) d\varphi \right) - r_j K_j,\end{aligned}$$

where $\tilde{T}_j = T_j^{1-\kappa_j}$, $\tilde{\eta}_j = \eta_j(1-\kappa_j)$, and $\tilde{\alpha}_j = \alpha_j + \kappa_j(1-\alpha_j)$. I now show that both perfect and monopolistic competition in the product market yield the same profit function. As a first step, I derive the first order condition of the firm's problem with respect to capital K_j. I write:

$$K_j = \left(\frac{r_j}{\tilde{\eta}_j \tilde{T}_j N_j^{1-\tilde{\alpha}_j}} \right)^{\frac{1}{\tilde{\eta}_j - 1}}.$$

Plugging this condition into the firm's profit function, I obtain the following:

$$\begin{aligned}\Pi_j &= \tilde{T}_j \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j N_j^{1-\tilde{\alpha}_j}} \right)^{\frac{\tilde{\eta}_j}{\tilde{\eta}_j - 1}} N_j^{1-\tilde{\alpha}_j} - \sum_{\chi\in\mathcal{X}} \left(\int W_j(\chi,\varphi) L_j(\chi,\varphi) d\varphi \right) - r \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j N_j^{1-\tilde{\alpha}_j}} \right)^{\frac{1}{\tilde{\eta}_j - 1}} \\ &= \left[\tilde{T}_j \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j} \right)^{\frac{\tilde{\eta}_j}{\tilde{\eta}_j - 1}} - r \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j} \right)^{\frac{1}{\tilde{\eta}_j - 1}} \right] N_j^{-\frac{1-\tilde{\alpha}_j}{\tilde{\eta}_j - 1}} - \sum_{\chi\in\mathcal{X}} \left(\int W_j(\chi,\varphi) L_j(\chi,\varphi) d\varphi \right).\end{aligned}$$

Note that this is just a reinterpretation of the original problem, where the firm's profit equals:

$$\Pi_j = \hat{T}_j N_j^{1-\hat{\alpha}_j} - \sum_{\chi\in\mathcal{X}} \left(\int W_j(\chi,\varphi) L_j(\chi,\varphi) d\varphi \right), \quad \text{where:} \quad \begin{cases} \hat{T}_j &= \tilde{T}_j \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j} \right)^{\frac{\tilde{\eta}_j}{\tilde{\eta}_j - 1}} - r \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j} \right)^{\frac{1}{\tilde{\eta}_j - 1}} \\ \hat{\alpha}_j &= \frac{\tilde{\eta}_j - \tilde{\alpha}_j}{\tilde{\eta}_j - 1}. \end{cases}.$$

[37] The term $\frac{a_i}{\eta}$ does not vary across firms. Therefore, it does not impact the worker's employment decisions.

B.2. Properties of the Labor Supply Curve

To understand how $S_j(X)$ depends on the wage $W_j(X)$, I first establish the following property.

Property A.9. The own-wage elasticity of labor supply, $\frac{\partial \log S_j(X)}{\partial \log W_j(X)}$, takes the following form:

$$\frac{\partial \log S_j(X)}{\partial \log W_j(X)} = \mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i) = j|\beta, X)}{\partial \log W_j(X)} \middle| X, j(i) = j \right).$$

Proof. The derivative of a firm's labor supply $\log S_j(X)$ with respect to $\log W_j(X)$ is given by:

$$\begin{aligned}
\frac{\partial \log S_j(X)}{\partial \log W_j(X)} &= \frac{W_j(X)}{S_j(X)} \times \frac{\partial}{\partial W_j(X)} \int \mathrm{P}(j(i) = j|\beta, X) f_{\beta,X}(\beta, X) d\beta \\
&= S_j^{-1}(X) \int \frac{\partial \log \mathrm{P}(j(i) = j|\beta, X)}{\partial \log W_j(X)} \mathrm{P}(j(i) = j|\beta, X) f_{\beta,X}(\beta, X) d\beta \\
&= \frac{\int \frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)} \mathrm{P}(j(i) = j|\beta, X) f_{\beta,X}(\beta|X) d\beta}{\mathrm{P}(j(i) = j|X)} \\
&= \frac{\mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)} \mathrm{P}(j(i) = j|\beta, X)|X \right)}{\mathrm{P}(j(i) = j|X)}.
\end{aligned}$$

By the Law of Iterated Expectations, this labor supply elasticity can be re-written as follows:

$$\begin{aligned}
\frac{\partial \log S_j(X)}{\partial \log W_j(X)} &= \frac{\sum_{k=1}^{J} \mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)} \mathrm{P}(j(i) = j|\beta, X)|X, j(i) = k \right) \times \mathrm{P}(j(i) = k|X)}{\mathrm{P}(j(i) = j|X)} \\
&= \frac{\mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)} |X, j(i) = j \right) \times \mathrm{P}(j(i) = j|X) + \sum_{k \neq j} 0 \times \mathrm{P}(j(i) = k|X)}{\mathrm{P}(j(i) = j|X)} \\
&= \mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i) = j|\beta, X)}{\partial \log W_j(X)} |X, j(i) = j \right).
\end{aligned}$$

□

This result shows that the elasticity of labor supply at firm j can be expressed as the average β-specific elasticity among incumbent workers at that firm. This representation is quite general, and it does not rely on assumptions about firm conduct, such as monopsonistic, Bertrand, or Cournot competition. To see how $\frac{\partial \log S_j(X)}{\partial \log W_j(X)}$ depends on $W_j(X)$, consider the next property.

Property A.10. The second derivative of $\log S_j(X)$ with respect to $\log W_j(X)$ takes the form:

$$\frac{\partial^2 \log S_j(X)}{\partial \left[\log W_j(X)\right]^2} = \mathrm{E}\left(\frac{\partial^2 \log \mathrm{P}(j(i) = j|\beta, X)}{\partial \left[\log W_j(X)\right]^2} \middle| X, j(i) = j \right) + \mathrm{Var}\left(\frac{\partial \log \mathrm{P}(j(i) = j|\beta, X)}{\partial \log W_j(X)} \middle| X, j(i) = j \right).$$

Proof. To simplify notation, let $\boldsymbol{\varepsilon}_j(X) = \frac{\partial \log S_j(X)}{\partial \log W_j(X)}$ and $\boldsymbol{\varepsilon}_j(\beta, X) = \frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)}$. I write:

$$
\begin{aligned}
\frac{\partial^2 \log S_j(X)}{\partial \left[\log W_j(X)\right]^2} &= \frac{\partial}{\partial \log W_j(X)} \left[\frac{\int \boldsymbol{\varepsilon}_j(\beta, X)\, \mathrm{P}(j(i) = j|\beta, X) f_{\beta,X}(\beta|X) d\beta}{\mathrm{P}(j(i) = j|X)} \right] \\
&= \frac{\int \frac{\partial \left[\boldsymbol{\varepsilon}_j(\beta,X)\, \mathrm{P}(j(i)=j|\beta,X)\right]}{\partial \log W_j(X)} f_{\beta|x}(\beta|X) d\beta - \boldsymbol{\varepsilon}_j(X) \int \boldsymbol{\varepsilon}_j(\beta, X)\, \mathrm{P}(j(i) = j|\beta, X) f_{\beta|x}(\beta|X) d\beta}{\mathrm{P}(j(i) = j|X)} \\
&= \frac{\int \frac{\partial \left[\boldsymbol{\varepsilon}_j(\beta,X)\, \mathrm{P}(j(i)=j|\beta,X)\right]}{\partial \log W_j(X)} f_{\beta|x}(\beta|X) d\beta}{\mathrm{P}(j(i) = j|X)} - \boldsymbol{\varepsilon}_j^2(X) \\
&= \frac{\int \frac{\partial \boldsymbol{\varepsilon}_j(\beta,X)}{\partial \log W_j(X)} \mathrm{P}(j(i) = j|\beta, X) f_{\beta|x}(\beta|X) d\beta}{\mathrm{P}(j(i) = j|X)} + \frac{\int \boldsymbol{\varepsilon}_j^2(\beta, X)\, \mathrm{P}(j(i) = j|\beta, X) f_{\beta|x}(\beta|X) d\beta}{\mathrm{P}(j(i) = j|X)} - \boldsymbol{\varepsilon}_j^2(X) \\
&= \mathrm{E}\left(\frac{\partial\, \boldsymbol{\varepsilon}_j(\beta, X)}{\partial \log W_j(X)} \middle| X, j(i) = j \right) + \mathrm{E}\left(\boldsymbol{\varepsilon}_j^2(\beta, X) \middle| X, j(i) = j \right) - \mathrm{E}\left(\boldsymbol{\varepsilon}_j(\beta, X) \middle| X, j(i) = j \right)^2 \\
&= \mathrm{E}\left(\frac{\partial\, \boldsymbol{\varepsilon}_j(\beta, X)}{\partial \log W_j(X)} \middle| X, j(i) = j \right) + \mathrm{Var}\left(\boldsymbol{\varepsilon}_j(\beta, X) \middle| X, j(i) = j \right).
\end{aligned}
$$

□

C. Proofs of Main Results

C.1. Equilibrium Properties of the Model

This appendix section establishes equilibrium properties relating to existence and uniqueness, which are necessary for the identification arguments. I begin by proving an equilibrium exists.

Lemma 1. There exists an equilibrium involving strictly positive wages and employment.

Proof. The taste shocks ϵ_{ij} have strictly positive density, so $\mathrm{P}(j(i) = j|\beta, X)$ is strictly positive for all β, X and j. Therefore, an equilibrium (if it exists) involves strictly positive employment.

I restrict attention to equilibria with non-negative wages, so that $W_j(X) \geq 0$ for all X and j. For a firm to be profitable, its wages cannot exceed the revenue that it earns per unit of labor. Since $\partial^2 Y_j / \partial L_j^2(X) < 0$ for all X and j, this means that there exists a strict upper bound on the wage vectors $\{W_j(\mathbf{X})\}_j$. Thus, the set of feasible wage vectors $\{W_j(\mathbf{X})\}_j$ is contained in a convex, compact subset of Euclidean space. Every equilibrium must lie within the interior of this subspace, since $\lim_{W_j(X)\to 0} \partial \Pi_j / \partial W_j(X) > 0$ and $\lim_{W_j(X)\to\infty} \partial \Pi_j / \partial W_j(X) < 0$ for all X and j, implying that firms can always raise their profit by deviating from a corner solution. Given these properties, I can restrict attention to wages that satisfy the first-order conditions:

$$
W_j(X) = \frac{\boldsymbol{\varepsilon}_j(X)}{1 + \boldsymbol{\varepsilon}_j(X)} \times \frac{\partial Y_j}{\partial L_j(X)}, \quad \text{for all } X \text{ and } j.
$$

The right-hand-side is a continuously differentiable function of $W_j(X)$ that is bounded within the set of feasible wages, since $0 < [\boldsymbol{\varepsilon}_j(X)/(1 + \boldsymbol{\varepsilon}_j(X))] \times \partial Y_j / \partial L_j(X) < \partial Y_j / \partial L_j(X)$, where $\partial Y_j / \partial L_j(X)$ is finite for all $L_j(X) > 0$. By Brouwer's fixed point theorem, there is a solution

to this system of equations, which corresponds to an equilibrium with strictly positive wages. □

Identification of firm-specific labor supply elasticities, as laid out in Section 4.A, requires that an infinitesimal, isolated productivity shock to one firm does not generate discrete jumps in that firm's wages and employment. One way to guarantee this is to show that, conditional on the wages at rival firms, $k \neq j$, each firm j's profit maximization problem admits a unique, locally stable solution.[38] In the following lemma, I show that this property holds almost surely.

Lemma 2. Conditional on the wages of rival firms, $\{W_k(\mathbf{X})\}_{k\neq j}$, firm j's profit-maximization problem will have a unique optimal wage schedule $W_j(\mathbf{X})$ for almost every value of (α_j, ρ_j).

Proof. Let $W_j(\mathbf{X})$ be firm j's wage vector. The first-order condition requires $\partial\Pi_j/\partial W_j(X) = 0$ for every X. If $W_j(\mathbf{X})$ satisfies this condition, then the Hessian $[\partial^2\Pi_j/\partial W_j(X)\partial W_j(X')]_{X,X'}$ is equal to $\mathrm{H}_{\Pi_j} = \mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A} + \mathbf{B}$, where H_{Y_j}, $\mathbf{A}$, and $\mathbf{B}$ are derived in Properties A.2 and A.3. This Hessian matrix satisfies two properties. First, $\mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A}$ and $\mathbf{B}$ are both nonsingular, as $\mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A}$ is symmetric, negative definite and $\mathbf{B}$ is diagonal. Second, given rival firms' wages $\{W_k(\mathbf{X})\}_{k\neq j}$, amenities $\{a_k(\mathbf{X})\}_k$, and worker distributions $(F_X, \{F_{\beta|X}\}_X, F_\epsilon)$, the matrices $\mathbf{A}$ and $\mathbf{B}$ are deterministic functions of $W_j(\mathbf{X})$. So, for any wage vector $W_j(\mathbf{X})$, H_{Π_j} depends on the firm's technology parameters only through H_{Y_j}. It follows that, for almost all (α_j, ρ_j) in $(0, 1) \times (-\infty, 1)$, the matrix H_{Π_j}, evaluated at $W_j(\mathbf{X})$, has a nonzero determinant and is therefore nonsingular. By the inverse function theorem, any solution $W_j(\mathbf{X})$ to the first-order condition is locally unique (and hence isolated) with probability one. Because all fixed points are isolated and the set of fixed points is compact, there can only exist finitely many of them.

Not every critical point $W_j(\mathbf{X})$ of Π_j is necessarily a global maximizer of the firm's profit function. I now show that, for almost all parameter values, there is a unique global maximizer, corresponding to a unique optimal wage schedule for the firm. Suppose that two distinct wage vectors both maximized Π_j. Since these vectors are (almost surely) locally stable fixed points of the first-order condition, the implicit function theorem delivers well-defined marginal effects of $\frac{1-\alpha_j}{\rho_j}$ on Π_j at each one. As these marginal effects differ, any arbitrarily small change in $\frac{1-\alpha_j}{\rho_j}$ yields different profit levels at these two wage vectors, so that they can no longer both be global maximizers. Thus, for almost all (α_j, ρ_j), the firm's problem has a unique solution. □

[38] This property is immediate if profit Π_j is concave in wages $\{W_j(X)\}_X$, as this guarantees a unique solution to the firm's first-order condition. As shown in appendix section A.4, concavity holds under two special cases: the competitive (Walrasian) benchmark and a model with constant wage–amenity tradeoffs, such that $\beta_i = \beta$. However, the property does not apply in general, which means that it cannot be leveraged for the proof. From inspection of the Hessian matrix of the profit function, H_{Π_j}, I can conclude that Π_j is concave in wages whenever $\partial^2 \log L_j(X)/\partial[\log W_j(X)]^2 < \boldsymbol{\varepsilon}_j(X)(\boldsymbol{\varepsilon}_j(X) + 1)$ for all X. In a logit model, $\boldsymbol{\varepsilon}_j(X)$ equals $\mathrm{E}(\beta|X, j(i) = j)$ and $\partial^2 \log L_j(X)/\partial[\log W_j(X)]^2$ equals $\mathrm{Var}(\beta|X, j(i) = j)$. Hence, this condition is more likely to hold if workers have higher marginal utilities of (log) earnings and/or if the dispersion of these marginal utilities is low.

Throughout the paper, I consider two key counterfactuals. First, I analyze the competitive (Walrasian) equilibrium, defined as the equilibrium in which firms are price-takers, acting as if they face perfectly elastic labor supply curves. Second, I analyze a version of the model in which workers' wage-amenity tradeoffs are homogeneous, so $\beta_i = \beta$ for all i. In both cases, it is important to show that the equilibrium is unique; otherwise, multiple counterfactual equilibria could exist, complicating the interpretation of the estimates. In the following lemma, I show that the equilibrium is indeed unique under either of these counterfactual specifications. For simplicity, I only provide a proof for the case where there is a single skill type, i.e., $|\chi| = 1$.

Lemma 3. Let $|\chi| = 1$. There exists a unique equilibrium under each of the two counterfactual specifications of the model: the Walrasian benchmark and the benchmark where β is constant.

Proof. Under each counterfactual specification, the wage markdown is constant: $\frac{\boldsymbol{\varepsilon}_j(X)}{1+\boldsymbol{\varepsilon}_j(X)} = \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}}$.[39] Thus, letting $|\chi| = 1$, I can define the effective wage at firm j as $W_j^{\text{eff}}(\varphi) = W_j(\varphi)/\varphi$, where:

$$\begin{aligned}
W_j^{\text{eff}}(\varphi) &= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)\left(\int \varphi L_j\big(\varphi, \mathbf{W}(\varphi)\big) d\varphi\right)^{-\alpha_j} \\
&= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)\left(\int \varphi \int \frac{\exp\big(\beta \log W_j(\varphi) + a_j(\varphi)\big)}{\sum_{k=1}^J \exp\big(\beta \log W_k(\varphi) + a_k(\varphi)\big)} f_{\beta,\varphi}(\beta,\varphi) d\beta d\varphi\right)^{-\alpha_j} \\
&= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)\left(\int \varphi \int \frac{\exp\big(\beta \log W_j^{\text{eff}}(\varphi) + a_j(\varphi)\big)}{\sum_{k=1}^J \exp\big(\beta \log W_k^{\text{eff}}(\varphi) + a_k(\varphi)\big)} f_{\beta,\varphi}(\beta,\varphi) d\beta d\varphi\right)^{-\alpha_j} \\
&= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)\left(\int \varphi L_j\big(\varphi, \mathbf{W}^{\text{eff}}(\varphi)\big) d\varphi\right)^{-\alpha_j}.
\end{aligned}$$

Because the right-hand-side is constant within each firm, I can write $W_j^{\text{eff}}(\varphi) = W_j^{\text{eff}}$ for all j. Hence, an equilibrium is characterized by a solution to $g(\mathbf{W}^{\text{eff}}) = \mathbf{0}$, where I define g so that:

$$g_j(\mathbf{W}^{\text{eff}}) = \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)[L_j^{\text{eff}}(\mathbf{W}^{\text{eff}})]^{-\alpha_j} - W_j^{\text{eff}}, \quad \text{for } j \in \{1, \ldots, J\}.$$

To prove that this equilibrium is unique, I apply Theorem 3.1 in Kennan (2001), which implies that there is at most one equilibrium if g is strictly radially quasi-concave and quasi-increasing. To show that g is strictly radially quasi-concave, fix any $\mathbf{W}^{\text{eff}} > \mathbf{0}$ where $g(\mathbf{W}^{\text{eff}}) = \mathbf{0}$. I write:

$$\begin{aligned}
g_j(\lambda \mathbf{W}^{\text{eff}}) &= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)[L_j^{\text{eff}}(\lambda \mathbf{W}^{\text{eff}})]^{-\alpha_j} - \lambda W_j^{\text{eff}} \\
&= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)[L_j^{\text{eff}}(\mathbf{W}^{\text{eff}})]^{-\alpha_j} - \lambda W_j^{\text{eff}} \\
&= (1-\lambda) W_j^{\text{eff}} > 0,
\end{aligned}$$

for any $\lambda \in (0, 1)$ and $j \in \{1, \ldots, J\}$. The second equality above holds since the labor supply

[39] For the Walrasian benchmark, this claim holds trivially: firms do not markdown wages, so $\frac{\boldsymbol{\varepsilon}_j(X)}{1+\boldsymbol{\varepsilon}_j(X)} = 1$. For the benchmark with constant β, it follows from Property A.1, which demonstrates $\boldsymbol{\varepsilon}_j(X) = \mathrm{E}(\beta | X, j(i) = j)$.

curves $L_j(\varphi, \mathbf{W})$ are homogeneous of degree zero. To show that g is quasi-increasing, notice that, for every firm j, $g_j(\lambda \mathbf{W}^{\text{eff}})$ is a decreasing function of effective labor $L_j^{\text{eff}}(\mathbf{W}^{\text{eff}})$, which is itself decreasing in the effective wages of j's rival firms, $\{W_k^{\text{eff}}\}_{k \neq j}$. Therefore, if $\tilde{W}_j^{\text{eff}} = W_j^{\text{eff}}$ and $\tilde{W}_k^{\text{eff}} \geq W_k^{\text{eff}}$ for all $k \neq j$, it must be that $g_j(\tilde{\mathbf{W}}^{\text{eff}}) \geq g_j(\mathbf{W}^{\text{eff}})$. The fact that g is radially quasi-concave, together with monotonicity, allows me to apply Kennan (2001), Theorem 3.1. So, I conclude that there is a unique equilibrium wage vector in each counterfactual economy. □

C.2. Identification Results

I prove a version of Proposition 1 for the case where workers' skill levels φ are unobserved. In this case, the moment conditions are defined in terms of effective wage for a given skill type χ. To understand how the effective wages are identified from data, see the proof of Proposition 2.

Proposition 1. Under Assumptions I and II, the following moment conditions are satisfied:

$$\mathrm{E}\left[\ell_{j\tau_1,0}(\chi) - \ell_{j\tau_0,0}(\chi) \middle| Z_j = 1, w_{j\tau_0}^{\text{eff}}(\chi) = w\right] = \mathrm{E}\left[\ell_{j\tau_1,0}(\chi) - \ell_{j\tau_0,0}(\chi) \middle| Z_j = 0, w_{j\tau_0}^{\text{eff}}(\chi) = w\right]$$

$$\mathrm{E}\left[w_{j\tau_1,0}^{\text{eff}}(\chi) - w_{j\tau_0,0}^{\text{eff}}(\chi) \middle| Z_j = 1, w_{j\tau_0}^{\text{eff}}(\chi) = w\right] = \mathrm{E}\left[w_{j\tau_1,0}^{\text{eff}}(\chi) - w_{j\tau_0,0}^{\text{eff}}(\chi) \middle| Z_j = 0, w_{j\tau_0}^{\text{eff}}(\chi) = w\right].$$

Proof. Let $\Delta w_{j0}^{\text{eff}}(\chi) = w_{j\tau_1,0}^{\text{eff}}(\chi) - w_{j\tau_0,0}^{\text{eff}}(\chi)$ and $\Delta \ell_{j0}(\chi) = \ell_{j\tau_1,0}(\chi) - \ell_{j\tau_0,0}(\chi)$ denote the change in untreated potential wage and labor outcomes from τ_0 and τ_1 for skill type χ at firm j. In the model, these untreated potential outcomes evolve due to spillover effects that result from the TFP shocks at treated firms. These spillovers operate through two channels: workers' wage indices $\{I(\beta, X)\}_{\beta,X}$ and the joint distribution of worker types $F_{\beta,X}$. So, for the common trends assumption to hold, I must show that treated and untreated firms would be affected in the same way, on average, by any change to $\{I(\beta, X)\}_{\beta,X}$ and $F_{\beta,X}$. I prove this in two steps.

<u>*Step 1.*</u> *Derive a sufficient statistic for $\Delta w_{j0}^{\text{eff}}(\chi)$ and $\Delta \ell_{j0}(\chi)$.*

First, I introduce some new notation. For $\Gamma_j = \left(\rho_j, \alpha_j, \{\theta_{j\chi}\}_\chi, \{a_{j\chi} - a_{j\chi'}\}_{\chi,\chi'}\right)'$, I define:

$$g_{\chi,\tau}(w|\Gamma_j) = \log(1-\alpha_j) + \log\theta_{j\chi} - (1-\rho_j)\log h_{\chi,\tau}(w|\Gamma_j) + \frac{1-\alpha_j-\rho_j}{\rho_j}\log\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(h_{\chi',\tau}(w|\Gamma_j)\right)^{\rho_j}$$

$$h_{\chi,\tau}(w|\Gamma_j) = \int\int \varphi\left(\frac{\exp\left(\beta w + a_{j\chi} - a_{j\chi^*}\right)}{I_\tau^{\text{eff}}(\beta,\chi)}\right) f_{\beta|\chi,\tau}(\beta|\chi) f_{\varphi|\chi,\tau}(\varphi) f_{\chi|\tau}(\chi) d\beta d\varphi,$$

for $\tau \in \{\tau_0, \tau_1\}$, where χ^* denotes some "reference" skill type. Since each firm j views itself as strategically small, it takes the wage index $I_\tau^{\text{eff}}(\beta, \chi)$ as given. Note that $g_{\chi,\tau}(w|\Gamma_j)$ can be interpreted as the equilibrium wage equation for a price-taking firm (with zero markdowns) where $\log(T_j) = 0$ and $a_{j\chi}$ is normalized relative to the reference type, i.e., $a_{j\chi} = a_{j\chi} - a_{j\chi^*}$. As shown in Appendix C.1, there is a unique fixed point solution to this system. Thus, I can write $g_{\chi,\tau}(w^*|\Gamma_j) = g_{\chi,\tau}^*(\Gamma_j)$. Given this property, the untreated potential wage $w_{j,\tau_1,0}^{\text{eff}}(\chi)$ is:

$$
\begin{aligned}
w^{\text{eff}}_{j\tau_1,0}(\chi) &= \log T_j + \log(1-\alpha_j) + \log\theta_{j\chi} - \log\left(1 + \frac{1}{\boldsymbol{\varepsilon}_{j\tau_1,0}(\chi)}\right) \\
&\quad - (1-\rho_j)\ell^{\text{eff}}_{j\tau_1,0}(\chi) + \frac{1-\alpha_j-\rho_j}{\rho_j}\log\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(\exp\ell^{\text{eff}}_{j\tau_1,0}(\chi)\right)^{\rho_j} \\
&= \log T_j - \alpha_j a_{j\chi^*} - \log\left(1 + \frac{1}{\boldsymbol{\varepsilon}_{j\tau_1,0}(\chi)}\right) + g^*_{\chi,\tau_1}(\Gamma_j),
\end{aligned}
$$

where the second equality follows from the fact that $\ell^{\text{eff}}_{j\tau_1,0}(\chi) = a_{j\chi^*} + \log h_{\chi,\tau}(w^{\text{eff}}_{j\tau_1,0}(\chi)|\Gamma_j)$. Because the wage changes are infinitesimal, I can write $\boldsymbol{\varepsilon}_{j\tau_1,0}(\chi) = \boldsymbol{\varepsilon}_{\tau_1,0}(\chi, w^{\text{eff}}_{j\tau_0,0}(\chi))$, where $\boldsymbol{\varepsilon}_{j\tau_1,0}(\chi, w^{\text{eff}}_{j\tau_0,0}(\chi))$ represents the labor supply elasticity faced by an untreated firm j at time period τ_1, evaluated at the pre-period effective wage $w^{\text{eff}}_{j\tau_0,0}(\chi)$. By this relationship, I obtain:

$$
\Delta w^{\text{eff}}_{j0}(\chi) = \underbrace{\log\left(1 + \frac{1}{\boldsymbol{\varepsilon}_{\tau_1,0}(\chi, w^{\text{eff}}_{j\tau_0,0}(\chi))}\right) - \log\left(1 + \frac{1}{\boldsymbol{\varepsilon}_{\tau_0,0}(\chi, w^{\text{eff}}_{j\tau_0,0}(\chi))}\right) + g^*_{\chi,\tau_1}(\Gamma_j) - g^*_{\chi,\tau_0}(\Gamma_j)}_{q_w(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)}.
$$

In addition, I can write the change in untreated potential labor outcomes, $\Delta\ell_{j0}(\chi)$, as follows:

$$
\Delta\ell_{j0}(\chi) = \underbrace{\log\left(\frac{\int \frac{\exp[\beta q_w(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi) + \beta w^{\text{eff}}_{j\tau_0,0}(\chi) + a_{j\chi} - a_{j\chi^*}]}{I_{\tau_1}(\beta,X)} f_{\beta|\chi,\tau_1}(\beta|\chi) f_{\chi,\varphi|\tau_1}(\chi,\varphi)d\beta}{\int \frac{\exp[\beta' w^{\text{eff}}_{j\tau_0,0}(\chi) + a_{j\chi} - a_{j\chi^*}]}{I_{\tau_0}(\beta',X)} f_{\beta|\chi,\tau_0}(\beta'|\chi) f_{\chi,\varphi|\tau_0}(\chi,\varphi)d\beta'}\right)}_{q_\ell(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)}.
$$

<u>*Step 2.*</u> *Show that Assumption I implies common trends.*

Plugging in the functions $q_w(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)$ and $q_\ell(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)$, I obtain:

$$
\begin{aligned}
\mathrm{E}\left[\Delta w^{\text{eff}}_{j0}(\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] &= \mathrm{E}\left[q_w(w^{\text{eff}}_{j\tau_0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] \\
&= \mathrm{E}\left[q_w(w^{\text{eff}}_{j\tau_0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] \\
&= \mathrm{E}\left[\Delta w^{\text{eff}}_{j0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{E}\left[\Delta\ell_{j0}(\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] &= \mathrm{E}\left[q_\ell(w^{\text{eff}}_{j\tau_0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] \\
&= \mathrm{E}\left[q_\ell(w^{\text{eff}}_{j\tau_0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] \\
&= \mathrm{E}\left[\Delta\ell_{j0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right].
\end{aligned}
$$

In both equations, the second equality holds by Assumption I. So, I can write $\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi)$ as:

$$\begin{aligned}
\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi) &= \frac{\mathrm{E}\left[\ell_{j\tau_1}(\chi)-\ell_{j\tau_0}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[\ell_{j\tau_1}(\chi)-\ell_{j\tau_0}(\chi)\middle|Z_j=0, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]}{\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1}(\chi)-w^{\mathrm{eff}}_{j\tau_0}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1}(\chi)-w^{\mathrm{eff}}_{j\tau_0}(\chi)\middle|Z_j=0, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]} \\
&= \frac{\mathrm{E}\left[\ell_{j\tau_1,1}(\chi)-\ell_{j\tau_0,1}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[\ell_{j\tau_1,0}(\chi)-\ell_{j\tau_0,0}(\chi)\middle|Z_j=0, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]}{\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1,1}(\chi)-w^{\mathrm{eff}}_{j\tau_0,1}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1,0}(\chi)-w^{\mathrm{eff}}_{j\tau_0,0}(\chi)\middle|Z_j=0, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]} \\
&= \frac{\mathrm{E}\left[\ell_{j\tau_1,1}(\chi)-\ell_{j\tau_0,1}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[\ell_{j\tau_1,0}(\chi)-\ell_{j\tau_0,0}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]}{\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1,1}(\chi)-w^{\mathrm{eff}}_{j\tau_0,1}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1,0}(\chi)-w^{\mathrm{eff}}_{j\tau_0,0}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]} \\
&= \frac{\mathrm{E}\left[\ell_{j\tau_1,1}(\chi)-\ell_{j\tau_1,0}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]}{\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1,1}(\chi)-w^{\mathrm{eff}}_{j\tau_1,0}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right]}.
\end{aligned}$$

Since the TFP shocks are infinitesimal, the difference $w^{\mathrm{eff}}_{j\tau_1,1}(\chi)-w^{\mathrm{eff}}_{j\tau_0,1}(\chi)$ is also infinitesimal for any firm j. Thus, I can conclude that $\boldsymbol{\varepsilon}_{\tau_1}(\chi,w)=\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi)$ for any wage w, because:

$$\begin{aligned}
\mathrm{E}\left[\ell_{j\tau_1,1}(\chi)-\ell_{j\tau_1,0}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right] &= \mathrm{E}\left[\boldsymbol{\varepsilon}_{j\tau_1}(\chi)\times\left(w^{\mathrm{eff}}_{j\tau_1,1}(\chi)-w^{\mathrm{eff}}_{j\tau_1,0}(\chi)\right)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right] \\
&= \mathrm{E}\left[\boldsymbol{\varepsilon}_{\tau_1}\left(\chi, w^{\mathrm{eff}}_{j\tau_0,1}(\chi)\right)\times\left(w^{\mathrm{eff}}_{j\tau_1,1}(\chi)-w^{\mathrm{eff}}_{j\tau_1,0}(\chi)\right)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right] \\
&= \boldsymbol{\varepsilon}_{\tau_1}(\chi,w)\times\mathrm{E}\left[w^{\mathrm{eff}}_{j\tau_1,1}(\chi)-w^{\mathrm{eff}}_{j\tau_1,0}(\chi)\middle|Z_j=1, w^{\mathrm{eff}}_{j\tau_0}(\chi)=w\right].
\end{aligned}$$

□

Proof of Proposition 2.

Proof. Let $W^{\mathrm{eff}}_{j\tau}(\chi,\varphi)=W_{j\tau}(\chi,\varphi)/\varphi$ denote the effective wage for workers with skill type χ at firm j in period τ. If Assumption II holds, then the equilibrium wage condition can be written as $W^{\mathrm{eff}}_{j\tau}(\chi,\varphi)=g_{j\chi\tau}(\{W^{\mathrm{eff}}_{k\tau}(\chi,\varphi)\}_k)$ for all (j,χ,τ), where $g_{j\chi\tau}$ does not depend on φ. I write:

$$W^{\mathrm{eff}}_{j\tau}(\chi,\varphi)=\frac{\boldsymbol{\varepsilon}_{j\tau}(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_{j\tau}(\chi,\varphi)}T_{j\tau}(1-\alpha_j)\theta_{j\chi}\left(L^{\mathrm{eff}}_{j\tau}(\chi)\right)^{\rho_j-1}\left(\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L^{\mathrm{eff}}_{j\tau}(\chi')\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-1},$$

where the labor supply elasticity $\boldsymbol{\varepsilon}_{j\tau}(\chi,\varphi)$ equals a (χ,τ)-specific function of $\{W^{\text{eff}}_{k\tau}(\chi,\varphi)\}_k$:

$$\boldsymbol{\varepsilon}_{j\tau}(\chi,\varphi) = \int \beta \times \frac{\frac{\exp\left(\beta \log W_{j\tau}(\chi,\varphi)+a_j(\chi,\varphi)\right)}{\sum_{k=1}^{J}\exp(\beta \log W_{k\tau}(\chi,\varphi)+a_k(\chi,\varphi))} f_{\beta|\chi,\tau}(\beta|\chi) f_{\chi,\varphi|\tau}(\chi,\varphi)}{\int \frac{\exp\left(\beta' \log W_{j\tau}(\chi,\varphi)+a_j(\chi,\varphi)\right)}{\sum_{k=1}^{J}\exp(\beta' \log W_{k\tau}(\chi,\varphi)+a_k(\chi,\varphi))} f_{\beta|\chi,\tau}(\beta'|\chi) f_{\chi,\varphi|\tau}(\chi,\varphi) d\beta'} d\beta$$

$$= \int \beta \times \frac{\frac{\exp\left(\beta \log W^{\text{eff}}_{j\tau}(\chi,\varphi)\right)}{\sum_{k=1}^{J}\exp\left(\beta \log W^{\text{eff}}_{k\tau}(\chi,\varphi)+a_{k\chi}\right)} f_{\beta|\chi,\tau}(\beta|\chi)}{\int \frac{\exp\left(\beta' \log W^{\text{eff}}_{j\tau}(\chi,\varphi)\right)}{\sum_{k=1}^{J}\exp\left(\beta' \log W^{\text{eff}}_{k\tau}(\chi,\varphi)+a_{k\chi}\right)} f_{\beta|\chi,\tau}(\beta'|\chi) d\beta'} d\beta.$$

As shown above, the equilibrium is fully characterized in terms of effective wages and labor, where φ does not enter any of the determining functions. Therefore, the effective wages do not depend on φ, so that $W^{\text{eff}}_{j\tau}(\chi,\varphi) = W^{\text{eff}}_{j\tau}(\chi)$, which delivers the desired log-additive separability. □

Proof of Proposition 3.

Proof. Consider two skill types χ and χ'. For each time period $\tilde{\tau} \in \{\tau,\tau'\}$, it must be that:

$$\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tilde{\tau}}(\chi)}{\text{MPL}^{\text{eff}}_{j\tilde{\tau}}(\chi')}\right) = \log\theta_{j\chi} - \log\theta_{j\chi'} - (1-\rho_j)\left[\log L^{\text{eff}}_{j\tilde{\tau}}(\chi) - \log L^{\text{eff}}_{j\tilde{\tau}}(\chi')\right].$$

Because ρ_j and $\{\theta_{j\chi}\}_\chi$ are constant over time, the elasticity of substitution may be recovered by computing inter-temporal shifts in the relative marginal products and effective labor shares:

$$(1-\rho_j)^{-1} = \frac{\log\left(\frac{L^{\text{eff}}_{j\tau}(\chi)}{L^{\text{eff}}_{j\tau}(\chi')}\right) - \log\left(\frac{L^{\text{eff}}_{j\tau'}(\chi)}{L^{\text{eff}}_{j\tau'}(\chi')}\right)}{\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau}(\chi')}\right) - \log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau'}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau'}(\chi')}\right)}.$$

□

Proof of Proposition 4.

Proof. I normalize the firm-specific efficiencies $\{\theta_{j\chi}\}_\chi$ so that $\theta_{j\chi^*} = 1$ for skill type $\chi^* \in \mathcal{X}$. Under this normalization, and given knowledge of ρ_j, these parameters may be computed as:

$$\theta_{j\chi} = \exp\left[\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau}(\chi^*)}\right) + (1-\rho_j)\log\left(\frac{L^{\text{eff}}_{j\tau}(\chi)}{L^{\text{eff}}_{j\tau}(\chi^*)}\right)\right].$$

A firm's returns to scale and total factor productivity can then be recovered from the effective

marginal products, the effective labor shares, and (log) value added. Specifically, I can write:

$$1-\alpha_j = \exp\left[\log \mathrm{MPL}^{\mathrm{eff}}_{j\tau}(\chi) - y_{j\tau} - \log(\theta_{j\chi}) + (1-\rho_j)\log L^{\mathrm{eff}}_{j\tau}(\chi) + \log \sum_{\chi'\in\mathcal{X}} \theta_{j\chi'}\left(L^{\mathrm{eff}}_{j\tau}(\chi')\right)^{\rho_j}\right]$$

$$T_{j\tau} = \exp\left[y_{j\tau} - \frac{1-\alpha_j}{\rho_j}\log \sum_{\chi\in\mathcal{X}} \theta_{j\chi}\left(L^{\mathrm{eff}}_{j\tau}(\chi)\right)^{\rho_j}\right].$$

□

Proof of Proposition 5.

Proof. For some firm j^*, set $a_{j^*\chi} = 0$. Under this normalization, the amenities $\{a_{j\chi}\}_{j\neq j^*}$ are:

$$a_{j\chi} = \log L_{j\tau}\big(\chi, w^{\mathrm{eff}}_{k\tau}(\chi)\big) - \log L_{j^*\tau}(\chi),$$

where $L_{j^*\tau}(\chi)$ is the labor supplied to firm j^* by skill type χ at time τ, and $L_{j\tau}\big(\chi, w^{\mathrm{eff}}_{k\tau}(\chi)\big)$ is the labor supplied to firm j if it posts the same log effective wage as firm k. If the elasticity $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ is known to the researcher for all wages w, then $\log L_j(\chi, w^{\mathrm{eff}}_{k\chi})$ is recovered from:

$$a_{j\chi} = \log L_{j\tau}(\chi) + \int_{w^{\mathrm{eff}}_{j\tau}(\chi)}^{w^{\mathrm{eff}}_{j^*\tau}(\chi)} \boldsymbol{\varepsilon}_{j\tau}(\chi, w)dw - \log L_{j^*\tau}(\chi).$$

□

Proof of Proposition 6.

Proof. Suppose that the elasticity curve $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ is known to the researcher for skill type χ. Then the firm-specific labor supply curves $\{L_{j\tau}(\chi, w)\}_j$ can be recovered through integration:

$$\log L_{j\tau}(\chi, w) = \log L_{j\tau}(\chi) + \int_{w^{\mathrm{eff}}_{j\tau}(\chi)}^{w} \boldsymbol{\varepsilon}_{j\tau}(\chi, \tilde{w})d\tilde{w}.$$

Each labor supply curve $L_{j\tau}(\chi, w)$ may be expressed as a Laplace transform $\mathcal{L}\{g\}(s)$, where:

$$\mathcal{L}\{g\}(s) = \int_0^\infty g(t)\exp(-st)dt, \quad \text{such that} \quad \begin{aligned} s &= -w \\ t &= \beta \\ g(t) &= \frac{\exp(a_{j\chi})}{\sum_{k=1}^{J}\exp\big(tw^{\mathrm{eff}}_{k\tau}(\chi) + a_{k\chi}\big)} f_{\beta|\chi,\tau}(t|\chi) f_{\chi|\tau}(\chi). \end{aligned}$$

The transform $\mathcal{L}\{g\}(s)$ is a one-to-one mapping of $g(t)$. Specifically, any two functions $g(t)$ can only share the same Laplace transform if they differ on a set of Lebesgue measure zero.

Therefore, when the elasticity $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ is point-identified, so is the function $g(\beta)$, where:

$$g(\beta) = \frac{\exp(a_{j\chi})}{\sum_{k=1}^{J} \exp\left(\beta w_{k\tau}^{\text{eff}}(\chi) + a_{k\chi}\right)} f_{\beta|\chi,\tau}(\beta|\chi) f_{\chi|\tau}(\chi).$$

If the amenities $\{a_{j\chi}\}_j$ are identified up-to-scale, i.e., relative to some reference amenity $a_{j^*\chi}$, then the density $f_{\beta|\chi}(\beta|\chi)$ is point-identified for any β-value from the following relationship:

$$f_{\beta|\chi}(\beta|\chi) = g(\beta) \times \frac{\sum_{k=1}^{J} \exp\left(\beta w_{k}^{\text{eff}}(\chi) + \tilde{a}_{k\chi} - \tilde{a}_{j^*\chi}\right)}{\exp\left(a_{j\chi} - a_{j^*\chi}\right) f_{\chi|\tau}(\chi)}.$$

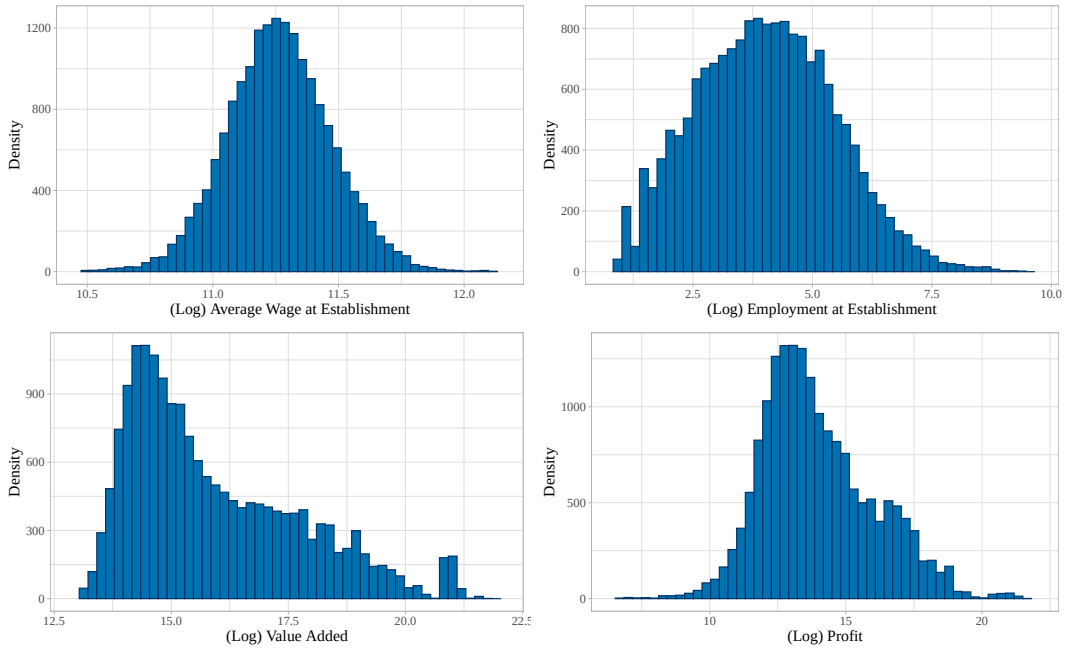
□

D. Data Preparation, Robustness Analyses, and Additional Results

D.1 Details about the Estimation Sample

Figure A.1 plots the empirical distributions of log wages, log labor, log value added, and log profits for the main estimation sample. Note that the distribution of log labor is truncated, as the estimation sample is restricted to firms that maintain at least five full-time employees.

Figure A.1: Empirical Distributions of Wages, Labor, Value Added, and Profits



Notes. This figure plots histograms of wages, labor, value added, and profits (in logs) for the estimation sample.

Additional Data Sources: External Instruments

Public procurement contracts in Norway are enforced by the Public Procurement Act and associated regulations. The laws apply to procurement of products, services, and construction

contracts valued at over NOK 100,000, and they cover all public sector entities in Norway. All public sector procurement is competitive, and procurement auctions are publicly announced on Doffin.no. The number and scale of announcements depends on the size of each contract.

H. de Frahan et al. (2024) collect data on contract award announcements from Doffin.no during the period from 2003 to 2018, encompassing over 60,000 announcements. The data collection process involves three main steps. First, the researchers obtain HTML files through the Doffin IT department, which contain the full history of contract awards for the specified period. Second, they employ standard scraping techniques to extract information from these HTML files. For each contract, various characteristics are collected, including the name of the winning firm, the size and date of the contract, and the product specifications. Notably, the Doffin files do not provide information on the names of losing bidders. Also, for 90% of the contract award announcements, the entire tax identifier of the winning firm is missing.

In the third step, the researchers utilize a fuzzy string matching procedure, following Raffo and Lhuillery (2009), to link firm names in the Doffin data with government registers that include both the names and tax identifiers of all Norwegian firms. This matching process yields results for approximately 30,000 contracts. The researchers assess the accuracy of the fuzzy matching procedure by analyzing the 10% of contract awards for which both the name and tax identifier of the winning firm are available. They find that the algorithm correctly assigns the tax identifier in 93% of cases. The accuracy of this matching procedure does not appear to be correlated with contract characteristics, such as the value or number of bidders.

The estimation sample is defined using the same data and following similar protocols as those used to analyze the internal instrument. Firms are classified as "treated" when they win their first procurement contracts through Doffin. All firms that receive treatment within the same calendar year τ are categorized into a "cohort". The control group for a given cohort consists of firms that win their first procurement contract in year $\tau' > \tau + 3$ or never. Future winner control firms are drawn from the Doffin data, and never winners are drawn from the full firm sample. Just as with the main sample, the data is restricted to full-time workers and firms that remain operational, maintaining at least five employees, for nine consecutive years. In total, the Doffin sample is composed of 25,714 unique firms and 901,811 unique workers.

D.2 Details about Estimation of Labor Supply Elasticities

To estimate the firm-specific labor supply elasticities, I use a nonparametric Kernel estimator:

$$\widehat{\mathrm{DiD}}_{\tau_0,\tau_1}(w|\chi) = \frac{\sum_j K_{1,j}(w)\mathbf{1}\{Z_j = 1\}\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big) - \sum_j K_{0,j}(w)\mathbf{1}\{Z_j = 0\}\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big)}{\sum_j K_{1,j}(w)\mathbf{1}\{Z_j = 1\}\big(w^{\mathrm{eff}}_{j\tau_1}(\chi) - w^{\mathrm{eff}}_{j\tau_0}(\chi)\big) - \sum_j K_{0,j}(w)\mathbf{1}\{Z_j = 0\}\big(w^{\mathrm{eff}}_{j\tau_1}(\chi) - w^{\mathrm{eff}}_{j\tau_0}(\chi)\big)},$$

where I define $K_{z,j}(w)$ to be the Kernel weight for a firm j that has treatment status $z \in \{0, 1\}$.

Examples of kernel functions include the Gaussian and Uniform kernel, defined as follows:

$$\textit{Gaussian:} \quad K_{z,j}(w) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{w^{\text{eff}}_{j\tau_0}(\chi) - w}{h}\right)^2\right].$$

$$\textit{Uniform:} \quad K_{z,j}(w) = \frac{\mathbb{1}\left\{w - h \leq w^{\text{eff}}_{j\tau_0}(\chi) \leq w + h\right\}}{\sum_j \mathbb{1}\left\{w - h \leq w^{\text{eff}}_{j\tau_0}(\chi) \leq w + h\right\}}.$$

$$\textit{Epanechnikov:} \quad K_{z,j}(w) = \frac{\frac{3}{4}\left[1 - \left(\left(w^{\text{eff}}_{j\tau_0}(\chi) - w\right)/h\right)^2\right]\mathbb{1}\left\{\left|\left(w^{\text{eff}}_{j\tau_0}(\chi) - w\right)/h\right| \leq 1\right\}}{\sum_j \frac{3}{4}\left[1 - \left(\left(w^{\text{eff}}_{j\tau_0}(\chi) - w\right)/h\right)^2\right]\mathbb{1}\left\{\left|\left(w^{\text{eff}}_{j\tau_0}(\chi) - w\right)/h\right| \leq 1\right\}}.$$

Here, the tuning parameter h determines the bandwidth of the estimator. As h tends to zero:

$$\lim_{h\to 0} \widehat{\text{DiD}}_{\tau_0,\tau_1}(w|\chi) = \frac{\frac{1}{N_1}\sum_j \mathbf{1}\{Z_j = 1\}\left(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\right) - \frac{1}{N_0}\sum_j \mathbf{1}\{Z_j = 0\}\left(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\right)}{\frac{1}{N_1}\sum_j \mathbf{1}\{Z_j = 1\}\left(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\right) - \frac{1}{N_0}\sum_j \mathbf{1}\{Z_j = 0\}\left(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\right)},$$

where $N_0 = \sum_j \mathbf{1}\{Z_j = 0\}$ and $N_1 = \sum_j \mathbf{1}\{Z_j = 1\}$. Also, by the weak law of large numbers:

$$\frac{1}{N_z}\sum_j \mathbf{1}\{Z_j = z\}\left(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\right) \xrightarrow{p} \text{E}\left[\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\middle|Z_j = z, w^{\text{eff}}_{j\tau_0}(\chi) = w\right], \quad \text{and:}$$

$$\frac{1}{N_z}\sum_j \mathbf{1}\{Z_j = z\}\left(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\right) \xrightarrow{p} \text{E}\left[w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\middle|Z_j = z, w^{\text{eff}}_{j\tau_0}(\chi) = w\right],$$

for $z \in \{0, 1\}$. Furthermore, using the continuous mapping theorem, I obtain the property:

$$\lim_{h\to 0} \widehat{\text{DiD}}_{\tau_0,\tau_1}(w|\chi) \xrightarrow{p} \text{DiD}_{\tau_0,\tau_1}(w|\chi) \quad \text{as } J \to \infty.$$

D.3. Robustness Analyses

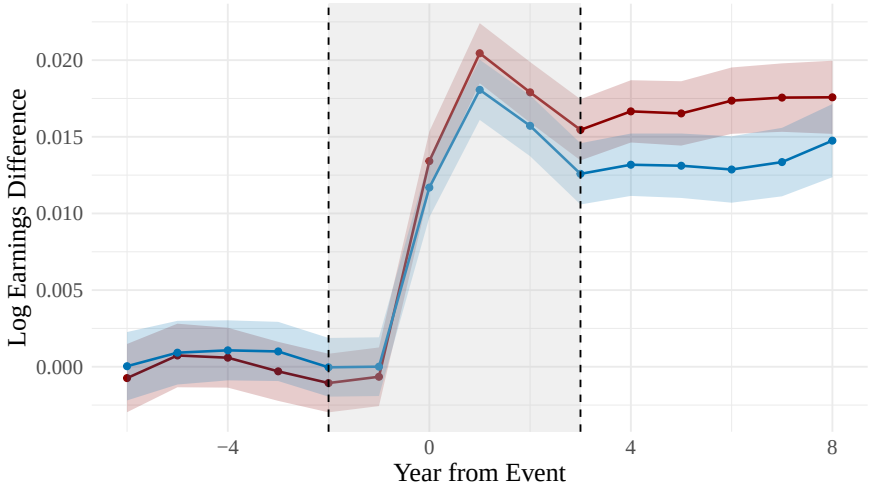
The first set of robustness analyses provide additional evidence to support the validity of the IV design. In Figure A.3, I plot the pass-through and reduced form estimates for the internal instrument by workers' college attainment and experience. In Figure A.2, I present the event study plots of the pass-through of internal instruments onto wages before and after the shock.

The second set of robustness analyses are misspecification tests. First, I examine whether the instrument Z affects labor supply on the intensive margin through hours worked, and find no statistically significant effect. Second, I test whether the IV estimates differ between firms with local market shares below versus above 0.05. I find no significant difference, supporting the assumption that firms do not internalize their market shares when setting workers' wages.

The third set of robustness analyses replicates the main IV estimates presented in Figure 5 under alternative specifications. In Figure A.4, I show a version of the estimates where I drop

market fixed effects, effectively treating Norway as one single labor market. In Figure A.5, I report estimates based on a restricted sample where firms only operate a single establishment. In Figures A.6, I give estimates for a restricted sample of firms with below-median union membership, addressing concerns that the wage-posting model may be less applicable in highly unionized settings. In Figures A.7 and A.8, I show estimates that use different Kernel estimators: Uniform and Epanechnikov kernel functions, respectively. In Figure A.9, I report estimates for a restricted sample of firms with local market shares below 0.05, addressing concerns about strategic wage-setting by high-share firms. In Figures A.10 and A.11, I show estimates computed using alternative bandwidth choices h. Lastly, in Figures A.12 and A.13, I present estimates based on the external instruments, which are described in Appendix D.1.

Figure A.2: Event Study—Pass-through of Productivity Shock to Wages



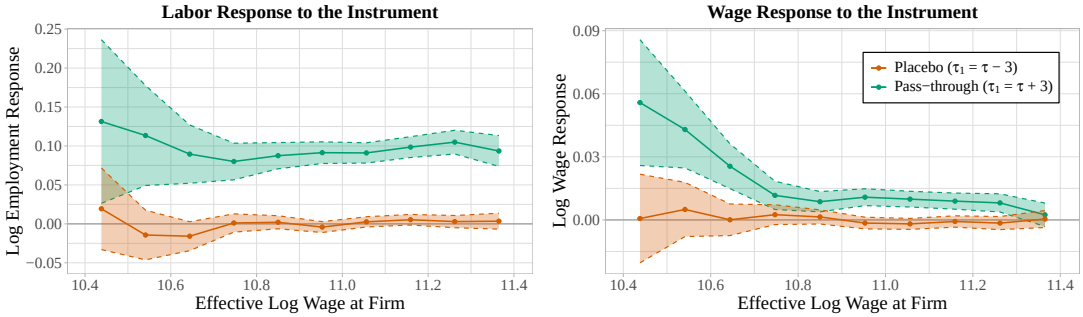
Notes. This figure displays the mean differences in log earnings (dotted lines) between firms with above-median versus below-median changes in log value added at event time zero. Estimates are shown with market fixed effects (red) and without market fixed effects (blue). The shaded area indicates time periods where the orthogonality condition need not hold in the identification of the permanent pass-through rate.

Table A.1: Two Misspecification Tests

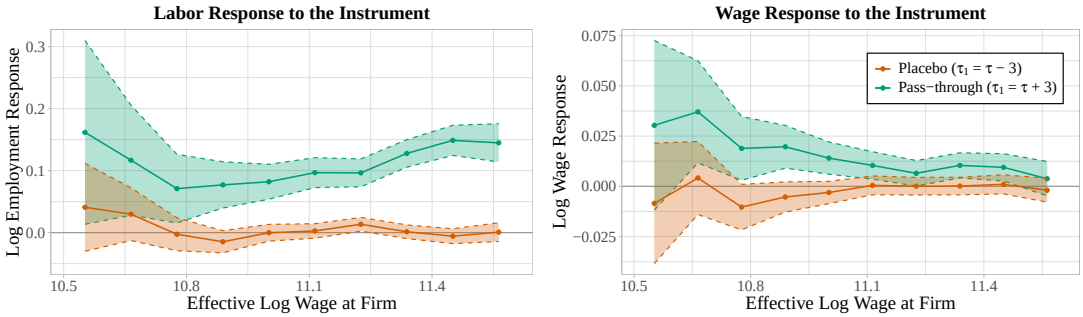
	Effect of Z on Work Hours		Market Share Effect	
	Estimated Effect	p-value	Estimated Effect	p-value
Market FEs	-0.011	0.105	1.503	0.183
	(0.007)		(1.666)	
No Market FEs	-0.003	0.624	-0.144	0.535
	(0.007)		(1.633)	

Notes. This table presents two misspecification tests. The first two columns report the estimated effect of Z on worker's scheduled hours. The third and fourth columns report the effect of having a local market share of above 0.05 on the labor supply elasticities. Both tests are conducted with and without market fixed effects.

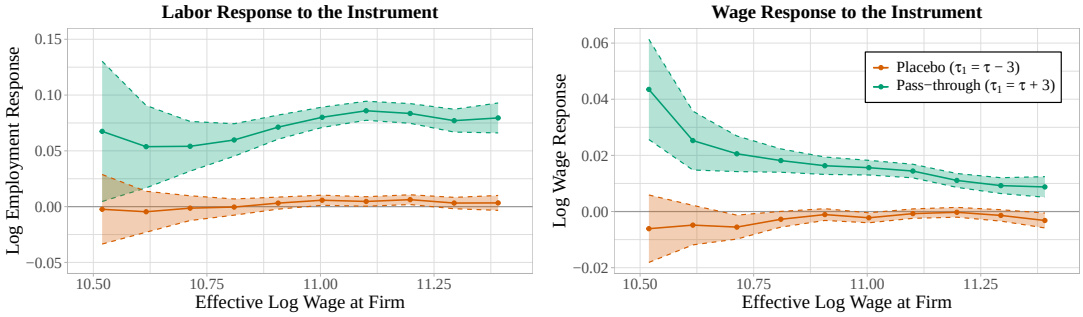
Figure A.3: Average Pass-Through by Education and Experience



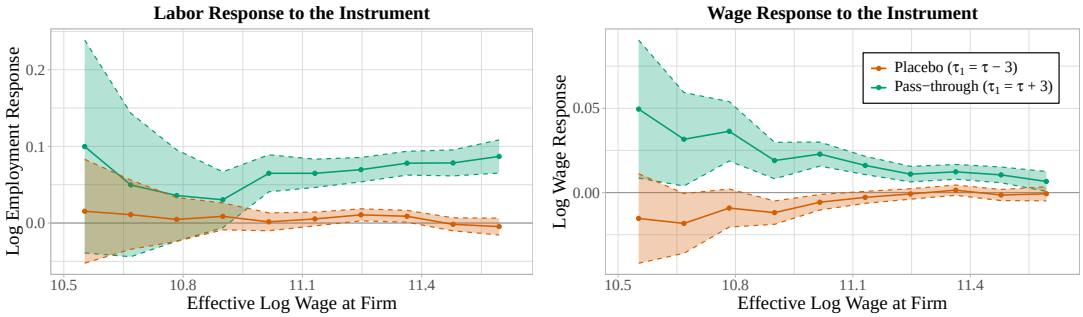
(a) No College Degree, < 10 Years Experience



(b) College Graduate, < 10 Years Experience



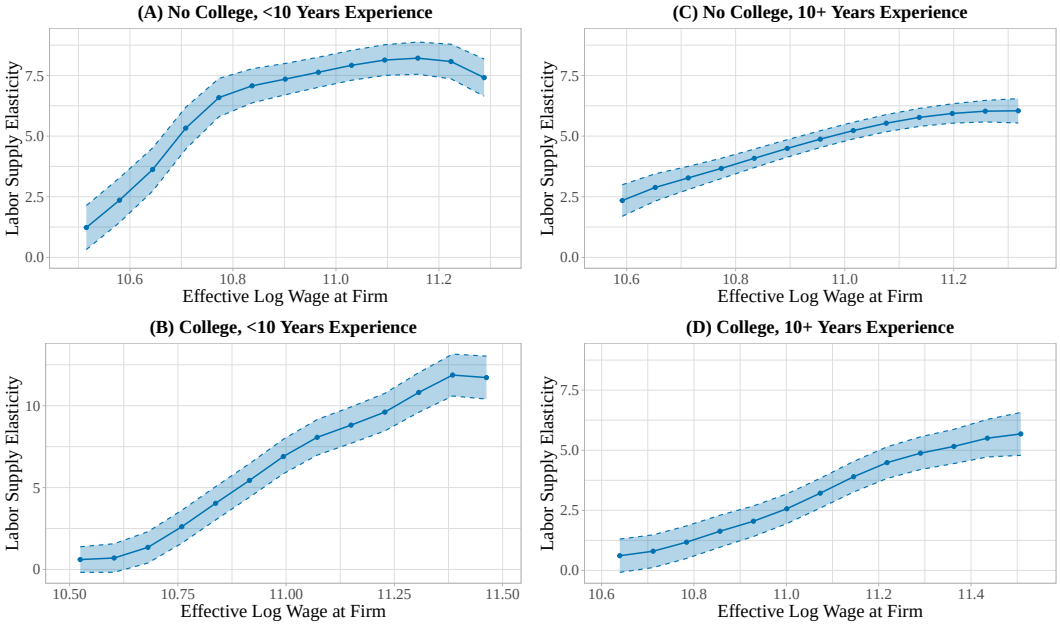
(c) No College Degree, ≥ 10 Years Experience



(d) College Graduate, ≥ 10 Years Experience

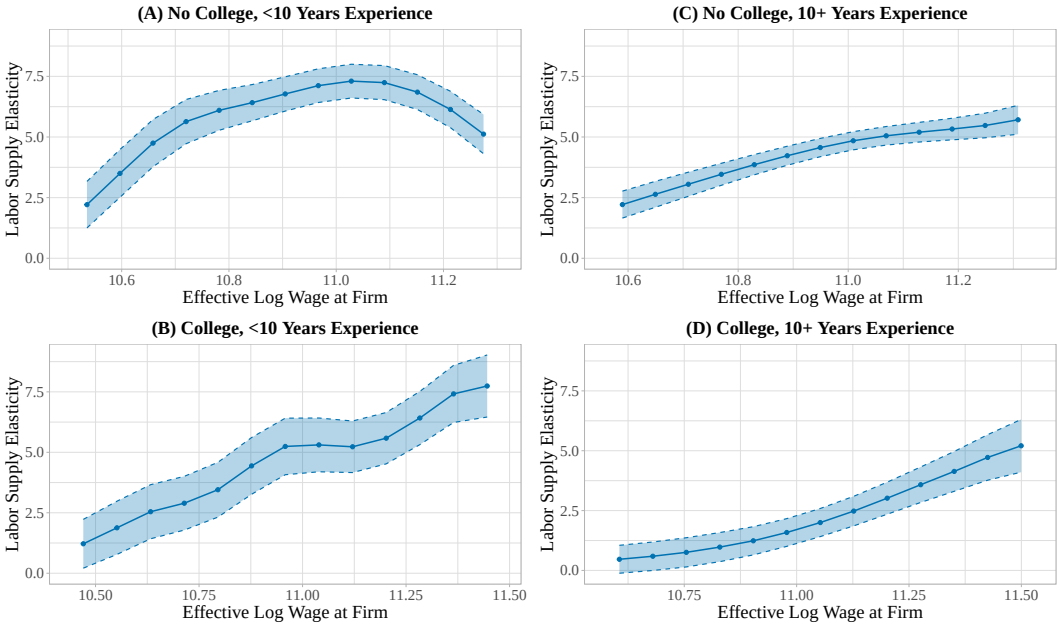
Notes. This figure plots the reduced form estimates by worker education and experience, averaged across years. This specification includes market fixed effects. Standard errors are bootstrapped using 500 bootstrap samples.

Figure A.4: IV Estimates of Labor Supply Elasticities—No Market Fixed Effects



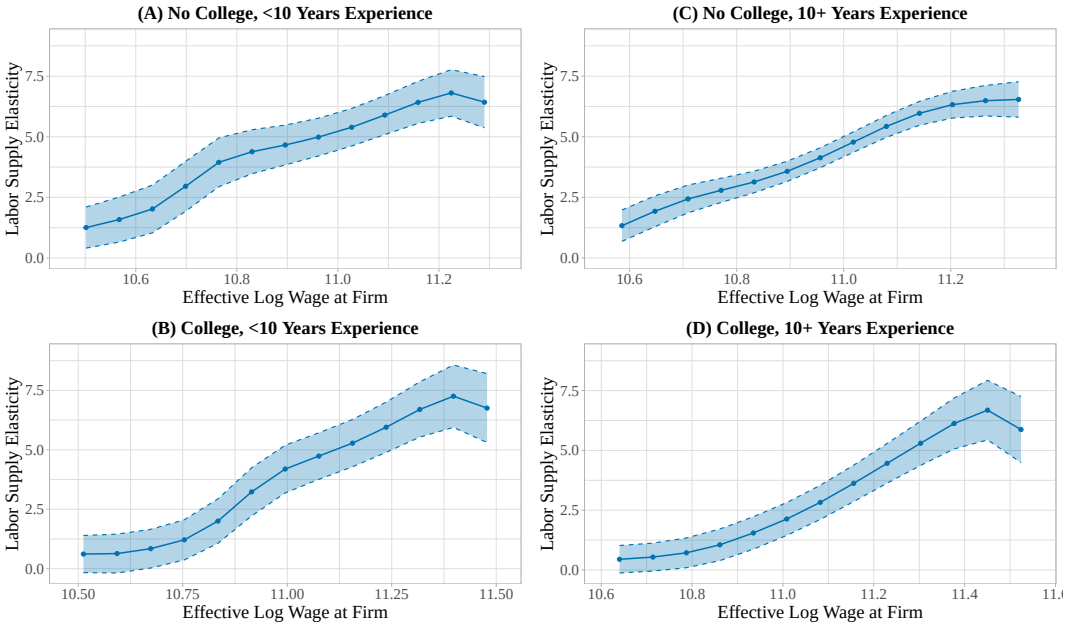
Notes. This figure presents IV estimates of firm-specific labor supply elasticities, excluding market fixed effects.

Figure A.5: IV Estimates of Labor Supply Elasticities—Single-Establishment Firms



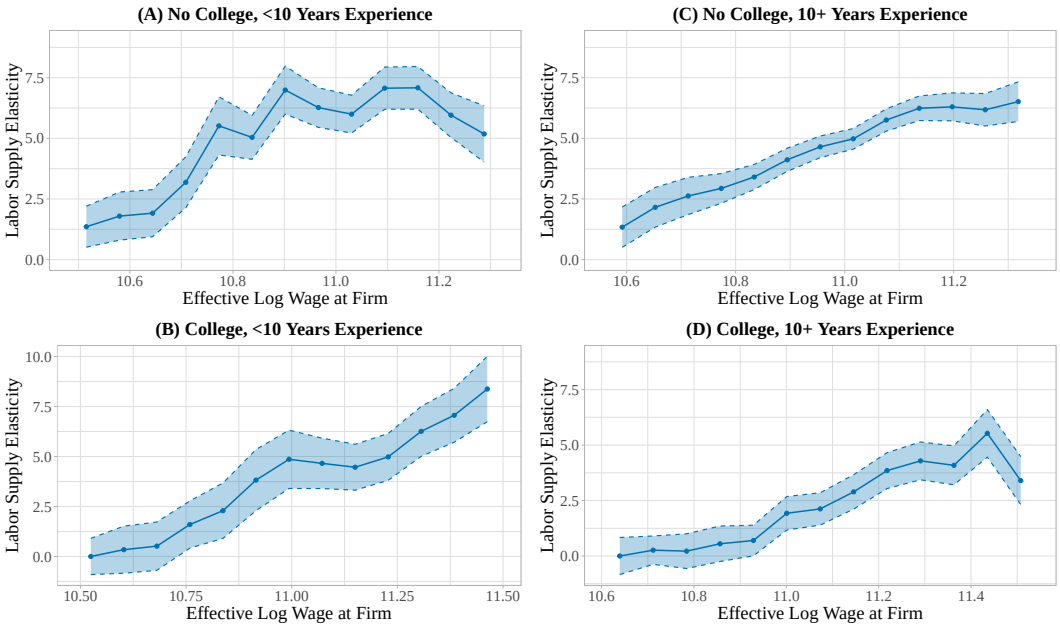
Notes. This figure plots IV estimates of firm-specific labor supply elasticities for single-establishment firms.

Figure A.6: IV Estimates of Labor Supply Elasticities—Low Union Membership Rate



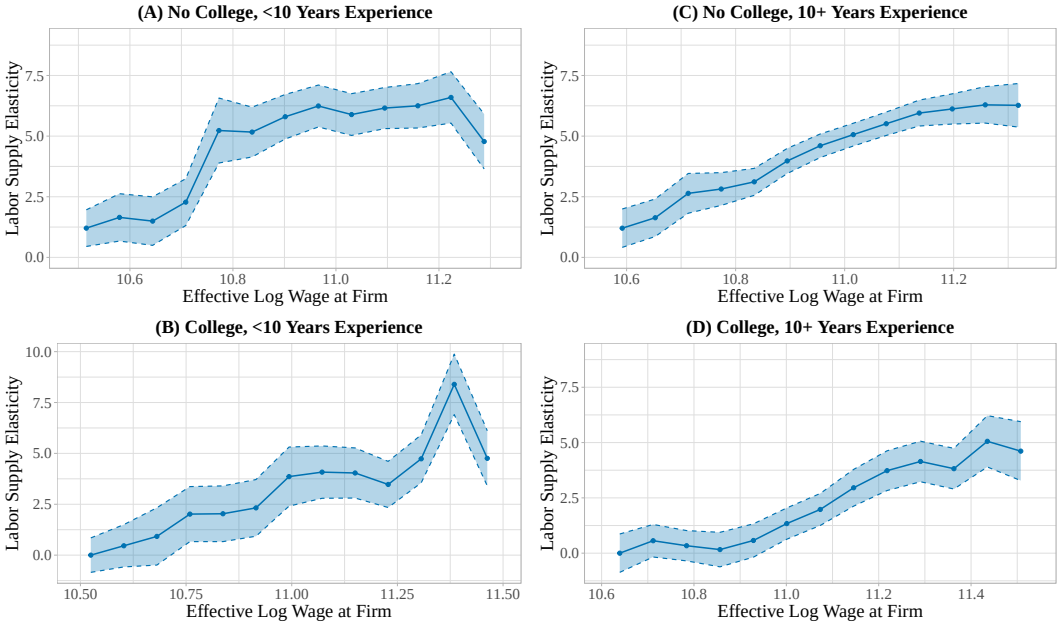
Notes. This figure plots IV estimates of firm-specific labor supply elasticities for a sample restricted to local labor markets with below-median union membership rates. This excludes education, healthcare, and manufacturing.

Figure A.7: IV Estimates of Labor Supply Elasticities—Uniform Kernel Estimator



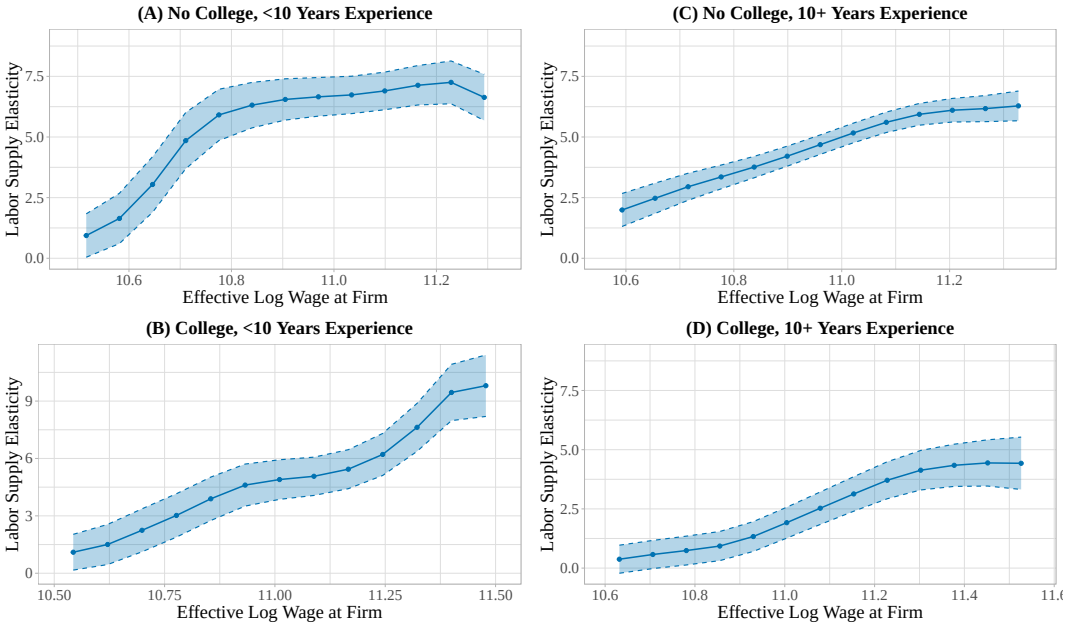
Notes. This figure plots IV estimates of firm-specific labor supply elasticities using a Uniform Kernel estimator.

Figure A.8: IV Estimates of Labor Supply Elasticities—Epanechnikov Kernel Estimator



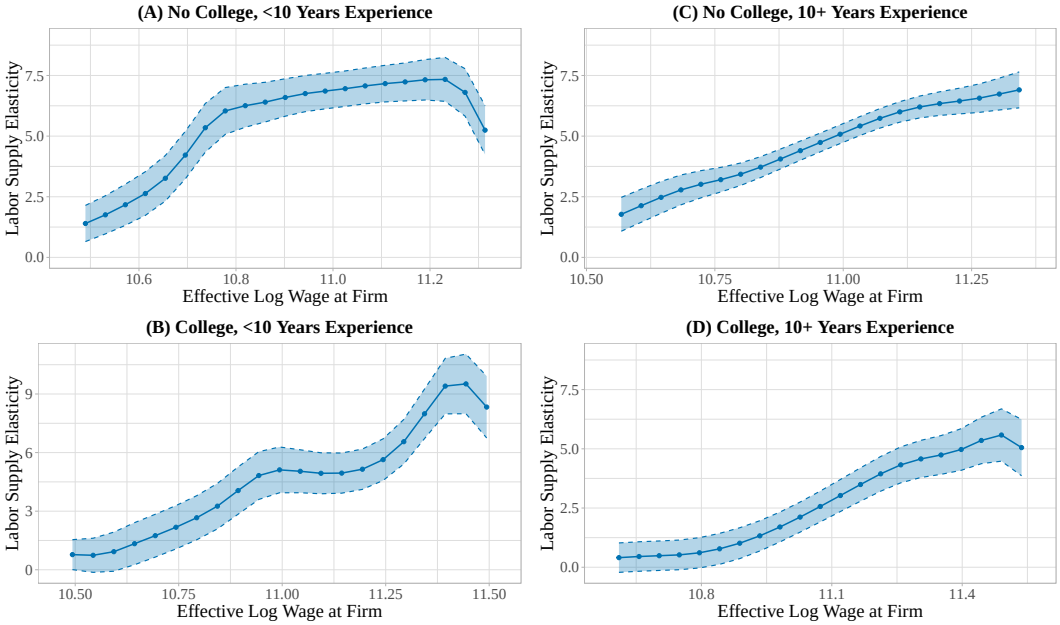
Notes. This figure plots IV estimates of firm-specific labor supply elasticities using an Epanechnikov Kernel.

Figure A.9: IV Estimates of Labor Supply Elasticities—Small Market Shares



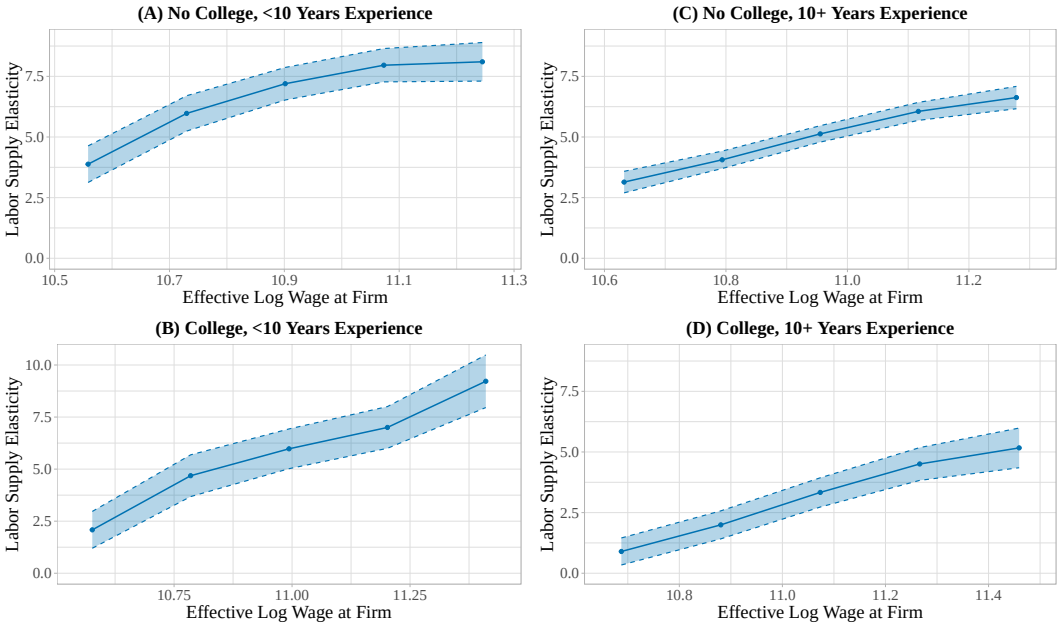
Notes. This figure plots IV estimates of firm-specific labor supply elasticities for firms with market shares < 0.05.

Figure A.10: IV Estimates of Labor Supply Elasticities—Large Kernel Bandwidth



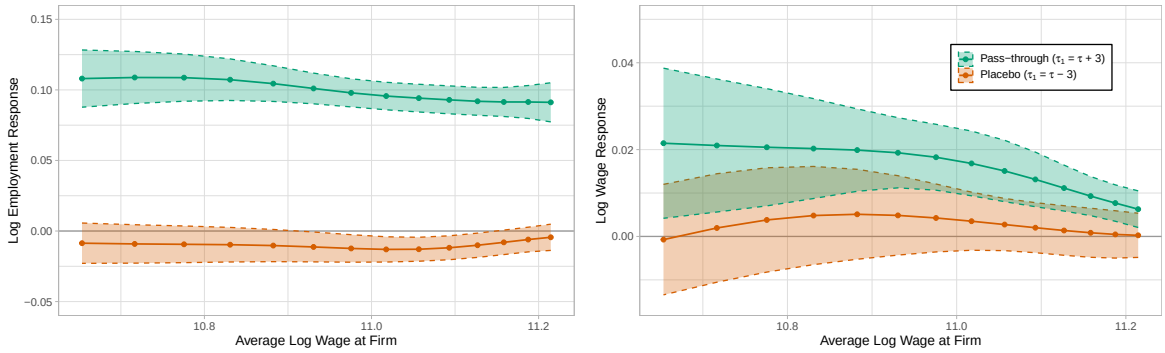
Notes. This figure plots IV estimates of firm-specific labor supply elasticities using a large kernel bandwidth.

Figure A.11: IV Estimates of Labor Supply Elasticities—Small Kernel Bandwidth



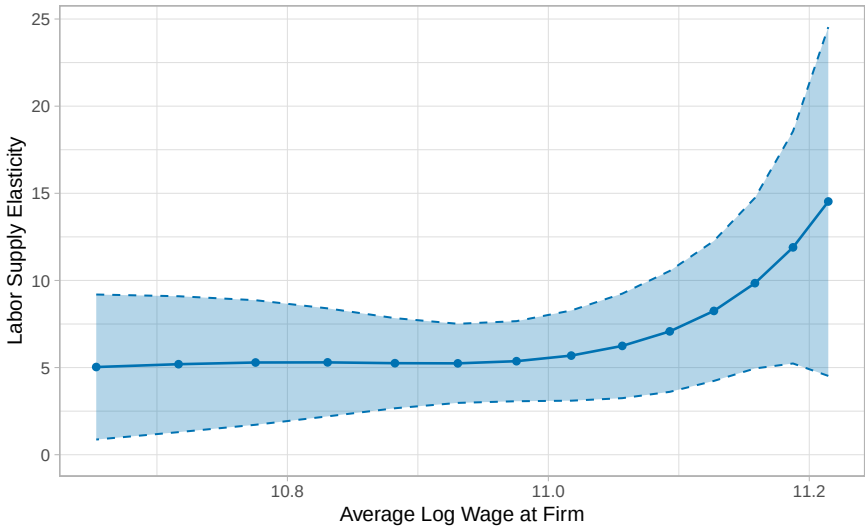
Notes. This figure plots IV estimates of firm-specific labor supply elasticities using a small kernel bandwidth.

Figure A.12: Average Pass-through Estimates—Procurement Auction Instrument



Notes. This figure plots the first stage and reduced form estimates for the external instrument, averaged across skill types and years. These estimates are constructed using the estimation strategy in H. de Frahan et al. (2024).

Figure A.13: IV Estimates of Labor Supply Elasticities—Procurement Auction Instrument

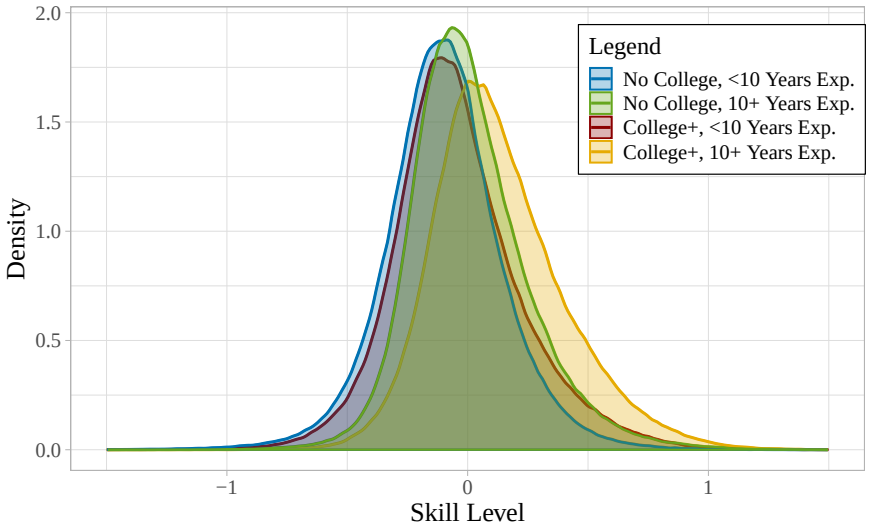


Notes. This figure plots the IV estimates of firm-specific labor supply elasticities for the external instrument, averaged across skills and years. These estimates follow the estimation strategy from H. de Frahan et al. (2024).

D.4. Additional Tables and Figures

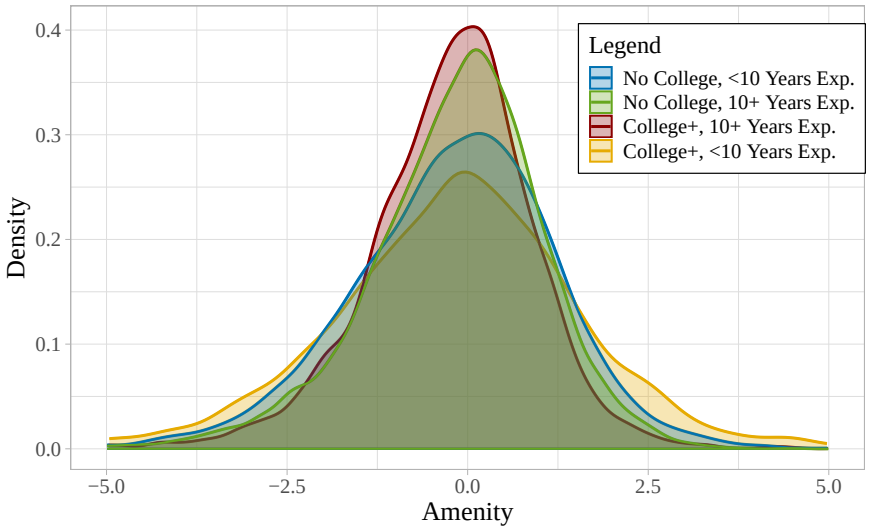
This Appendix section provides additional tables and figures not included in the main text.

Figure A.14: Density of Worker Skill Levels by Education and Experience



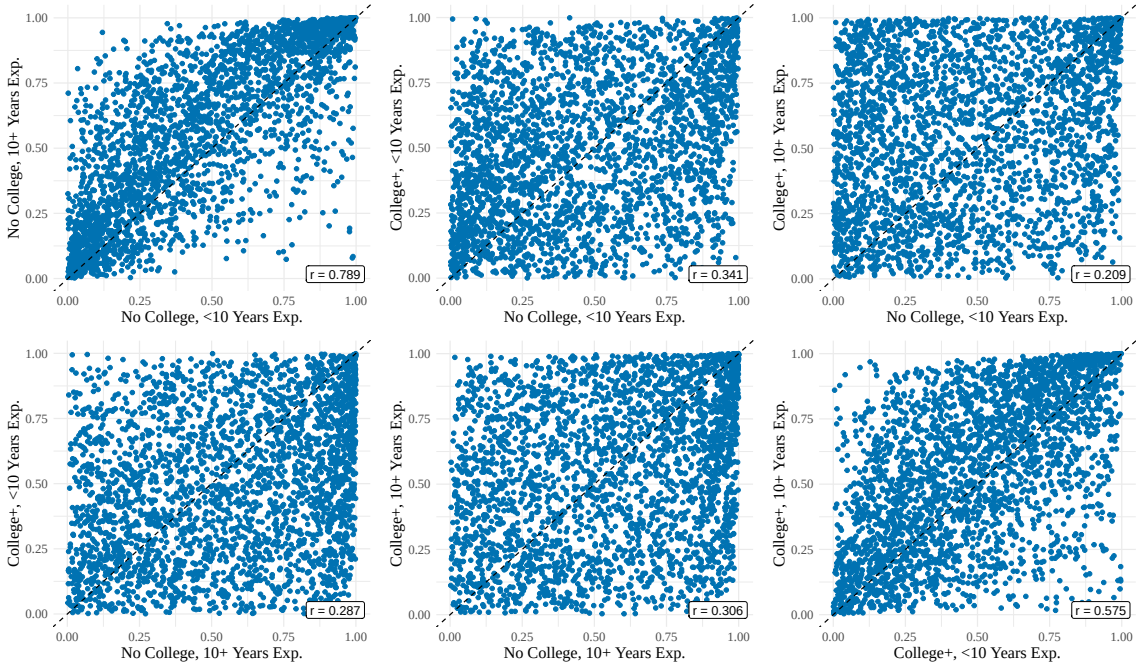
Notes. This figure plots the the estimated distributions of worker skill levels by college attainment and experience.

Figure A.15: Density of Non-Wage Amenities by Education and Experience



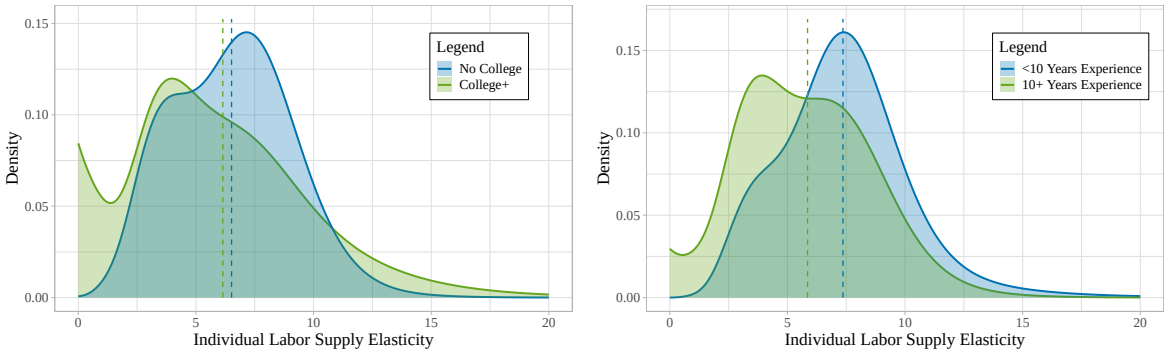
Notes. This figure plots the estimated distributions of non-wage amenities by college attainment and experience.

Figure A.16: Rank-Rank Plots of Amenities by Education and Experience



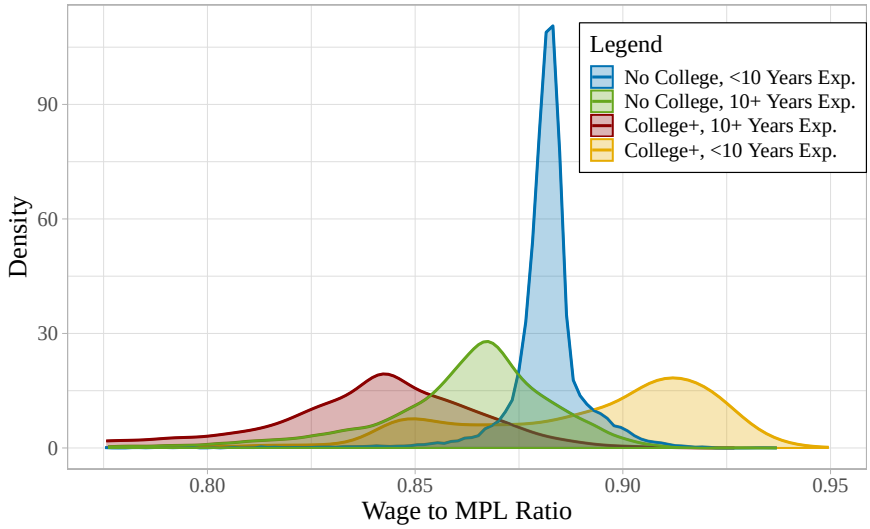
Notes. This figure presents rank–rank plots of non-wage employer amenities by workers' college attainment and experience, along with the Spearman rank correlations (r) that summarize the similarity of these rankings.

Figure A.17: Density of Individual Labor Supply Elasticities by Education and Experience



Notes. This figure plots the estimated densities f_β of workers' wage-amenity trade-offs by education and experience. To compute f_β, I first estimate the firm-specific labor supply curves by skill type, and then I implement the estimator from equation (30). The dashed lines in each figure indicate the means of these distributions.

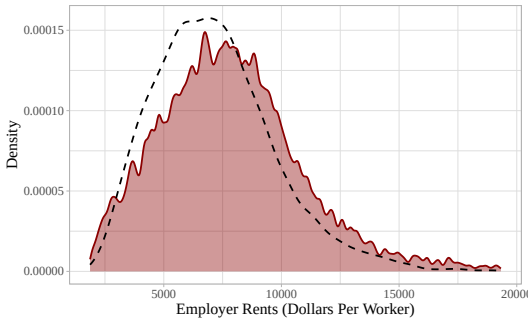
Figure A.18: Density of Wage Markdowns by Education and Experience



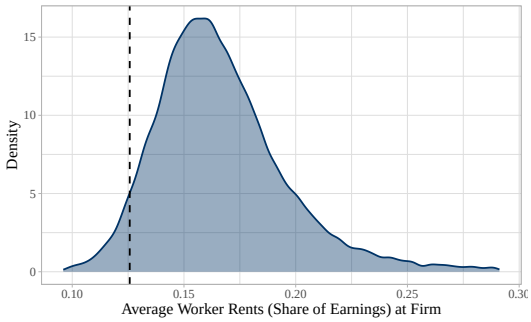
Notes. This figure plots the estimated distributions of wage markdowns by college attainment and experience.

Figure A.19: Estimated Distributions of Worker and Firm Rents

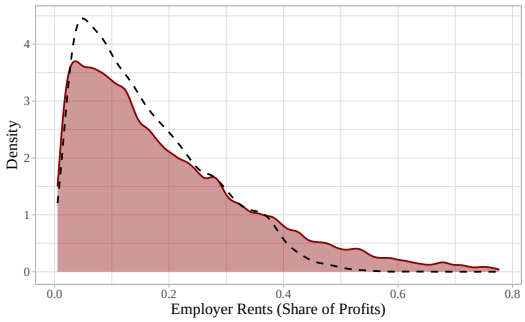
(a) Worker Rents



(b) Firm Rents



(c) Worker Rents (as Share of Earnings)



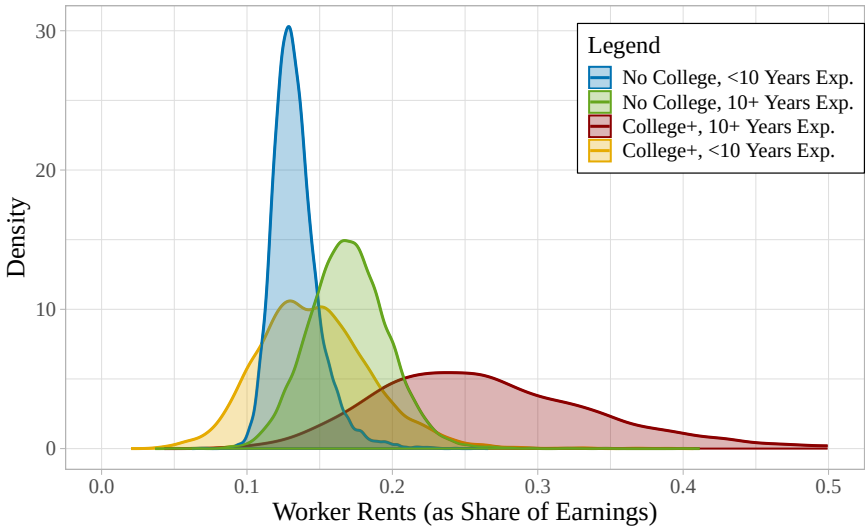
(d) Firm Rents (as Share of Profits)

Notes. This figure plots the distributions of estimated firm rents and estimated average worker rents at a firm. The dotted black lines display the estimates implied by a model with homogeneous β, i.e., where $\text{Var}(\beta) = 0$.

Table A.2: Estimates of Wage Markdowns by Industry and Region

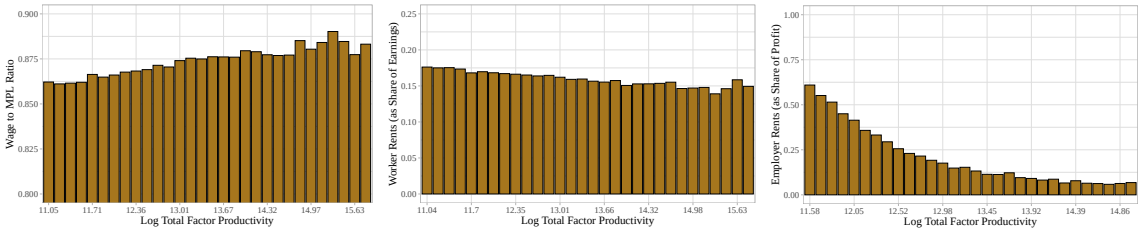
	Firm-Specific Labor Supply Elasticity		Implied Wage-to-MPL Ratio	
	Mean	Std. Dev.	Mean	Std. Dev.
Panel A. Broad Industry Group				
Business/Finance	7.745	2.402	0.877	0.037
Construction	7.053	1.088	0.873	0.022
Education	7.088	1.755	0.870	0.037
Healthcare	6.411	1.745	0.857	0.039
Manufacturing	6.840	1.401	0.868	0.027
Services	6.530	1.499	0.861	0.038
Trade	7.003	1.454	0.871	0.028
Utilities/Transport	7.100	1.345	0.873	0.023
Panel B. Broad Region				
Rural Region	5.725	1.153	0.846	0.031
Urban Region	7.726	3.070	0.874	0.035

Figure A.20: Density of Worker Rents (as Share of Earnings) by Education and Experience



Notes. This figure plots the estimated distributions of worker rent shares by college attainment and experience.

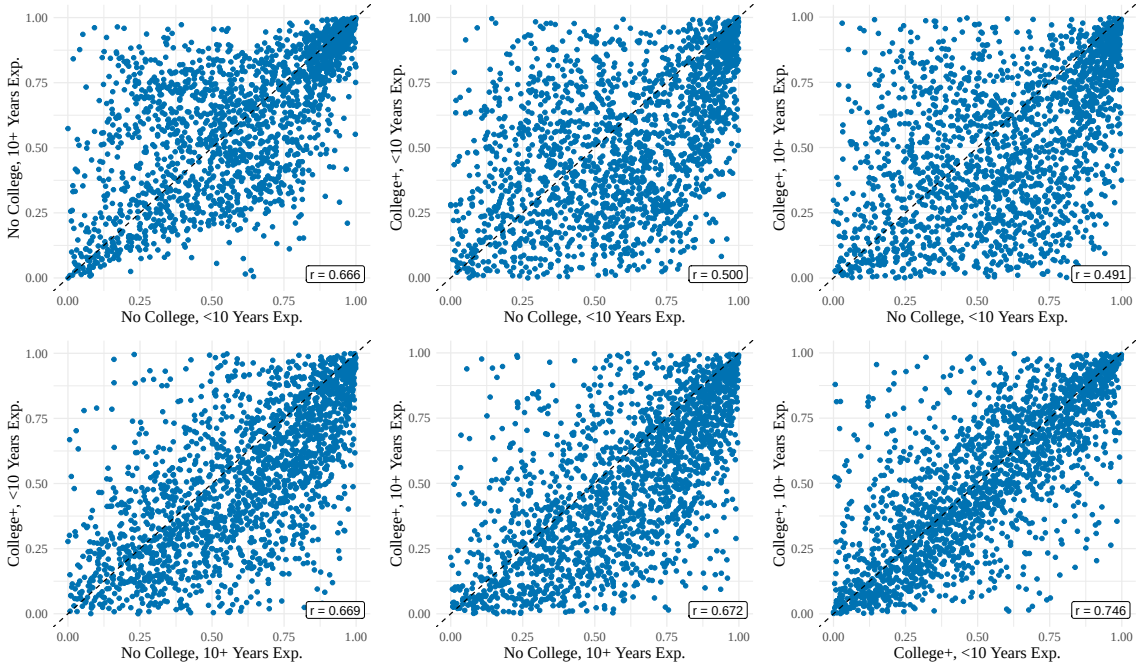
Figure A.21: Incidence of Wage Markdowns and Rents by Firm TFP and Amenities



(a) Wage Markdowns (Wage to MPL Ratio) (b) Worker Rents (as Share of Earnings) (c) Firm Rents (as Share of Profit)

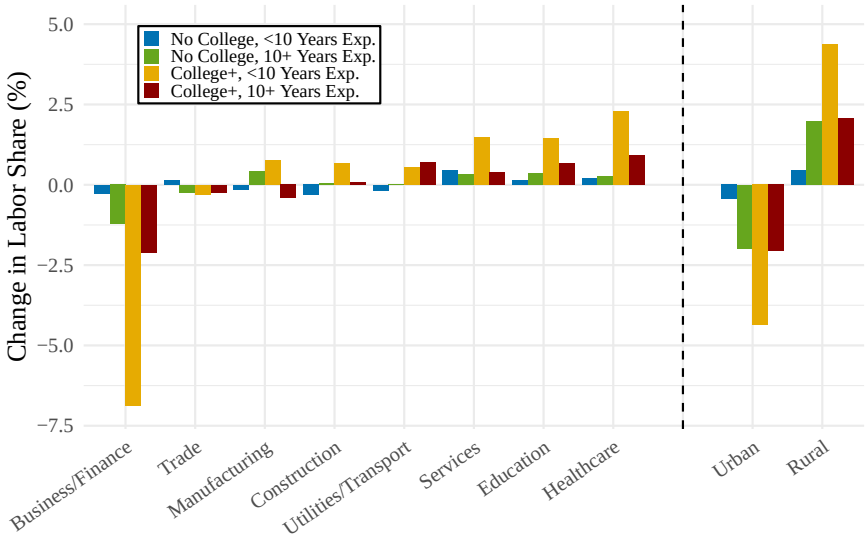
Notes. Each row shows the relationship between the outcome (wage markdown, worker rents, or firm rents) and TFP at the firm level. Firm rents are measured as a share of profits; worker rents as a share of earnings.

Figure A.22: Rank-Rank Plots of Wage Markdowns by Education and Experience



Notes. This figure presents rank–rank plots of wage-to-marginal-product ratios by workers' college attainment and experience, along with the Spearman rank correlations (r) that summarize the similarity of these rankings.

Figure A.23: Estimated Reallocation of Labor by Eliminating Labor Wedges



Notes. This figure shows the change in labor shares by sector and region resulting from eliminating labor wedges.

Figure A.24: Wage Markdown Distribution by Workers' Earnings

Notes. This figure plots the estimated mean wage markdown across the realized log-wage distribution, with workers grouped into wage bins. Bars indicate the 1st and 99th percentiles of the wage markdown distribution in each bin.